GRADE 8 COMMON CORE MATH

Answer Key

Main • Backward Branches • Forward 1 (Grade 9) • Forward 2 (Grade 10)

58 question sets • 1450 questions

Page numbers match the Question Collection.

Every page shows its question ID (for example S40-F2-Q3).

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SET 1 • 8.NS.A.1 • THE NUMBER SYSTEM

Classify numbers as rational or irrational

MAIN	Grade 8 main question	**8.NS.A.1**
BACKWARD 1	Write a fraction as a terminating or repeating decimal	**7.NS.A.2.d**
BACKWARD 2	Write rational numbers as quotients of integers	**7.NS.A.2.b**
FORWARD 1	Decide whether sums and products are rational or irrational	**HSN-RN.B.3**
FORWARD 2	Extend the number system to complex numbers using i	**HSN-CN.A.1**

SET 1 • 8.NS.A.1

MAIN

Grade 8 main question • Grade 8 • The Number System • 8.NS.A.1

Which number is irrational?

A. $\sqrt{49}$

B. $\frac{-7}{3}$

C. 0.1212... (the digits 12 repeat)

D. $\sqrt{15}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q1

Grade 8 • The Number System • 8.NS.A.1 • Set 1 • MAIN • Q1/5

8

QUESTION ID

S1-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

D. √15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q1

Grade 8 • The Number System • 8.NS.A.1 • Set 1 • MAIN • Q1/5

Is 0.375 rational or irrational? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q2

Grade 8 • The Number System • 8.NS.A.1 • Set 1 • MAIN • Q2/5

9

QUESTION ID

S1-M-Q2

TYPE

Short answer

CORRECT ANSWER

Rational. It equals 3/8, a quotient of integers (its decimal ends).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q2

Grade 8 • The Number System • 8.NS.A.1 • Set 1 • MAIN • Q2/5

Is √50 rational or irrational? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q3

Grade 8 • The Number System • 8.NS.A.1 • Set 1 • MAIN • Q3/5

10

QUESTION ID

S1-M-Q3

TYPE

Short answer

CORRECT ANSWER

Irrational. 50 is not a perfect square, so √50 cannot be written as a quotient of integers; its decimal neither ends nor repeats.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q3

Grade 8 • The Number System • 8.NS.A.1 • Set 1 • MAIN • Q3/5

True or false?

Every number whose decimal expansion repeats is rational.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q4

Grade 8 • The Number System • 8.NS.A.1 • Set 1 • MAIN • Q4/5

11

QUESTION ID

S1-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q4

Grade 8 • The Number System • 8.NS.A.1 • Set 1 • MAIN • Q4/5

True or false?

π is rational because it is often written as 3.14.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q5

Grade 8 • The Number System • 8.NS.A.1 • Set 1 • MAIN • Q5/5

12

QUESTION ID

S1-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 3.14 is only an approximation; π is irrational.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q5

Grade 8 • The Number System • 8.NS.A.1 • Set 1 • MAIN • Q5/5

SET 1 • 8.NS.A.1

BACKWARD 1

Write a fraction as a terminating or repeating decimal • Grade 7 • The Number System • 7.NS.A.2.d

Write $\frac{5}{8}$ as a decimal.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q1

Grade 7 • The Number System • 7.NS.A.2.d • Set 1 • BACKWARD 1 • Q1/5

14

QUESTION ID

S1-B1-Q1

TYPE

Short answer

CORRECT ANSWER

0.625

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q1

Grade 7 • The Number System • 7.NS.A.2.d • Set 1 • BACKWARD 1 • Q1/5

Write $\frac{2}{3}$ as a decimal.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q2

Grade 7 • The Number System • 7.NS.A.2.d • Set 1 • BACKWARD 1 • Q2/5

15

QUESTION ID

S1-B1-Q2

TYPE

Short answer

CORRECT ANSWER

0.666... (the 6 repeats)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q2

Grade 7 • The Number System • 7.NS.A.2.d • Set 1 • BACKWARD 1 • Q2/5

Which fraction has a repeating decimal?

A. $\frac{3}{5}$

B. $\frac{1}{6}$

C. $\frac{1}{4}$

D. $\frac{7}{8}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q3

Grade 7 • The Number System • 7.NS.A.2.d • Set 1 • BACKWARD 1 • Q3/5

16

QUESTION ID

S1-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 1/6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q3

Grade 7 • The Number System • 7.NS.A.2.d • Set 1 • BACKWARD 1 • Q3/5

True or false?

$\frac{4}{11}$ = 0.3636... (the digits 36 repeat)

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q4

Grade 7 • The Number System • 7.NS.A.2.d • Set 1 • BACKWARD 1 • Q4/5

17

QUESTION ID

S1-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q4

Grade 7 • The Number System • 7.NS.A.2.d • Set 1 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\frac{1}{3} = 0.3$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q5

Grade 7 • The Number System • 7.NS.A.2.d • Set 1 • BACKWARD 1 • Q5/5

18

QUESTION ID

S1-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 1/3 = 0.333..., a repeating decimal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q5

Grade 7 • The Number System • 7.NS.A.2.d • Set 1 • BACKWARD 1 • Q5/5

SET 1 • 8.NS.A.1

BACKWARD 2

Write rational numbers as quotients of integers • Grade 7 • The Number System • 7.NS.A.2.b

Write $-2\frac{1}{4}$ as a quotient of two integers.

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 1 • BACKWARD 2 • Q1/5

20

QUESTION ID

S1-B2-Q1

TYPE

Short answer

CORRECT ANSWER

−9/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 1 • BACKWARD 2 • Q1/5

Write 0.7 as a quotient of two integers.

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 1 • BACKWARD 2 • Q2/5

21

QUESTION ID

S1-B2-Q2

TYPE

Short answer

CORRECT ANSWER

7/10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 1 • BACKWARD 2 • Q2/5

Which number is equal to $\frac{-5}{6}$?

A. $\frac{5}{6}$

B. $\frac{-5}{-6}$

C. $\frac{6}{-5}$

D. $\frac{5}{-6}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 1 • BACKWARD 2 • Q3/5

22

QUESTION ID

S1-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 5/(−6)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 1 • BACKWARD 2 • Q3/5

True or false?

-12 can be written as $\frac{-12}{1}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 1 • BACKWARD 2 • Q4/5

23

QUESTION ID

S1-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 1 • BACKWARD 2 • Q4/5

True or false?

$\frac{-3}{0}$ is a rational number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 1 • BACKWARD 2 • Q5/5

24

QUESTION ID

S1-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Division by 0 is undefined.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 1 • BACKWARD 2 • Q5/5

SET 1 • 8.NS.A.1

FORWARD 1

Decide whether sums and products are rational or irrational • High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.B.3

Is 4 + √3 rational or irrational? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q1

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 1 • FORWARD 1 • Q1/5

26

QUESTION ID

S1-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Irrational. A rational number plus an irrational number is irrational.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q1

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 1 • FORWARD 1 • Q1/5

Is √8 × √2 rational or irrational? Give its value.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q2

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 1 • FORWARD 1 • Q2/5

27

QUESTION ID

S1-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Rational; √16 = 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q2

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 1 • FORWARD 1 • Q2/5

Which product is irrational?

A. $\sqrt{3} \times \sqrt{12}$

B. $3\sqrt{5}$

C. $0 \times \sqrt{7}$

D. $\frac{2}{3} \times 6$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q3

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 1 • FORWARD 1 • Q3/5

28

QUESTION ID

S1-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $3\sqrt{5}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q3

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 1 • FORWARD 1 • Q3/5

True or false?

The sum of two rational numbers is always rational.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q4

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 1 • FORWARD 1 • Q4/5

29

QUESTION ID

S1-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q4

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 1 • FORWARD 1 • Q4/5

True or false?

The product of two irrational numbers is always irrational.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q5

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 1 • FORWARD 1 • Q5/5

30

QUESTION ID

S1-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\sqrt{2} \times \sqrt{2} = 2$, which is rational.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S1-F1-Q5**

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 1 • FORWARD 1 • Q5/5

SET 1 • 8.NS.A.1

FORWARD 2

Extend the number system to complex numbers using i • High School (Grade 10 in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1

The number i has $i^2 = -1$.

Write $\sqrt{-16}$ in the form bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q1

High School (Grade 10 in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 2 • Q1/5

32

QUESTION ID

S1-F2-Q1

TYPE

Short answer

CORRECT ANSWER

4i

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S1-F2-Q1**

High School (Grade 10 in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 2 • Q1/5

The number i has $i^2 = -1$.

What is i^4?

$i^4 =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q2

High School (Grade 10 in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 2 • Q2/5

33

QUESTION ID

S1-F2-Q2

TYPE

Short answer

CORRECT ANSWER

1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S1-F2-Q2**

High School (Grade 10 in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 2 • Q2/5

The number i has $i^2 = -1$.

Which number is a solution of $x^2 = -9$?

A. 3

B. -3

C. 9i

D. 3i

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q3

High School (Grade 10 in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 2 • Q3/5

34

QUESTION ID

S1-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 3i

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q3

High School (Grade 10 in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 2 • Q3/5

True or false?

Every real number a is also a complex number, a + 0i.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q4

High School (Grade 10 in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 2 • Q4/5

35

QUESTION ID

S1-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q4

High School (Grade 10 in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 2 • Q4/5

True or false?

$\sqrt{-25}$ is a real number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q5

High School (Grade 10 in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 2 • Q5/5

36

QUESTION ID

S1-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\sqrt{-25} = 5i$, which is not real.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q5

High School (Grade 10 in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 2 • Q5/5

SET 2 • 8.NS.A.1 • THE NUMBER SYSTEM

Convert a repeating decimal into a fraction

MAIN	Grade 8 main question	**8.NS.A.1**
BACKWARD 1	Use long division to write a fraction as a repeating decimal	**7.NS.A.2.d**
BACKWARD 2	Multiply decimals by powers of 10	**5.NBT.A.2**
FORWARD 1	Solve linear equations whose solutions are fractions	**HSA-REI.B.3**
FORWARD 2	Write a geometric sequence from a repeating decimal	**HSF-BF.A.2**

SET 2 • 8.NS.A.1

MAIN

Grade 8 main question • Grade 8 • The Number System • 8.NS.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Write 0.444... (the 4 repeats) as a fraction. Show your work.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q1

Grade 8 • The Number System • 8.NS.A.1 • Set 2 • MAIN • Q1/5

39

QUESTION ID

S2-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

4/9

GRADING NOTE REQUIRED WORK: Any valid method. Example: let x = 0.444...; then 10x = 4.444...; subtracting gives 9x = 4, so x = 4/9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q1

Grade 8 • The Number System • 8.NS.A.1 • Set 2 • MAIN • Q1/5

Write 0.2727... (the digits 27 repeat) as a fraction in simplest form.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q2

Grade 8 • The Number System • 8.NS.A.1 • Set 2 • MAIN • Q2/5

40

QUESTION ID

S2-M-Q2

TYPE

Short answer

CORRECT ANSWER

3/11

GRADING NOTE 27/99 is equal but is not in simplest form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q2

Grade 8 • The Number System • 8.NS.A.1 • Set 2 • MAIN • Q2/5

Which fraction is equal to 0.1666... (the 6 repeats)?

A. $\frac{1}{16}$

B. $\frac{1}{6}$

C. $\frac{16}{100}$

D. $\frac{6}{10}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q3

Grade 8 • The Number System • 8.NS.A.1 • Set 2 • MAIN • Q3/5

41

QUESTION ID

S2-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 1/6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q3

Grade 8 • The Number System • 8.NS.A.1 • Set 2 • MAIN • Q3/5

True or false?

2.333... (the 3 repeats) is equal to $\frac{7}{3}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q4

Grade 8 • The Number System • 8.NS.A.1 • Set 2 • MAIN • Q4/5

42

QUESTION ID

S2-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q4

Grade 8 • The Number System • 8.NS.A.1 • Set 2 • MAIN • Q4/5

True or false?

0.0909... (the digits 09 repeat) is equal to $\frac{9}{100}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q5

Grade 8 • The Number System • 8.NS.A.1 • Set 2 • MAIN • Q5/5

43

QUESTION ID

S2-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is equal to 9/99 = 1/11.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q5

Grade 8 • The Number System • 8.NS.A.1 • Set 2 • MAIN • Q5/5

SET 2 • 8.NS.A.1

BACKWARD 1

Use long division to write a fraction as a repeating decimal • Grade 7 • The Number System • 7.NS.A.2.d

Write $\frac{7}{9}$ as a decimal.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q1

Grade 7 • The Number System • 7.NS.A.2.d • Set 2 • BACKWARD 1 • Q1/5

45

QUESTION ID

S2-B1-Q1

TYPE

Short answer

CORRECT ANSWER

0.777... (the 7 repeats)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q1

Grade 7 • The Number System • 7.NS.A.2.d • Set 2 • BACKWARD 1 • Q1/5

Write $\frac{5}{12}$ as a decimal.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q2

Grade 7 • The Number System • 7.NS.A.2.d • Set 2 • BACKWARD 1 • Q2/5

46

QUESTION ID

S2-B1-Q2

TYPE

Short answer

CORRECT ANSWER

0.41666... (the 6 repeats)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q2

Grade 7 • The Number System • 7.NS.A.2.d • Set 2 • BACKWARD 1 • Q2/5

Which decimal is equal to $\frac{3}{11}$?

A. 0.27 (the decimal ends)

B. 0.3

C. 0.2727... (the digits 27 repeat)

D. 0.311

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q3

Grade 7 • The Number System • 7.NS.A.2.d • Set 2 • BACKWARD 1 • Q3/5

47

QUESTION ID

S2-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 0.2727... (the digits 27 repeat)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q3

Grade 7 • The Number System • 7.NS.A.2.d • Set 2 • BACKWARD 1 • Q3/5

True or false?

$\frac{1}{9} = 0.111...$ (the 1 repeats)

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q4

Grade 7 • The Number System • 7.NS.A.2.d • Set 2 • BACKWARD 1 • Q4/5

48

QUESTION ID

S2-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q4

Grade 7 • The Number System • 7.NS.A.2.d • Set 2 • BACKWARD 1 • Q4/5

True or false?

$\frac{5}{6} = 0.56$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q5

Grade 7 • The Number System • 7.NS.A.2.d • Set 2 • BACKWARD 1 • Q5/5

49

QUESTION ID

S2-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 5/6 = 0.8333...

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q5

Grade 7 • The Number System • 7.NS.A.2.d • Set 2 • BACKWARD 1 • Q5/5

SET 2 • 8.NS.A.1

BACKWARD 2

Multiply decimals by powers of 10 • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2

Multiply.

0.454 × 100

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 2 • BACKWARD 2 • Q1/5

51

QUESTION ID

S2-B2-Q1

TYPE

Short answer

CORRECT ANSWER

45.4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 2 • BACKWARD 2 • Q1/5

Multiply.

3.27 × 10

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 2 • BACKWARD 2 • Q2/5

52

QUESTION ID

S2-B2-Q2

TYPE

Short answer

CORRECT ANSWER

32.7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 2 • BACKWARD 2 • Q2/5

Multiply.

0.0081 × 1,000

A. 81

B. 8.1

C. 0.81

D. 0.081

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 2 • BACKWARD 2 • Q3/5

53

QUESTION ID

S2-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 8.1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 2 • BACKWARD 2 • Q3/5

True or false?

$2.5 \times 100 = 250$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 2 • BACKWARD 2 • Q4/5

54

QUESTION ID

S2-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 2 • BACKWARD 2 • Q4/5

True or false?

$0.36 \times 10 = 36$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 2 • BACKWARD 2 • Q5/5

55

QUESTION ID

S2-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 2 • BACKWARD 2 • Q5/5

SET 2 • 8.NS.A.1

FORWARD 1

Solve linear equations whose solutions are fractions • High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

Solve for x. Write the answer as a fraction in simplest form.

$100x - x = 45$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 2 • FORWARD 1 • Q1/5

57

QUESTION ID

S2-F1-Q1

TYPE

Short answer

CORRECT ANSWER

5/11

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F1-Q1**

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 2 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve for x.

$10x = x + 6.3$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 2 • FORWARD 1 • Q2/5

58

QUESTION ID

S2-F1-Q2

TYPE

Short answer

CORRECT ANSWER

0.7

GRADING NOTE 7/10 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F1-Q2**

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 2 • FORWARD 1 • Q2/5

Solve for x.

$1{,}000x - 10x = 270$

A. $x = \frac{3}{11}$

B. $x = \frac{1}{37}$

C. $x = \frac{27}{100}$

D. $x = 27$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 2 • FORWARD 1 • Q3/5

59

QUESTION ID

S2-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. x = 3/11

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 2 • FORWARD 1 • Q3/5

True or false?

$x = \frac{4}{9}$ is the solution of $10x - x = 4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 2 • FORWARD 1 • Q4/5

60

QUESTION ID

S2-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 2 • FORWARD 1 • Q4/5

True or false?

$x = 0.12$ is the solution of $100x - x = 12$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 2 • FORWARD 1 • Q5/5

61

QUESTION ID

S2-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The solution is 12/99 = 4/33.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 2 • FORWARD 1 • Q5/5

SET 2 • 8.NS.A.1

FORWARD 2

Write a geometric sequence from a repeating decimal • High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2

0.333... = 0.3 + 0.03 + 0.003 + ... The terms 0.3, 0.03, 0.003, ... form a geometric sequence.

What is the common ratio?

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q1

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 2 • FORWARD 2 • Q1/5

63

QUESTION ID

S2-F2-Q1

TYPE

Short answer

CORRECT ANSWER

0.1

GRADING NOTE 1/10 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q1

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 2 • FORWARD 2 • Q1/5

Write an explicit formula for the nth term of the sequence 0.3, 0.03, 0.003, ...

a(n) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q2

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 2 • FORWARD 2 • Q2/5

64

QUESTION ID

S2-F2-Q2

TYPE

Short answer

CORRECT ANSWER

0.3(0.1)^(n − 1)

GRADING NOTE Equivalent forms such as 3(0.1)^n are correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F2-Q2**

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 2 • FORWARD 2 • Q2/5

Which recursive formula gives the sequence 0.27, 0.0027, 0.000027, ...?

A. $a(1) = 0.27;\ a(n) = 100 \cdot a(n-1)$

B. $a(1) = 0.27;\ a(n) = 0.01 \cdot a(n-1)$

C. $a(1) = 0.27;\ a(n) = a(n-1) - 0.01$

D. $a(1) = 0.0027;\ a(n) = 0.01 \cdot a(n-1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q3

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 2 • FORWARD 2 • Q3/5

65

QUESTION ID

S2-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $a(1) = 0.27;\ a(n) = 0.01 \cdot a(n-1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q3

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 2 • FORWARD 2 • Q3/5

True or false?

The sequence 0.5, 0.05, 0.005, ... is geometric with a common ratio of $\frac{1}{10}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q4

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 2 • FORWARD 2 • Q4/5

66

QUESTION ID

S2-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q4

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 2 • FORWARD 2 • Q4/5

True or false?

The sequence 0.4, 0.04, 0.004, ... is arithmetic.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q5

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 2 • FORWARD 2 • Q5/5

67

QUESTION ID

S2-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The terms have a common ratio, not a common difference.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q5

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 2 • FORWARD 2 • Q5/5

SET 3 • 8.NS.A.2 • THE NUMBER SYSTEM

Estimate the value of an irrational number

MAIN	Grade 8 main question	**8.NS.A.2**
BACKWARD 1	Find squares of whole numbers and decimals	**6.EE.A.1**
BACKWARD 2	Round decimals to the nearest tenth	**5.NBT.A.4**
FORWARD 1	Solve quadratic equations by taking square roots, exactly and ap…	**HSA-REI.B.4.b**
FORWARD 2	Find arc lengths and sector areas exactly in terms of π and appro…	**HSG-C.B.5**

SET 3 • 8.NS.A.2

MAIN

Grade 8 main question • Grade 8 • The Number System • 8.NS.A.2

Between which two whole numbers is √40?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q1

Grade 8 • The Number System • 8.NS.A.2 • Set 3 • MAIN • Q1/5

70

QUESTION ID

S3-M-Q1

TYPE

Short answer

CORRECT ANSWER

6 and 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q1

Grade 8 • The Number System • 8.NS.A.2 • Set 3 • MAIN • Q1/5

Estimate √20 to the nearest tenth. Explain how you know.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q2

Grade 8 • The Number System • 8.NS.A.2 • Set 3 • MAIN • Q2/5

71

QUESTION ID

S3-M-Q2

TYPE

Short answer

CORRECT ANSWER

4.5. $4.4^2 = 19.36$ and $4.5^2 = 20.25$, and 20 is closer to 20.25.

GRADING NOTE Grade both: 4.5, and a reason that compares squares of tenths.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q2

Grade 8 • The Number System • 8.NS.A.2 • Set 3 • MAIN • Q2/5

Which is the best estimate of √75 to the nearest tenth?

A. 8.7

B. 8.6

C. 37.5

D. 9.1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q3

Grade 8 • The Number System • 8.NS.A.2 • Set 3 • MAIN • Q3/5

72

QUESTION ID

S3-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 8.7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q3

Grade 8 • The Number System • 8.NS.A.2 • Set 3 • MAIN • Q3/5

True or false?

√2 is between 1.4 and 1.5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q4

Grade 8 • The Number System • 8.NS.A.2 • Set 3 • MAIN • Q4/5

73

QUESTION ID

S3-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q4

Grade 8 • The Number System • 8.NS.A.2 • Set 3 • MAIN • Q4/5

True or false?

√90 is between 8 and 9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q5

Grade 8 • The Number System • 8.NS.A.2 • Set 3 • MAIN • Q5/5

74

QUESTION ID

S3-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $9^2 = 81$ and $10^2 = 100$, so √90 is between 9 and 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q5

Grade 8 • The Number System • 8.NS.A.2 • Set 3 • MAIN • Q5/5

SET 3 • 8.NS.A.2

BACKWARD 1

Find squares of whole numbers and decimals • Grade 6 • Expressions & Equations • 6.EE.A.1

Find the value.

7^2

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 3 • BACKWARD 1 • Q1/5

76

QUESTION ID

S3-B1-Q1

TYPE

Short answer

CORRECT ANSWER

49

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 3 • BACKWARD 1 • Q1/5

Find the value.

1.5^2

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 3 • BACKWARD 1 • Q2/5

77

QUESTION ID

S3-B1-Q2

TYPE

Short answer

CORRECT ANSWER

2.25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 3 • BACKWARD 1 • Q2/5

Find the value.

12^2

A. 24

B. 121

C. 122

D. 144

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 3 • BACKWARD 1 • Q3/5

78

QUESTION ID

S3-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 144

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 3 • BACKWARD 1 • Q3/5

True or false?

$8^2 = 64$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 3 • BACKWARD 1 • Q4/5

79

QUESTION ID

S3-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 3 • BACKWARD 1 • Q4/5

True or false?

$3^2 = 6$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 3 • BACKWARD 1 • Q5/5

80

QUESTION ID

S3-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 3 • BACKWARD 1 • Q5/5

SET 3 • 8.NS.A.2

BACKWARD 2

Round decimals to the nearest tenth • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4

Round 4.472 to the nearest tenth.

Rounded: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 3 • BACKWARD 2 • Q1/5

82

QUESTION ID

S3-B2-Q1

TYPE

Short answer

CORRECT ANSWER

4.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 3 • BACKWARD 2 • Q1/5

Round 8.660 to the nearest tenth.

Rounded: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 3 • BACKWARD 2 • Q2/5

83

QUESTION ID

S3-B2-Q2

TYPE

Short answer

CORRECT ANSWER

8.7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 3 • BACKWARD 2 • Q2/5

Round 3.162 to the nearest tenth.

A. 3.1

B. 3.0

C. 3.2

D. 3.16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 3 • BACKWARD 2 • Q3/5

84

QUESTION ID

S3-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 3.2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 3 • BACKWARD 2 • Q3/5

True or false?

6.325 rounded to the nearest tenth is 6.3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 3 • BACKWARD 2 • Q4/5

85

QUESTION ID

S3-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 3 • BACKWARD 2 • Q4/5

True or false?

2.449 rounded to the nearest tenth is 2.5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 3 • BACKWARD 2 • Q5/5

86

QUESTION ID

S3-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 3 • BACKWARD 2 • Q5/5

SET 3 • 8.NS.A.2

FORWARD 1

Solve quadratic equations by taking square roots, exactly and approximately • High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b

Solve $x^2 = 30$. Give the exact solutions, then round each to the nearest tenth.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 3 • FORWARD 1 • Q1/5

88

QUESTION ID

S3-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$x = \sqrt{30}$ and $x = -\sqrt{30}$; about 5.5 and −5.5

GRADING NOTE Both solutions are required, exact and rounded.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-F1-Q1**

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 3 • FORWARD 1 • Q1/5

Solve $x^2 - 7 = 0$. Give the exact solutions.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 3 • FORWARD 1 • Q2/5

89

QUESTION ID

S3-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$x = \sqrt{7}$ and $x = -\sqrt{7}$

GRADING NOTE Both solutions are required; $x = \pm\sqrt{7}$ is accepted.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-F1-Q2**

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 3 • FORWARD 1 • Q2/5

Solve $(x - 1)^2 = 5$.

A. $x = 1 + \sqrt{5}$ or $x = 1 - \sqrt{5}$

B. $x = 1 + \sqrt{5}$ only

C. $x = 6$ or $x = -4$

D. $x = \sqrt{6}$ or $x = -\sqrt{6}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 3 • FORWARD 1 • Q3/5

90

QUESTION ID

S3-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $x = 1 + \sqrt{5}$ or $x = 1 - \sqrt{5}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-F1-Q3**

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 3 • FORWARD 1 • Q3/5

True or false?

The solutions of $x^2 = 12$ are $x = 2\sqrt{3}$ and $x = -2\sqrt{3}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 3 • FORWARD 1 • Q4/5

91

QUESTION ID

S3-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 3 • FORWARD 1 • Q4/5

True or false?

$x^2 = -9$ has the real solutions 3 and −3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 3 • FORWARD 1 • Q5/5

92

QUESTION ID

S3-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE No real number squared is negative, so there is no real solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 3 • FORWARD 1 • Q5/5

SET 3 • 8.NS.A.2

FORWARD 2

Find arc lengths and sector areas exactly in terms of π and approximately • High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.B.5

A circle has a radius of 6 cm.

Find the length of a 90° arc exactly in terms of π, then to the nearest tenth.

Arc length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q1

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.B.5 • Set 3 • FORWARD 2 • Q1/5

94

QUESTION ID

S3-F2-Q1

TYPE

Short answer

CORRECT ANSWER

3π cm, about 9.4 cm

GRADING NOTE Both the exact and the rounded value are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q1

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.B.5 • Set 3 • FORWARD 2 • Q1/5

A circle has a radius of 4 inches.

Find the area of a 45° sector exactly in terms of π, then to the nearest tenth.

Sector area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q2

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.B.5 • Set 3 • FORWARD 2 • Q2/5

95

QUESTION ID

S3-F2-Q2

TYPE

Short answer

CORRECT ANSWER

2π square inches, about 6.3 square inches

GRADING NOTE Both the exact and the rounded value are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q2

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.B.5 • Set 3 • FORWARD 2 • Q2/5

A circle has a radius of 3 m.

To the nearest tenth, how long is a 60° arc?

A. 6.3 m

B. 9.4 m

C. 3.1 m

D. 18.8 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q3

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.B.5 • Set 3 • FORWARD 2 • Q3/5

96

QUESTION ID

S3-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 3.1 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q3

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.B.5 • Set 3 • FORWARD 2 • Q3/5

True or false?

The arc of a semicircle with a radius of 1 is π units long, about 3.14 units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q4

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.B.5 • Set 3 • FORWARD 2 • Q4/5

97

QUESTION ID

S3-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q4

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.B.5 • Set 3 • FORWARD 2 • Q4/5

True or false?

A 180° sector of a circle with a radius of 2 has an area of 4π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q5

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.B.5 • Set 3 • FORWARD 2 • Q5/5

98

QUESTION ID

S3-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Its area is half of 4π, which is 2π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q5

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.B.5 • Set 3 • FORWARD 2 • Q5/5

SET 4 • 8.NS.A.2 • THE NUMBER SYSTEM

Locate and compare irrational numbers on a number line

MAIN	Grade 8 main question	**8.NS.A.2**
BACKWARD 1	Read inequalities as positions on a number line	**6.NS.C.7.a**
BACKWARD 2	Position decimals on a number line	**6.NS.C.6.c**
FORWARD 1	Report irrational values to a level of accuracy that fits the situati…	**HSN-Q.A.3**
FORWARD 2	Use coordinates to find exact lengths and perimeters with radicals	**HSG-GPE.B.7**

SET 4 • 8.NS.A.2

MAIN

Grade 8 main question • Grade 8 • The Number System • 8.NS.A.2

Plot √5 on the number line. Use an estimate to the nearest tenth.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q1

Grade 8 • The Number System • 8.NS.A.2 • Set 4 • MAIN • Q1/5

101

QUESTION ID

S4-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

A point at about 2.2: between 2 and 2.5, closer to 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q1

Grade 8 • The Number System • 8.NS.A.2 • Set 4 • MAIN • Q1/5

Which list is in order from least to greatest?

A. 2, √5, √3, 1.8

B. √3, 1.8, 2, √5

C. √3, 2, 1.8, √5

D. 1.8, √3, 2, √5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q2

Grade 8 • The Number System • 8.NS.A.2 • Set 4 • MAIN • Q2/5

102

QUESTION ID

S4-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. √3, 1.8, 2, √5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q2

Grade 8 • The Number System • 8.NS.A.2 • Set 4 • MAIN • Q2/5

Write <, >, or =.

π ___ √10

Symbol: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q3

Grade 8 • The Number System • 8.NS.A.2 • Set 4 • MAIN • Q3/5

103

QUESTION ID

S4-M-Q3

TYPE

Short answer

CORRECT ANSWER

<

GRADING NOTE π is about 3.14 and √10 is about 3.16.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q3

Grade 8 • The Number System • 8.NS.A.2 • Set 4 • MAIN • Q3/5

Which number is the best name for point P: √2, √3, or √6?



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q4

Grade 8 • The Number System • 8.NS.A.2 • Set 4 • MAIN • Q4/5

104

QUESTION ID

S4-M-Q4

TYPE

Short answer

CORRECT ANSWER

√3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q4

Grade 8 • The Number System • 8.NS.A.2 • Set 4 • MAIN • Q4/5

True or false?

√30 is to the left of 5.5 on a number line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q5

Grade 8 • The Number System • 8.NS.A.2 • Set 4 • MAIN • Q5/5

105

QUESTION ID

S4-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE √30 is about 5.48.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q5

Grade 8 • The Number System • 8.NS.A.2 • Set 4 • MAIN • Q5/5

SET 4 • 8.NS.A.2

BACKWARD 1

Read inequalities as positions on a number line • Grade 6 • The Number System • 6.NS.C.7.a

2.4 is to the right of 2.25 on a number line.

Write an inequality that shows this.

Inequality: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q1

Grade 6 • The Number System • 6.NS.C.7.a • Set 4 • BACKWARD 1 • Q1/5

107

QUESTION ID

S4-B1-Q1

TYPE

Short answer

CORRECT ANSWER

2.4 > 2.25

GRADING NOTE 2.25 < 2.4 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q1

Grade 6 • The Number System • 6.NS.C.7.a • Set 4 • BACKWARD 1 • Q1/5

Which number is farthest to the left on a number line?

A. 0.5

B. −1.9

C. −2.05

D. −2.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q2

Grade 6 • The Number System • 6.NS.C.7.a • Set 4 • BACKWARD 1 • Q2/5

108

QUESTION ID

S4-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. −2.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q2

Grade 6 • The Number System • 6.NS.C.7.a • Set 4 • BACKWARD 1 • Q2/5

What does 1.7 < 1.75 tell you about the positions of the two numbers on a number line?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q3

Grade 6 • The Number System • 6.NS.C.7.a • Set 4 • BACKWARD 1 • Q3/5

109

QUESTION ID

S4-B1-Q3

TYPE

Short answer

CORRECT ANSWER

1.7 is to the left of 1.75.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q3

Grade 6 • The Number System • 6.NS.C.7.a • Set 4 • BACKWARD 1 • Q3/5

True or false?

$-1.4 > -1.41$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q4

Grade 6 • The Number System • 6.NS.C.7.a • Set 4 • BACKWARD 1 • Q4/5

110

QUESTION ID

S4-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q4

Grade 6 • The Number System • 6.NS.C.7.a • Set 4 • BACKWARD 1 • Q4/5

True or false?

3.09 > 3.1

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q5

Grade 6 • The Number System • 6.NS.C.7.a • Set 4 • BACKWARD 1 • Q5/5

111

QUESTION ID

S4-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q5

Grade 6 • The Number System • 6.NS.C.7.a • Set 4 • BACKWARD 1 • Q5/5

SET 4 • 8.NS.A.2

BACKWARD 2

Position decimals on a number line • Grade 6 • The Number System • 6.NS.C.6.c

Plot 1.4 and 2.75 on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 4 • BACKWARD 2 • Q1/5

113

QUESTION ID

S4-B2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

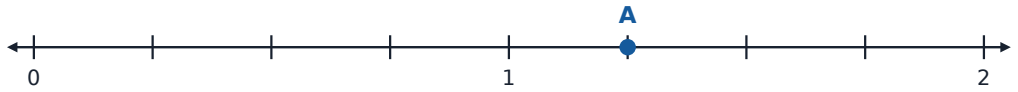
Points at 1.4 (a little past the 1.25 mark) and 2.75

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 4 • BACKWARD 2 • Q1/5

What decimal is at point A?



Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 4 • BACKWARD 2 • Q2/5

114

QUESTION ID

S4-B2-Q2

TYPE

Short answer

CORRECT ANSWER

1.25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 4 • BACKWARD 2 • Q2/5

Which decimal is between 1.7 and 1.8?

A. 1.81

B. 1.73

C. 1.69

D. 1.08

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 4 • BACKWARD 2 • Q3/5

115

QUESTION ID

S4-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 1.73

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 4 • BACKWARD 2 • Q3/5

True or false?

2.5 is halfway between 2 and 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 4 • BACKWARD 2 • Q4/5

116

QUESTION ID

S4-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 4 • BACKWARD 2 • Q4/5

True or false?

0.9 is to the right of 1.1 on a number line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 4 • BACKWARD 2 • Q5/5

117

QUESTION ID

S4-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 4 • BACKWARD 2 • Q5/5

SET 4 • 8.NS.A.2

FORWARD 1

Report irrational values to a level of accuracy that fits the situation • High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3

A square garden has an area of 50 square meters, so its side is √50 m. A tape measure reads to the nearest centimeter.

Report the side length to that accuracy, in meters.

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q1

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 4 • FORWARD 1 • Q1/5

119

QUESTION ID

S4-F1-Q1

TYPE

Short answer

CORRECT ANSWER

7.07 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q1

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 4 • FORWARD 1 • Q1/5

A coin has a diameter of 4 cm. Its circumference is 4π cm.

Report the circumference to the nearest tenth of a centimeter.

Circumference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q2

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 4 • FORWARD 1 • Q2/5

120

QUESTION ID

S4-F1-Q2

TYPE

Short answer

CORRECT ANSWER

12.6 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q2

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 4 • FORWARD 1 • Q2/5

Which value of √2 is given to the nearest thousandth?

A. 1.41

B. 1.414

C. 1.4142

D. 1.42

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q3

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 4 • FORWARD 1 • Q3/5

121

QUESTION ID

S4-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 1.414

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q3

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 4 • FORWARD 1 • Q3/5

True or false?

Reporting √3 as 1.7 is accurate to the nearest tenth.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q4

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 4 • FORWARD 1 • Q4/5

122

QUESTION ID

S4-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q4

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 4 • FORWARD 1 • Q4/5

True or false?

Writing more digits of √5 always makes a measured length more accurate.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q5

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 4 • FORWARD 1 • Q5/5

123

QUESTION ID

S4-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A reported length cannot be more accurate than the measuring tool.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q5

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 4 • FORWARD 1 • Q5/5

SET 4 • 8.NS.A.2

FORWARD 2

Use coordinates to find exact lengths and perimeters with radicals • High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7

Find the exact perimeter of the triangle with vertices (0, 0), (2, 0), and (0, 2).

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q1

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 4 • FORWARD 2 • Q1/5

125

QUESTION ID

S4-F2-Q1

TYPE

Short answer

CORRECT ANSWER

4 + 2√2 units

GRADING NOTE About 6.8 units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S4-F2-Q1**

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 4 • FORWARD 2 • Q1/5

Find the exact length of the segment from (1, 1) to (4, 3).

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q2

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 4 • FORWARD 2 • Q2/5

126

QUESTION ID

S4-F2-Q2

TYPE

Short answer

CORRECT ANSWER

√13 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S4-F2-Q2**

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 4 • FORWARD 2 • Q2/5

A square has vertices (0, 0), (1, 2), (−1, 3), and (−2, 1).

What is its perimeter?

A. 2√5 units

B. 20 units

C. 4√5 units

D. 4√3 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q3

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 4 • FORWARD 2 • Q3/5

127

QUESTION ID

S4-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 4√5 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S4-F2-Q3**

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 4 • FORWARD 2 • Q3/5

True or false?

The distance from $(-1, -1)$ to $(2, 3)$ is 5 units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q4

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 4 • FORWARD 2 • Q4/5

128

QUESTION ID

S4-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q4

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 4 • FORWARD 2 • Q4/5

True or false?

A rectangle with vertices (0, 0), (3, 0), (3, 1), and (0, 1) has a diagonal 4 units long.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q5

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 4 • FORWARD 2 • Q5/5

129

QUESTION ID

S4-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The diagonal is √10 units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q5

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 4 • FORWARD 2 • Q5/5

SET 5 • 8.EE.A.1 • EXPRESSIONS & EQUATIONS

Use the product and quotient properties of integer exponents

MAIN	Grade 8 main question	**8.EE.A.1**
BACKWARD 1	Evaluate whole-number exponents	**6.EE.A.1**
BACKWARD 2	Write powers of 10 with whole-number exponents	**5.NBT.A.2**
FORWARD 1	Multiply and divide monomials using exponent properties	**HSA-SSE.A.2**
FORWARD 2	Use exponent properties to rewrite and interpret exponential funct	**HSF-IF.C.8.b**

SET 5 • 8.EE.A.1

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.A.1

Write as a single power.

$3^4 \times 3^5$

Power: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 5 • MAIN • Q1/5

132

QUESTION ID

S5-M-Q1

TYPE

Short answer

CORRECT ANSWER

3^9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 5 • MAIN • Q1/5

Write as a single power.

$7^{10} \div 7^{4}$

Power: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 5 • MAIN • Q2/5

133

QUESTION ID

S5-M-Q2

TYPE

Short answer

CORRECT ANSWER

7^{6}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 5 • MAIN • Q2/5

Which expression is equal to 2^6?

A. $2^{12} \div 2^2$

B. 2×6

C. $2^2 \times 2^3$

D. $2^2 \times 2^4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 5 • MAIN • Q3/5

134

QUESTION ID

S5-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $2^2 \times 2^4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 5 • MAIN • Q3/5

True or false?

$5^3 \times 5^2 = 5^5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 5 • MAIN • Q4/5

135

QUESTION ID

S5-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 5 • MAIN • Q4/5

True or false?

$4^8 \div 4^2 = 4^4$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 5 • MAIN • Q5/5

136

QUESTION ID

S5-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $4^8 \div 4^2 = 4^6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 5 • MAIN • Q5/5

SET 5 • 8.EE.A.1

BACKWARD 1

Evaluate whole-number exponents • Grade 6 • Expressions & Equations • 6.EE.A.1

Find the value.

2^5

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 5 • BACKWARD 1 • Q1/5

138

QUESTION ID

S5-B1-Q1

TYPE

Short answer

CORRECT ANSWER

32

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 5 • BACKWARD 1 • Q1/5

Find the value.

10^4

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 5 • BACKWARD 1 • Q2/5

139

QUESTION ID

S5-B1-Q2

TYPE

Short answer

CORRECT ANSWER

10,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 5 • BACKWARD 1 • Q2/5

Find the value.

3^4

A. 81

B. 64

C. 12

D. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 5 • BACKWARD 1 • Q3/5

140

QUESTION ID

S5-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 81

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 5 • BACKWARD 1 • Q3/5

True or false?

$6^2 = 36$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 5 • BACKWARD 1 • Q4/5

141

QUESTION ID

S5-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 5 • BACKWARD 1 • Q4/5

True or false?

$2^3 = 6$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 5 • BACKWARD 1 • Q5/5

142

QUESTION ID

S5-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 5 • BACKWARD 1 • Q5/5

SET 5 • 8.EE.A.1

BACKWARD 2

Write powers of 10 with whole-number exponents • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2

Write $10 \times 10 \times 10 \times 10$ as a power of 10.

Power: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 5 • BACKWARD 2 • Q1/5

144

QUESTION ID

S5-B2-Q1

TYPE

Short answer

CORRECT ANSWER

10^4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 5 • BACKWARD 2 • Q1/5

Write 1,000,000 as a power of 10.

Power: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 5 • BACKWARD 2 • Q2/5

145

QUESTION ID

S5-B2-Q2

TYPE

Short answer

CORRECT ANSWER

10^6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 5 • BACKWARD 2 • Q2/5

What is the value of 10^5?

A. 10,000

B. 50

C. 100,000

D. 1,000,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 5 • BACKWARD 2 • Q3/5

146

QUESTION ID

S5-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 100,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 5 • BACKWARD 2 • Q3/5

True or false?

$10^3 = 1{,}000$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 5 • BACKWARD 2 • Q4/5

147

QUESTION ID

S5-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 5 • BACKWARD 2 • Q4/5

True or false?

$10^2 \times 10 = 10^2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 5 • BACKWARD 2 • Q5/5

148

QUESTION ID

S5-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $10^2 \times 10 = 1{,}000 = 10^3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 5 • BACKWARD 2 • Q5/5

SET 5 • 8.EE.A.1

FORWARD 1

Multiply and divide monomials using exponent properties • High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

Simplify.

$(3x^4)(5x^2)$

Simplified: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q1

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • FORWARD 1 • Q1/5

150

QUESTION ID

S5-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$15x^6$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q1

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • FORWARD 1 • Q1/5

Simplify. Assume $x \neq 0$.

$$\frac{12x^7}{4x^3}$$

Simplified: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q2

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • FORWARD 1 • Q2/5

151

QUESTION ID

S5-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$3x^4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q2

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • FORWARD 1 • Q2/5

Simplify.

$x^5 \cdot x \cdot x^2$

A. x^7

B. $3x^8$

C. x^{10}

D. x^8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q3

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • FORWARD 1 • Q3/5

152

QUESTION ID

S5-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. x^8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q3

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • FORWARD 1 • Q3/5

True or false?

$(2a^3)(4a^2) = 8a^5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q4

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • FORWARD 1 • Q4/5

153

QUESTION ID

S5-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q4

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • FORWARD 1 • Q4/5

True or false?

Assume $y \neq 0$. $\frac{y^9}{y^3} = y^3$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q5

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • FORWARD 1 • Q5/5

154

QUESTION ID

S5-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $y^9 \div y^3 = y^6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q5

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • FORWARD 1 • Q5/5

SET 5 • 8.EE.A.1

FORWARD 2

Use exponent properties to rewrite and interpret exponential functions • High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.b

f(t) = 2^(t + 3)

Use a property of exponents to write f(t) in the form a · 2^t.

f(t) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q1

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 5 • FORWARD 2 • Q1/5

156

QUESTION ID

S5-F2-Q1

TYPE

Short answer

CORRECT ANSWER

8 · 2^t

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q1

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 5 • FORWARD 2 • Q1/5

A population is P(t) = 200 · 2^t after t years.

By what factor does the population grow each year, and what is P(5)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q2

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 5 • FORWARD 2 • Q2/5

157

QUESTION ID

S5-F2-Q2

TYPE

Short answer

CORRECT ANSWER

It doubles (factor 2) each year;
P(5) = 6,400

GRADING NOTE Both parts are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q2

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 5 • FORWARD 2 • Q2/5

$g(t) = 5^{(t - 1)}$

Which expression is equal to g(t)?

A. $5^t - 5$

B. $5 \div 5^t$

C. $5^t \div 5$

D. $5^t - 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q3

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 5 • FORWARD 2 • Q3/5

158

QUESTION ID

S5-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $5^t \div 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q3

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 5 • FORWARD 2 • Q3/5

True or false?

If f(x) = 5 · 2^x, then f(x + 1) = 2 · f(x).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q4

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 5 • FORWARD 2 • Q4/5

159

QUESTION ID

S5-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q4

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 5 • FORWARD 2 • Q4/5

True or false?

If f(x) = 2^x, then f(3) · f(4) = f(12).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q5

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 5 • FORWARD 2 • Q5/5

160

QUESTION ID

S5-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE f(3) · f(4) = $2^3 \cdot 2^4 = 2^7$ = f(7).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q5

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 5 • FORWARD 2 • Q5/5

SET 6 • 8.EE.A.1 • EXPRESSIONS & EQUATIONS

Use the power-of-a-power property and zero and negative exponents

MAIN	Grade 8 main question	**8.EE.A.1**
BACKWARD 1	Find powers of fractions and decimals	**6.EE.A.1**
BACKWARD 2	Multiply signed numbers	**7.NS.A.2.a**
FORWARD 1	Explain rational exponents with the properties of exponents	**HSN-RN.A.1**
FORWARD 2	Rewrite exponential expressions with the properties of exponents	**HSA-SSE.B.3.c**

SET 6 • 8.EE.A.1

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.A.1

Write as a single power.

$(5^2)^3$

Power: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 6 • MAIN • Q1/5

163

QUESTION ID

S6-M-Q1

TYPE

Short answer

CORRECT ANSWER

5^6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 6 • MAIN • Q1/5

Find the value.

4^{-2}

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 6 • MAIN • Q2/5

164

QUESTION ID

S6-M-Q2

TYPE

Short answer

CORRECT ANSWER

1/16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 6 • MAIN • Q2/5

Which expression is equal to 1?

A. 0^9

B. 9^0

C. $1^{-9} \times 9$

D. 9^{-1}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 6 • MAIN • Q3/5

QUESTION ID

S6-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 9^0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 6 • MAIN • Q3/5

True or false?

$3^{-2} \times 3^{5} = 27$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 6 • MAIN • Q4/5

166

QUESTION ID

S6-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 6 • MAIN • Q4/5

True or false?

$(2^3)^2 = 2^5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 6 • MAIN • Q5/5

167

QUESTION ID

S6-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(2^3)^2 = 2^6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 6 • MAIN • Q5/5

SET 6 • 8.EE.A.1

BACKWARD 1

Find powers of fractions and decimals • Grade 6 • Expressions & Equations • 6.EE.A.1

Find the value.

$(\frac{1}{2})^3$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 6 • BACKWARD 1 • Q1/5

169

QUESTION ID

S6-B1-Q1

TYPE

Short answer

CORRECT ANSWER

1/8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 6 • BACKWARD 1 • Q1/5

Find the value.

0.3^2

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 6 • BACKWARD 1 • Q2/5

170

QUESTION ID

S6-B1-Q2

TYPE

Short answer

CORRECT ANSWER

0.09

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 6 • BACKWARD 1 • Q2/5

Find the value.

$(\frac{2}{3})^2$

A. $\frac{4}{3}$

B. $\frac{4}{9}$

C. $\frac{2}{9}$

D. $\frac{4}{6}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 6 • BACKWARD 1 • Q3/5

171

QUESTION ID

S6-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 4/9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 6 • BACKWARD 1 • Q3/5

True or false?

$0.1^3 = 0.001$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 6 • BACKWARD 1 • Q4/5

172

QUESTION ID

S6-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 6 • BACKWARD 1 • Q4/5

True or false?

$(\frac{3}{4})^2 = \frac{6}{8}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 6 • BACKWARD 1 • Q5/5

173

QUESTION ID

S6-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(3/4)^2 = 9/16$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 6 • BACKWARD 1 • Q5/5

SET 6 • 8.EE.A.1

BACKWARD 2

Multiply signed numbers • Grade 7 • The Number System • 7.NS.A.2.a

Multiply.

$(-2)(-2)(-2)$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q1

Grade 7 • The Number System • 7.NS.A.2.a • Set 6 • BACKWARD 2 • Q1/5

175

QUESTION ID

S6-B2-Q1

TYPE

Short answer

CORRECT ANSWER

−8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q1

Grade 7 • The Number System • 7.NS.A.2.a • Set 6 • BACKWARD 2 • Q1/5

Multiply.

$(-3)(-3)$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q2

Grade 7 • The Number System • 7.NS.A.2.a • Set 6 • BACKWARD 2 • Q2/5

176

QUESTION ID

S6-B2-Q2

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q2

Grade 7 • The Number System • 7.NS.A.2.a • Set 6 • BACKWARD 2 • Q2/5

Multiply.

$(-1)(-1)(-1)(-1)$

A. -1

B. -4

C. 1

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q3

Grade 7 • The Number System • 7.NS.A.2.a • Set 6 • BACKWARD 2 • Q3/5

177

QUESTION ID

S6-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q3

Grade 7 • The Number System • 7.NS.A.2.a • Set 6 • BACKWARD 2 • Q3/5

True or false?

$(-5)(-5) = 25$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q4

Grade 7 • The Number System • 7.NS.A.2.a • Set 6 • BACKWARD 2 • Q4/5

178

QUESTION ID

S6-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q4

Grade 7 • The Number System • 7.NS.A.2.a • Set 6 • BACKWARD 2 • Q4/5

True or false?

$(-2)(-2)(-2)(-2) = -16$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q5

Grade 7 • The Number System • 7.NS.A.2.a • Set 6 • BACKWARD 2 • Q5/5

179

QUESTION ID

S6-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q5

Grade 7 • The Number System • 7.NS.A.2.a • Set 6 • BACKWARD 2 • Q5/5

SET 6 • 8.EE.A.1

FORWARD 1

Explain rational exponents with the properties of exponents • High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.1

$(5^{(1/2)})^2 = 5^{((1/2) \cdot 2)} = 5.$

Which radical is equal to $5^{(1/2)}$?

Radical: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q1

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 6 • FORWARD 1 • Q1/5

181

QUESTION ID

S6-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$\sqrt{5}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q1

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 6 • FORWARD 1 • Q1/5

Write 7^(1/3) as a radical.

Radical: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q2

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 6 • FORWARD 1 • Q2/5

182

QUESTION ID

S6-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$\sqrt[3]{7}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q2

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 6 • FORWARD 1 • Q2/5

Which number is equal to 9^(3/2)?

A. 6

B. 27

C. 13.5

D. 81

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q3

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 6 • FORWARD 1 • Q3/5

183

QUESTION ID

S6-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 27

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q3

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 6 • FORWARD 1 • Q3/5

True or false?

Because $(8^{(1/3)})^3 = 8$, $8^{(1/3)} = 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q4

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 6 • FORWARD 1 • Q4/5

184

QUESTION ID

S6-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q4

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 6 • FORWARD 1 • Q4/5

True or false?

x^(1/2) means x ÷ 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q5

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 6 • FORWARD 1 • Q5/5

185

QUESTION ID

S6-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE x^(1/2) is the square root of x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q5

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 6 • FORWARD 1 • Q5/5

SET 6 • 8.EE.A.1

FORWARD 2

Rewrite exponential expressions with the properties of exponents • High School (Grade 10 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.c

Rewrite 1.21^t as a power of 1.1.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q1

High School (Grade 10 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.c • Set 6 • FORWARD 2 • Q1/5

187

QUESTION ID

S6-F2-Q1

TYPE

Short answer

CORRECT ANSWER

1.1^(2t)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q1

High School (Grade 10 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.c • Set 6 • FORWARD 2 • Q1/5

A value grows by a factor of 1.04 each month, so after t years it has grown by (1.04^12)^t.

Write the yearly growth factor, rounded to the nearest hundredth.

Factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q2

High School (Grade 10 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.c • Set 6 • FORWARD 2 • Q2/5

188

QUESTION ID

S6-F2-Q2

TYPE

Short answer

CORRECT ANSWER

1.60

GRADING NOTE 1.04^12 ≈ 1.601, about 60% growth per year.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q2

High School (Grade 10 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.c • Set 6 • FORWARD 2 • Q2/5

Which expression is equivalent to 8^x?

A. 2^x ÷ 3

B. 3^(2x)

C. 2^(3x)

D. 2^(x + 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q3

High School (Grade 10 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.c • Set 6 • FORWARD 2 • Q3/5

189

QUESTION ID

S6-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 2^(3x)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q3

High School (Grade 10 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.c • Set 6 • FORWARD 2 • Q3/5

True or false?

$(0.5)^{(-t)} = 2^t$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q4

High School (Grade 10 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.c • Set 6 • FORWARD 2 • Q4/5

190

QUESTION ID

S6-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q4

High School (Grade 10 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.c • Set 6 • FORWARD 2 • Q4/5

True or false?

$4^x \times 4^x = 16^{(2x)}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q5

High School (Grade 10 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.c • Set 6 • FORWARD 2 • Q5/5

191

QUESTION ID

S6-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $4^x \times 4^x = 4^{(2x)} = 16^x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q5

High School (Grade 10 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.c • Set 6 • FORWARD 2 • Q5/5

SET 7 • 8.EE.A.2 • EXPRESSIONS & EQUATIONS

Solve equations of the form $x^2 = p$ and $x^3 = p$

MAIN	Grade 8 main question	**8.EE.A.2**
BACKWARD 1	Solve one-step multiplication equations	**6.EE.B.7**
BACKWARD 2	Find squares and cubes of whole numbers	**6.EE.A.1**
FORWARD 1	Solve quadratic equations by taking square roots	**HSA-REI.B.4.b**
FORWARD 2	Find a dimension from a volume with square and cube roots	**HSG-GMD.A.3**

SET 7 • 8.EE.A.2

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.A.2

Solve for x. Give all solutions.

$x^2 = 49$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 7 • MAIN • Q1/5

194

QUESTION ID

S7-M-Q1

TYPE

Short answer

CORRECT ANSWER

x = 7 and x = −7

GRADING NOTE Both solutions are required; x = ±7 is accepted.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 7 • MAIN • Q1/5

Solve for x.

$x^3 = 64$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 7 • MAIN • Q2/5

195

QUESTION ID

S7-M-Q2

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 7 • MAIN • Q2/5

Solve for x.

$x^2 = \frac{9}{16}$

A. $x = \frac{3}{4}$ only

B. $x = \frac{3}{4}$ or $x = -\frac{3}{4}$

C. $x = \frac{9}{32}$

D. $x = \frac{81}{256}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 7 • MAIN • Q3/5

196

QUESTION ID

S7-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. x = 3/4 or x = −3/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 7 • MAIN • Q3/5

Solve for x.

$x^3 = -27$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 7 • MAIN • Q4/5

197

QUESTION ID

S7-M-Q4

TYPE

Short answer

CORRECT ANSWER

−3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 7 • MAIN • Q4/5

True or false?

$x^2 = 10$ has the two solutions $x = \sqrt{10}$ and $x = -\sqrt{10}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 7 • MAIN • Q5/5

198

QUESTION ID

S7-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 7 • MAIN • Q5/5

SET 7 • 8.EE.A.2

BACKWARD 1

Solve one-step multiplication equations • Grade 6 • Expressions & Equations • 6.EE.B.7

Solve for x.

4x = 64

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 7 • BACKWARD 1 • Q1/5

200

QUESTION ID

S7-B1-Q1

TYPE

Short answer

CORRECT ANSWER

16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 7 • BACKWARD 1 • Q1/5

Solve for x.

$\frac{x}{3} = 9$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 7 • BACKWARD 1 • Q2/5

201

QUESTION ID

S7-B1-Q2

TYPE

Short answer

CORRECT ANSWER

27

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 7 • BACKWARD 1 • Q2/5

Solve for x.

9x = 45

A. x = 36

B. x = 5

C. x = 405

D. x = 54

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 7 • BACKWARD 1 • Q3/5

QUESTION ID

S7-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. x = 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 7 • BACKWARD 1 • Q3/5

True or false?

x = 12 is the solution of 3x = 36.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 7 • BACKWARD 1 • Q4/5

203

QUESTION ID

S7-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 7 • BACKWARD 1 • Q4/5

True or false?

x = 5 is the solution of 7x = 42.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 7 • BACKWARD 1 • Q5/5

204

QUESTION ID

S7-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 7 • BACKWARD 1 • Q5/5

SET 7 • 8.EE.A.2

BACKWARD 2

Find squares and cubes of whole numbers • Grade 6 • Expressions & Equations • 6.EE.A.1

Find the value.

9^2

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 7 • BACKWARD 2 • Q1/5

206

QUESTION ID

S7-B2-Q1

TYPE

Short answer

CORRECT ANSWER

81

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 7 • BACKWARD 2 • Q1/5

Find the value.

5^3

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 7 • BACKWARD 2 • Q2/5

207

QUESTION ID

S7-B2-Q2

TYPE

Short answer

CORRECT ANSWER

125

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 7 • BACKWARD 2 • Q2/5

Which whole number cubed is 8?

A. 3

B. 4

C. 8

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 7 • BACKWARD 2 • Q3/5

208

QUESTION ID

S7-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 7 • BACKWARD 2 • Q3/5

True or false?

$10^4 = 10{,}000$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 7 • BACKWARD 2 • Q4/5

209

QUESTION ID

S7-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 7 • BACKWARD 2 • Q4/5

True or false?

$4^2 = 8$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 7 • BACKWARD 2 • Q5/5

210

QUESTION ID

S7-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 7 • BACKWARD 2 • Q5/5

SET 7 • 8.EE.A.2

FORWARD 1

Solve quadratic equations by taking square roots • High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b

ANSWER KEY Reduced copy of the question page, with its answer

Solve for x. Give all solutions.

$2x^2 = 50$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 7 • FORWARD 1 • Q1/5

212

QUESTION ID

S7-F1-Q1

TYPE

Short answer

CORRECT ANSWER

x = 5 and x = −5

GRADING NOTE Both solutions are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 7 • FORWARD 1 • Q1/5

Solve for x. Give all solutions.

$(x + 2)^2 = 36$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 7 • FORWARD 1 • Q2/5

213

QUESTION ID

S7-F1-Q2

TYPE

Short answer

CORRECT ANSWER

x = 4 and x = −8

GRADING NOTE Both solutions are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S7-F1-Q2**

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 7 • FORWARD 1 • Q2/5

Solve for x.

$x^2 - 9 = 0$

A. x = 9 or x = −9

B. x = 3 only

C. x = 3 or x = −3

D. x = 4.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 7 • FORWARD 1 • Q3/5

214

QUESTION ID

S7-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. x = 3 or x = −3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 7 • FORWARD 1 • Q3/5

True or false?

The solutions of $3x^2 = 27$ are 3 and −3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 7 • FORWARD 1 • Q4/5

215

QUESTION ID

S7-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 7 • FORWARD 1 • Q4/5

True or false?

The solutions of $(x - 1)^2 = 16$ are 5 and 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 7 • FORWARD 1 • Q5/5

216

QUESTION ID

S7-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The solutions are 5 and −3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 7 • FORWARD 1 • Q5/5

SET 7 • 8.EE.A.2

FORWARD 2

Find a dimension from a volume with square and cube roots • High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

A cube has a volume of 343 cubic centimeters.

How long is each edge?

Edge: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q1

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 7 • FORWARD 2 • Q1/5

218

QUESTION ID

S7-F2-Q1

TYPE

Short answer

CORRECT ANSWER

7 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q1

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 7 • FORWARD 2 • Q1/5

A sphere has a volume of 36π cubic centimeters.

($V = \frac{4}{3}\pi r^3$)

What is its radius?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q2

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 7 • FORWARD 2 • Q2/5

219

QUESTION ID

S7-F2-Q2

TYPE

Short answer

CORRECT ANSWER

3 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q2

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 7 • FORWARD 2 • Q2/5

A cylinder has a volume of 200π cubic units and a height of 8 units.

What is its radius?

A. 10 units

B. 25 units

C. 5 units

D. √200 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q3

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 7 • FORWARD 2 • Q3/5

220

QUESTION ID

S7-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 5 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q3

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 7 • FORWARD 2 • Q3/5

True or false?

A cone with a volume of 12π cubic units and a height of 4 units has a radius of 3 units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q4

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 7 • FORWARD 2 • Q4/5

221

QUESTION ID

S7-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q4

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 7 • FORWARD 2 • Q4/5

True or false?

A cube with a volume of 1,000 cubic centimeters has edges 100 cm long.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q5

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 7 • FORWARD 2 • Q5/5

222

QUESTION ID

S7-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Each edge is 10 cm.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q5

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 7 • FORWARD 2 • Q5/5

SET 8 • 8.EE.A.2 • EXPRESSIONS & EQUATIONS

Evaluate square roots and cube roots and recognize irrational roots

MAIN	Grade 8 main question	**8.EE.A.2**
BACKWARD 1	Find factor pairs and square numbers	**4.OA.B.4**
BACKWARD 2	Find the volume of a cube	**5.MD.C.5.b**
FORWARD 1	Simplify square roots by factoring out perfect squares	**HSN-RN.A.2**
FORWARD 2	Solve quadratic equations by completing the square	**HSA-REI.B.4.a**

SET 8 • 8.EE.A.2

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.A.2

Find the value.

√144

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • MAIN • Q1/5

225

QUESTION ID

S8-M-Q1

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • MAIN • Q1/5

Find the value.

$\sqrt[3]{125}$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • MAIN • Q2/5

226

QUESTION ID

S8-M-Q2

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • MAIN • Q2/5

Which square root is irrational?

A. $\sqrt{\frac{1}{4}}$

B. $\sqrt{0.09}$

C. $\sqrt{20}$

D. $\sqrt{36}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • MAIN • Q3/5

QUESTION ID

S8-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. √20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • MAIN • Q3/5

True or false?

$\sqrt[3]{(-8)} = -2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • MAIN • Q4/5

228

QUESTION ID

S8-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • MAIN • Q4/5

True or false?

√2 can be written as a fraction of two whole numbers.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • MAIN • Q5/5

229

QUESTION ID
S8-M-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE √2 is irrational.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • MAIN • Q5/5

SET 8 • 8.EE.A.2

BACKWARD 1

Find factor pairs and square numbers • Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4

List all the factor pairs of 16.

Factor pairs: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 8 • BACKWARD 1 • Q1/5

231

QUESTION ID

S8-B1-Q1

TYPE

Short answer

CORRECT ANSWER

1 × 16, 2 × 8, and 4 × 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 8 • BACKWARD 1 • Q1/5

What whole number times itself is 49?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 8 • BACKWARD 1 • Q2/5

232

QUESTION ID

S8-B1-Q2

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 8 • BACKWARD 1 • Q2/5

Which number has a factor pair made of two equal factors?

A. 30

B. 24

C. 18

D. 25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 8 • BACKWARD 1 • Q3/5

233

QUESTION ID

S8-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 8 • BACKWARD 1 • Q3/5

True or false?

81 has a factor pair made of two equal factors.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 8 • BACKWARD 1 • Q4/5

234

QUESTION ID

S8-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 8 • BACKWARD 1 • Q4/5

True or false?

12 has a factor pair made of two equal factors.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 8 • BACKWARD 1 • Q5/5

235

QUESTION ID

S8-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 8 • BACKWARD 1 • Q5/5

SET 8 • 8.EE.A.2

BACKWARD 2

Find the volume of a cube • Grade 5 • Measurement & Data • 5.MD.C.5.b

A cube has edges 3 units long.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q1

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 8 • BACKWARD 2 • Q1/5

237

QUESTION ID

S8-B2-Q1

TYPE

Short answer

CORRECT ANSWER

27 cubic units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q1

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 8 • BACKWARD 2 • Q1/5

A cube has edges 5 cm long.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q2

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 8 • BACKWARD 2 • Q2/5

238

QUESTION ID

S8-B2-Q2

TYPE

Short answer

CORRECT ANSWER

125 cubic centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q2

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 8 • BACKWARD 2 • Q2/5

Which cube has a volume of 64 cubic units?

A. A cube with 32-unit edges

B. A cube with 4-unit edges

C. A cube with 16-unit edges

D. A cube with 8-unit edges

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q3

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 8 • BACKWARD 2 • Q3/5

239

QUESTION ID

S8-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. A cube with 4-unit edges

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q3

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 8 • BACKWARD 2 • Q3/5

True or false?

A cube with 2-unit edges has a volume of 8 cubic units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q4

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 8 • BACKWARD 2 • Q4/5

240

QUESTION ID

S8-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q4

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 8 • BACKWARD 2 • Q4/5

True or false?

A cube with 10-unit edges has a volume of 30 cubic units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q5

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 8 • BACKWARD 2 • Q5/5

241

QUESTION ID

S8-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q5

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 8 • BACKWARD 2 • Q5/5

SET 8 • 8.EE.A.2

FORWARD 1

Simplify square roots by factoring out perfect squares • High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2

Simplify.

√48

Simplified: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q1

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 8 • FORWARD 1 • Q1/5

243

QUESTION ID

S8-F1-Q1

TYPE

Short answer

CORRECT ANSWER

4√3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q1

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 8 • FORWARD 1 • Q1/5

Simplify.

√200

Simplified: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q2

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 8 • FORWARD 1 • Q2/5

244

QUESTION ID

S8-F1-Q2

TYPE

Short answer

CORRECT ANSWER

10√2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-F1-Q2**

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 8 • FORWARD 1 • Q2/5

Which expression is equal to √18?

A. 2√3

B. 9√2

C. 3√2

D. 6√3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q3

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 8 • FORWARD 1 • Q3/5

245

QUESTION ID

S8-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 3√2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q3

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 8 • FORWARD 1 • Q3/5

True or false?

$\sqrt{50} = 5\sqrt{2}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q4

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 8 • FORWARD 1 • Q4/5

246

QUESTION ID

S8-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q4

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 8 • FORWARD 1 • Q4/5

True or false?

$\sqrt{27} = 9\sqrt{3}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q5

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 8 • FORWARD 1 • Q5/5

247

QUESTION ID

S8-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\sqrt{27} = 3\sqrt{3}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-F1-Q5**

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 8 • FORWARD 1 • Q5/5

SET 8 • 8.EE.A.2

FORWARD 2

Solve quadratic equations by completing the square • High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a

What number goes in the blank?

$x^2 + 6x + 9 = (x + ___)^2$

Blank: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q1

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 8 • FORWARD 2 • Q1/5

249

QUESTION ID

S8-F2-Q1

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-F2-Q1**

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 8 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve by completing the square. Give exact solutions.

$x^2 + 4x = 1$

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q2

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 8 • FORWARD 2 • Q2/5

250

QUESTION ID

S8-F2-Q2

TYPE

Short answer

CORRECT ANSWER

x = −2 + √5 and
x = −2 − √5

GRADING NOTE Both solutions are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q2

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 8 • FORWARD 2 • Q2/5

Which number completes the square?

$x^2 - 10x + ___$

A. 5

B. −25

C. 100

D. 25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q3

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 8 • FORWARD 2 • Q3/5

251

QUESTION ID

S8-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q3

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 8 • FORWARD 2 • Q3/5

True or false?

$x^2 + 8x + 16 = (x + 4)^2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q4

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 8 • FORWARD 2 • Q4/5

252

QUESTION ID

S8-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q4

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 8 • FORWARD 2 • Q4/5

True or false?

Completing the square on $x^2 + 2x = 3$ gives $(x + 1)^2 = 3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q5

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 8 • FORWARD 2 • Q5/5

253

QUESTION ID

S8-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It gives $(x + 1)^2 = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q5

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 8 • FORWARD 2 • Q5/5

SET 9 • 8.EE.A.3 • EXPRESSIONS & EQUATIONS

Estimate very large and very small quantities as a single digit times a power of 10

MAIN	Grade 8 main question	**8.EE.A.3**
BACKWARD 1	Multiply and divide by powers of 10	**5.NBT.A.2**
BACKWARD 2	Round whole numbers to any place	**4.NBT.A.3**
FORWARD 1	Choose a level of accuracy for very large and very small quantities	**HSN-Q.A.3**
FORWARD 2	Use base-10 logarithms to describe orders of magnitude	**HSF-LE.A.4**

SET 9 • 8.EE.A.3

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.A.3

Estimate 412,000,000 as a single digit times a power of 10.

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q1

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 9 • MAIN • Q1/5

256

QUESTION ID

S9-M-Q1

TYPE

Short answer

CORRECT ANSWER

4×10^8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q1

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 9 • MAIN • Q1/5

Estimate 0.00079 as a single digit times a power of 10.

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q2

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 9 • MAIN • Q2/5

257

QUESTION ID

S9-M-Q2

TYPE

Short answer

CORRECT ANSWER

8×10^{-4}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q2

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 9 • MAIN • Q2/5

A country has about 3×10^8 people. A city has about 6×10^5 people.

About how many times as many people live in the country?

A. 2

B. 50

C. 500

D. 5,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q3

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 9 • MAIN • Q3/5

258

QUESTION ID

S9-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 500

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q3

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 9 • MAIN • Q3/5

A whale has a mass of about 2×10^5 kg. A mouse has a mass of about 2×10^{-2} kg.

About how many times as large is the whale's mass?

Times as large: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q4

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 9 • MAIN • Q4/5

259

QUESTION ID

S9-M-Q4

TYPE

Short answer

CORRECT ANSWER

10,000,000

GRADING NOTE 10^7 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q4

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 9 • MAIN • Q4/5

True or false?

6×10^{-3} is greater than 9×10^{-4}.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q5

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 9 • MAIN • Q5/5

260

QUESTION ID

S9-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q5

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 9 • MAIN • Q5/5

SET 9 • 8.EE.A.3

BACKWARD 1

Multiply and divide by powers of 10 • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2

Multiply.

4×10^3

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 9 • BACKWARD 1 • Q1/5

262

QUESTION ID

S9-B1-Q1

TYPE

Short answer

CORRECT ANSWER

4,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 9 • BACKWARD 1 • Q1/5

Divide.

$7 \div 10^2$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 9 • BACKWARD 1 • Q2/5

263

QUESTION ID

S9-B1-Q2

TYPE

Short answer

CORRECT ANSWER

0.07

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 9 • BACKWARD 1 • Q2/5

What is the value of 3×10^5?

A. 15

B. 3,000,000

C. 300,000

D. 30,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 9 • BACKWARD 1 • Q3/5

264

QUESTION ID

S9-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 300,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 9 • BACKWARD 1 • Q3/5

True or false?

$6 \div 10^3 = 0.006$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 9 • BACKWARD 1 • Q4/5

265

QUESTION ID

S9-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 9 • BACKWARD 1 • Q4/5

True or false?

$8 \times 10^2 = 8{,}000$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 9 • BACKWARD 1 • Q5/5

266

QUESTION ID

S9-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 9 • BACKWARD 1 • Q5/5

SET 9 • 8.EE.A.3

BACKWARD 2

Round whole numbers to any place • Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3

Round 412,000,000 to the nearest hundred million.

Rounded: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 9 • BACKWARD 2 • Q1/5

268

QUESTION ID

S9-B2-Q1

TYPE

Short answer

CORRECT ANSWER

400,000,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 9 • BACKWARD 2 • Q1/5

Round 6,750 to the nearest thousand.

Rounded: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 9 • BACKWARD 2 • Q2/5

269

QUESTION ID

S9-B2-Q2

TYPE

Short answer

CORRECT ANSWER

7,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 9 • BACKWARD 2 • Q2/5

Round 58,300 to the nearest ten thousand.

A. 50,000

B. 59,000

C. 60,000

D. 58,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 9 • BACKWARD 2 • Q3/5

270

QUESTION ID

S9-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 60,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 9 • BACKWARD 2 • Q3/5

True or false?

245,000 rounded to the nearest hundred thousand is 200,000.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 9 • BACKWARD 2 • Q4/5

271

QUESTION ID

S9-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 9 • BACKWARD 2 • Q4/5

True or false?

8,500 rounded to the nearest thousand is 8,000.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 9 • BACKWARD 2 • Q5/5

272

QUESTION ID

S9-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 9 • BACKWARD 2 • Q5/5

SET 9 • 8.EE.A.3

FORWARD 1

Choose a level of accuracy for very large and very small quantities • High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3

Earth's mean radius is about 6,371 km.

Write it to one significant digit as a single digit times a power of 10.

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q1

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 9 • FORWARD 1 • Q1/5

274

QUESTION ID

S9-F1-Q1

TYPE

Short answer

CORRECT ANSWER

6×10^3 km

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q1

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 9 • FORWARD 1 • Q1/5

A bacterium is 0.0000023 m long.

Write its length in micrometers. (1 μm = 10^{-6} m)

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q2

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 9 • FORWARD 1 • Q2/5

275

QUESTION ID

S9-F1-Q2

TYPE

Short answer

CORRECT ANSWER

2.3 μm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q2

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 9 • FORWARD 1 • Q2/5

A sheet of paper is about 0.0001 m thick.

Which unit best expresses this thickness?

A. Millimeters (0.1 mm)

B. Megameters

C. Meters

D. Kilometers

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q3

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 9 • FORWARD 1 • Q3/5

276

QUESTION ID

S9-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Millimeters (0.1 mm)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q3

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 9 • FORWARD 1 • Q3/5

True or false?

Reporting 7,900,000,000 people as 8×10^9 is accurate to one significant digit.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q4

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 9 • FORWARD 1 • Q4/5

277

QUESTION ID

S9-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q4

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 9 • FORWARD 1 • Q4/5

True or false?

A distance known to be about 3×10^8 m is reported more accurately as 300,000,000.0 m.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q5

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 9 • FORWARD 1 • Q5/5

278

QUESTION ID

S9-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Writing more digits does not make an estimate more accurate.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q5

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 9 • FORWARD 1 • Q5/5

SET 9 • 8.EE.A.3

FORWARD 2

Use base-10 logarithms to describe orders of magnitude • High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

Find the value.

$\log_{10}(1{,}000{,}000)$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q1

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 9 • FORWARD 2 • Q1/5

280

QUESTION ID

S9-F2-Q1

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S9-F2-Q1**

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 9 • FORWARD 2 • Q1/5

Solve for x.

10^x = 0.001

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q2

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 9 • FORWARD 2 • Q2/5

281

QUESTION ID

S9-F2-Q2

TYPE

Short answer

CORRECT ANSWER

−3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S9-F2-Q2**

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 9 • FORWARD 2 • Q2/5

Between which two whole numbers is $\log_{10}(4{,}500)$?

A. 2 and 3

B. 45 and 46

C. 4 and 5

D. 3 and 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q3

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 9 • FORWARD 2 • Q3/5

282

QUESTION ID

S9-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 3 and 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S9-F2-Q3**

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 9 • FORWARD 2 • Q3/5

True or false?

$\log_{10}(10^{-5}) = -5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q4

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 9 • FORWARD 2 • Q4/5

283

QUESTION ID

S9-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q4

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 9 • FORWARD 2 • Q4/5

True or false?

$\log_{10}(50)$ is between 5 and 6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q5

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 9 • FORWARD 2 • Q5/5

284

QUESTION ID

S9-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $10^1 = 10$ and $10^2 = 100$, so $\log_{10}(50)$ is between 1 and 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q5

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 9 • FORWARD 2 • Q5/5

SET 10 • 8.EE.A.4 • EXPRESSIONS & EQUATIONS

Multiply and divide numbers written in scientific notation

MAIN	Grade 8 main question	**8.EE.A.4**
BACKWARD 1	Multiply decimals	**5.NBT.B.7**
BACKWARD 2	Multiply by powers of 10	**5.NBT.A.2**
FORWARD 1	Use units to guide multistep calculations in scientific notation	**HSN-Q.A.1**
FORWARD 2	Model exponential growth and decay with scientific notation	**HSF-LE.A.1.c**

SET 10 • 8.EE.A.4

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.A.4

Multiply. Write the answer in scientific notation.

$(4 \times 10^{6})(2.5 \times 10^{3})$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q1

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 10 • MAIN • Q1/5

287

QUESTION ID

S10-M-Q1

TYPE

Short answer

CORRECT ANSWER

1×10^{10}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q1

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 10 • MAIN • Q1/5

Divide. Write the answer in scientific notation.

$(6.3 \times 10^{8}) \div (9 \times 10^{2})$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q2

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 10 • MAIN • Q2/5

288

QUESTION ID

S10-M-Q2

TYPE

Short answer

CORRECT ANSWER

7×10^{5}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q2

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 10 • MAIN • Q2/5

Light travels about 3×10^5 km per second.

How far does it travel in 5×10^2 seconds?

A. 1.5×10^7 km

B. 6×10^2 km

C. 1.5×10^8 km

D. 8×10^7 km

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q3

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 10 • MAIN • Q3/5

289

QUESTION ID

S10-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 1.5×10^8 km

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q3

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 10 • MAIN • Q3/5

True or false?

$(2 \times 10^{-3})(3 \times 10^{-4}) = 6 \times 10^{-7}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q4

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 10 • MAIN • Q4/5

290

QUESTION ID

S10-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q4

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 10 • MAIN • Q4/5

True or false?

$(8 \times 10^9) \div (4 \times 10^3) = 2 \times 10^3$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q5

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 10 • MAIN • Q5/5

291

QUESTION ID

S10-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The quotient is 2×10^6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q5

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 10 • MAIN • Q5/5

SET 10 • 8.EE.A.4

BACKWARD 1

Multiply decimals • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7

Multiply.

2.5 × 4

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 10 • BACKWARD 1 • Q1/5

293

QUESTION ID

S10-B1-Q1

TYPE

Short answer

CORRECT ANSWER

10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 10 • BACKWARD 1 • Q1/5

Multiply.

1.2 × 3.5

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 10 • BACKWARD 1 • Q2/5

294

QUESTION ID

S10-B1-Q2

TYPE

Short answer

CORRECT ANSWER

4.2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 10 • BACKWARD 1 • Q2/5

Multiply.

6×0.5

A. 6.5

B. 3

C. 0.3

D. 30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 10 • BACKWARD 1 • Q3/5

295

QUESTION ID

S10-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 10 • BACKWARD 1 • Q3/5

True or false?

$2.4 \times 0.5 = 1.2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 10 • BACKWARD 1 • Q4/5

296

QUESTION ID

S10-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 10 • BACKWARD 1 • Q4/5

True or false?

$3.5 \times 2 = 6$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 10 • BACKWARD 1 • Q5/5

297

QUESTION ID

S10-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 10 • BACKWARD 1 • Q5/5

SET 10 • 8.EE.A.4

BACKWARD 2

Multiply by powers of 10 • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2

Multiply.

2.5×10^4

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 10 • BACKWARD 2 • Q1/5

299

QUESTION ID

S10-B2-Q1

TYPE

Short answer

CORRECT ANSWER

25,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 10 • BACKWARD 2 • Q1/5

Multiply.

3.1×10^2

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 10 • BACKWARD 2 • Q2/5

300

QUESTION ID

S10-B2-Q2

TYPE

Short answer

CORRECT ANSWER

310

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 10 • BACKWARD 2 • Q2/5

Multiply.

0.6×10^3

A. 60

B. 0.0006

C. 6,000

D. 600

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 10 • BACKWARD 2 • Q3/5

301

QUESTION ID

S10-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 600

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 10 • BACKWARD 2 • Q3/5

True or false?

$7 \times 10^3 = 7{,}000$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 10 • BACKWARD 2 • Q4/5

302

QUESTION ID

S10-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 10 • BACKWARD 2 • Q4/5

True or false?

$4.2 \times 10 = 4.20$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 10 • BACKWARD 2 • Q5/5

303

QUESTION ID

S10-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 10 • BACKWARD 2 • Q5/5

SET 10 • 8.EE.A.4

FORWARD 1

Use units to guide multistep calculations in scientific notation • High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1

A drive transfers 2×10^8 bytes per second.

How many seconds does it take to transfer 6×10^{10} bytes?

Seconds: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q1

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 10 • FORWARD 1 • Q1/5

305

QUESTION ID

S10-F1-Q1

TYPE

Short answer

CORRECT ANSWER

300

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q1

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 10 • FORWARD 1 • Q1/5

Light travels 3×10^8 meters per second.

How many meters does it travel in 1 hour (3.6×10^3 seconds)? Use scientific notation.

Meters: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q2

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 10 • FORWARD 1 • Q2/5

306

QUESTION ID

S10-F1-Q2

TYPE

Short answer

CORRECT ANSWER

1.08×10^{12} m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S10-F1-Q2**

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 10 • FORWARD 1 • Q2/5

The Sun is about 1.5×10^{11} m from Earth. Light travels 3×10^{8} m per second.

About how long does sunlight take to reach Earth?

A. 500 seconds

B. 4.5×10^{19} seconds

C. 5×10^{19} seconds

D. 50 seconds

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q3

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 10 • FORWARD 1 • Q3/5

307

QUESTION ID

S10-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 500 seconds

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q3

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 10 • FORWARD 1 • Q3/5

True or false?

A flow of 5×10^3 liters per minute is 3×10^5 liters per hour.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q4

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 10 • FORWARD 1 • Q4/5

308

QUESTION ID

S10-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q4

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 10 • FORWARD 1 • Q4/5

True or false?

6×10^{-3} km is equal to 6×10^{-6} m.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q5

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 10 • FORWARD 1 • Q5/5

309

QUESTION ID

S10-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 6×10^{-3} km = 6 m.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q5

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 10 • FORWARD 1 • Q5/5

SET 10 • 8.EE.A.4

FORWARD 2

Model exponential growth and decay with scientific notation • High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c

A colony of 2×10^3 bacteria triples every hour.

How many bacteria are there after 4 hours? Use scientific notation.

Bacteria: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q1

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 10 • FORWARD 2 • Q1/5

311

QUESTION ID

S10-F2-Q1

TYPE

Short answer

CORRECT ANSWER

1.62×10^5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q1

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 10 • FORWARD 2 • Q1/5

A quantity of 5×10^9 is cut in half each year.

What is it after 3 years? Use scientific notation.

Amount: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q2

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 10 • FORWARD 2 • Q2/5

312

QUESTION ID

S10-F2-Q2

TYPE

Short answer

CORRECT ANSWER

6.25×10^8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q2

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 10 • FORWARD 2 • Q2/5

A population of 4×10^6 grows by 10% each year.

What is the population after 2 years?

A. 4.84×10^6

B. 8×10^6

C. 4.4×10^6

D. 4.8×10^6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q3

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 10 • FORWARD 2 • Q3/5

313

QUESTION ID

S10-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 4.84×10^6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q3

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 10 • FORWARD 2 • Q3/5

True or false?

Doubling 3×10^5 ten times gives about 3.07×10^8.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q4

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 10 • FORWARD 2 • Q4/5

QUESTION ID

S10-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q4

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 10 • FORWARD 2 • Q4/5

True or false?

A quantity that grows by a constant percent rate increases by the same amount each year.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q5

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 10 • FORWARD 2 • Q5/5

315

QUESTION ID

S10-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q5

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 10 • FORWARD 2 • Q5/5

SET 11 • 8.EE.A.4 • EXPRESSIONS & EQUATIONS

Add and subtract numbers written in scientific notation

MAIN	Grade 8 main question	**8.EE.A.4**
BACKWARD 1	Add decimals	**5.NBT.B.7**
BACKWARD 2	Write a number times a power of 10 in standard form	**5.NBT.A.2**
FORWARD 1	Add and subtract polynomials	**HSA-APR.A.1**
FORWARD 2	Add and subtract functions	**HSF-BF.A.1.b**

SET 11 • 8.EE.A.4

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.A.4

Add. Write the answer in scientific notation.

$(3.2 \times 10^5) + (4.5 \times 10^4)$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q1

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 11 • MAIN • Q1/5

318

QUESTION ID

S11-M-Q1

TYPE

Short answer

CORRECT ANSWER

3.65×10^5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q1

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 11 • MAIN • Q1/5

Subtract. Write the answer in scientific notation.

$(9 \times 10^{7}) - (6 \times 10^{6})$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q2

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 11 • MAIN • Q2/5

319

QUESTION ID

S11-M-Q2

TYPE

Short answer

CORRECT ANSWER

8.4×10^{7}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q2

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 11 • MAIN • Q2/5

Add. Write the answer in scientific notation.

$(1.5 \times 10^{-3}) + (2 \times 10^{-4})$

A. 1.7×10^{-4}

B. 3.5×10^{-3}

C. 1.7×10^{-3}

D. 3.5×10^{-7}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q3

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 11 • MAIN • Q3/5

320

QUESTION ID

S11-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 1.7×10^{-3}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q3

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 11 • MAIN • Q3/5

True or false?

$(5 \times 10^8) + (5 \times 10^8) = 1 \times 10^9$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q4

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 11 • MAIN • Q4/5

321

QUESTION ID

S11-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q4

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 11 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$(4 \times 10^6) - (3 \times 10^5) = 1 \times 10^1$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q5

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 11 • MAIN • Q5/5

322

QUESTION ID

S11-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The difference is 3.7×10^6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q5

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 11 • MAIN • Q5/5

SET 11 • 8.EE.A.4

BACKWARD 1

Add decimals • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7

Add.

3.2 + 0.45

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 11 • BACKWARD 1 • Q1/5

QUESTION ID

S11-B1-Q1

TYPE

Short answer

CORRECT ANSWER

3.65

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 11 • BACKWARD 1 • Q1/5

Add.

8.4 + 1.6

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 11 • BACKWARD 1 • Q2/5

325

QUESTION ID

S11-B1-Q2

TYPE

Short answer

CORRECT ANSWER

10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 11 • BACKWARD 1 • Q2/5

Add.

1.5 + 0.2

A. 0.17

B. 1.7

C. 1.52

D. 3.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 11 • BACKWARD 1 • Q3/5

326

QUESTION ID

S11-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 1.7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 11 • BACKWARD 1 • Q3/5

True or false?

2.75 + 0.25 = 3

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 11 • BACKWARD 1 • Q4/5

327

QUESTION ID

S11-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 11 • BACKWARD 1 • Q4/5

True or false?

$4.6 + 0.04 = 5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 11 • BACKWARD 1 • Q5/5

328

QUESTION ID

S11-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 11 • BACKWARD 1 • Q5/5

SET 11 • 8.EE.A.4

BACKWARD 2

Write a number times a power of 10 in standard form • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2

Write 6×10^6 in standard form.

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 11 • BACKWARD 2 • Q1/5

330

QUESTION ID

S11-B2-Q1

TYPE

Short answer

CORRECT ANSWER

6,000,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 11 • BACKWARD 2 • Q1/5

Write 0.6×10^7 in standard form.

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 11 • BACKWARD 2 • Q2/5

331

QUESTION ID

S11-B2-Q2

TYPE

Short answer

CORRECT ANSWER

6,000,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 11 • BACKWARD 2 • Q2/5

Which number is equal to 45×10^4?

A. 450,000

B. 4,500

C. 45,000

D. 4,500,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 11 • BACKWARD 2 • Q3/5

332

QUESTION ID

S11-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 450,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 11 • BACKWARD 2 • Q3/5

True or false?

$3.2 \times 10^2 = 320$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 11 • BACKWARD 2 • Q4/5

333

QUESTION ID

S11-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 11 • BACKWARD 2 • Q4/5

True or false?

$0.5 \times 10^3 = 50$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 11 • BACKWARD 2 • Q5/5

334

QUESTION ID

S11-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 11 • BACKWARD 2 • Q5/5

SET 11 • 8.EE.A.4

FORWARD 1

Add and subtract polynomials • High School (Grade 9 in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Add.

$(3x^2 + 5x) + (4x^2 - 2x)$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q1

High School (Grade 9 in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 11 • FORWARD 1 • Q1/5

336

QUESTION ID

S11-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$7x^2 + 3x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S11-F1-Q1**

High School (Grade 9 in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 11 • FORWARD 1 • Q1/5

Subtract.

$(6a^3 - 2a + 1) - (2a^3 + 5a - 4)$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q2

High School (Grade 9 in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 11 • FORWARD 1 • Q2/5

337

QUESTION ID

S11-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$4a^3 - 7a + 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q2

High School (Grade 9 in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 11 • FORWARD 1 • Q2/5

Add.

$(x^2 - 4) + (3x^2 + x)$

A. $4x^2 + x - 4$

B. $3x^4 + x - 4$

C. $4x^2 - 3$

D. $4x^2 + x + 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q3

High School (Grade 9 in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 11 • FORWARD 1 • Q3/5

338

QUESTION ID

S11-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $4x^2 + x - 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q3

High School (Grade 9 in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 11 • FORWARD 1 • Q3/5

True or false?

$(2y^2 + 3) - (y^2 - 3) = y^2 + 6$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q4

High School (Grade 9 in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 11 • FORWARD 1 • Q4/5

339

QUESTION ID

S11-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q4

High School (Grade 9 in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 11 • FORWARD 1 • Q4/5

True or false?

$(5n - 2) - (3n - 4) = 2n - 6$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q5

High School (Grade 9 in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 11 • FORWARD 1 • Q5/5

340

QUESTION ID

S11-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The difference is 2n + 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q5

High School (Grade 9 in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 11 • FORWARD 1 • Q5/5

SET 11 • 8.EE.A.4

FORWARD 2

Add and subtract functions • High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.1.b

Revenue is R(x) = 25x and cost is C(x) = 4,000 + 9x for x items.

Write the profit function P(x) = R(x) − C(x).

P(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q1

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.1.b • Set 11 • FORWARD 2 • Q1/5

342

QUESTION ID

S11-F2-Q1

TYPE

Short answer

CORRECT ANSWER

16x − 4,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q1

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.1.b • Set 11 • FORWARD 2 • Q1/5

The profit function is P(x) = 16x − 4,000.

Find P(1,000).

P(1,000) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q2

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.1.b • Set 11 • FORWARD 2 • Q2/5

343

QUESTION ID

S11-F2-Q2

TYPE

Short answer

CORRECT ANSWER

12,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q2

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.1.b • Set 11 • FORWARD 2 • Q2/5

$f(x) = 3x^2$ and $g(x) = x^2 - 5$.

Which expression is $(f + g)(x)$?

A. $2x^2 + 5$

B. $4x^2 - 5$

C. $3x^4 - 5$

D. $4x^2 + 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q3

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.1.b • Set 11 • FORWARD 2 • Q3/5

344

QUESTION ID

S11-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $4x^2 - 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q3

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.1.b • Set 11 • FORWARD 2 • Q3/5

True or false?

If f(x) = 2x and g(x) = x + 1, then (f − g)(3) = 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q4

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.1.b • Set 11 • FORWARD 2 • Q4/5

345

QUESTION ID

S11-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q4

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.1.b • Set 11 • FORWARD 2 • Q4/5

True or false?

If $f(x) = x + 4$ and $g(x) = 2x$, then $(f + g)(x) = 2x^2 + 4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q5

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.1.b • Set 11 • FORWARD 2 • Q5/5

346

QUESTION ID

S11-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(f + g)(x) = 3x + 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q5

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.1.b • Set 11 • FORWARD 2 • Q5/5

SET 12 • 8.EE.A.4 • EXPRESSIONS & EQUATIONS

Choose units for scientific notation and read calculator notation

MAIN	Grade 8 main question	**8.EE.A.4**
BACKWARD 1	Convert metric units	**5.MD.A.1**
BACKWARD 2	Divide by powers of 10	**5.NBT.A.2**
FORWARD 1	Convert units of very large and very small quantities	**HSN-Q.A.1**
FORWARD 2	Define the quantities and units a model needs	**HSN-Q.A.2**

SET 12 • 8.EE.A.4

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.A.4

ANSWER KEY Reduced copy of the question page, with its answer

A calculator shows 4.7E8.

Write the number in scientific notation and in standard form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q1

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 12 • MAIN • Q1/5

349

QUESTION ID

S12-M-Q1

TYPE

Short answer

CORRECT ANSWER

4.7×10^8

$= 470{,}000{,}000$

GRADING NOTE Grade both forms.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q1

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 12 • MAIN • Q1/5

A calculator shows 2.05E-6.

Write the number in standard form.

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q2

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 12 • MAIN • Q2/5

350

QUESTION ID

S12-M-Q2

TYPE

Short answer

CORRECT ANSWER

0.00000205

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q2

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 12 • MAIN • Q2/5

The distance from Earth to the Moon is about 3.84×10^8 m.

Which unit is the most sensible for this distance?

A. Kilometers

B. Centimeters

C. Millimeters

D. Micrometers

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q3

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 12 • MAIN • Q3/5

351

QUESTION ID

S12-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Kilometers

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q3

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 12 • MAIN • Q3/5

A dust particle has a mass of 7.53×10^{-10} kg.

Write the mass in micrograms. (1 μg = 10^{-9} kg)

Mass: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q4

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 12 • MAIN • Q4/5

352

QUESTION ID

S12-M-Q4

TYPE

Short answer

CORRECT ANSWER

0.753 μg

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q4

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 12 • MAIN • Q4/5

True or false?

A calculator display of 3E-4 means −30,000.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q5

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 12 • MAIN • Q5/5

353

QUESTION ID

S12-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 3E-4 means $3 \times 10^{-4} = 0.0003$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q5

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 12 • MAIN • Q5/5

SET 12 • 8.EE.A.4

BACKWARD 1

Convert metric units • Grade 5 • Measurement & Data • 5.MD.A.1

How many grams are in 3.5 kilograms?

Grams: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q1

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 12 • BACKWARD 1 • Q1/5

355

QUESTION ID

S12-B1-Q1

TYPE

Short answer

CORRECT ANSWER

3,500

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q1

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 12 • BACKWARD 1 • Q1/5

How many meters are in 4,200 millimeters?

Meters: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q2

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 12 • BACKWARD 1 • Q2/5

356

QUESTION ID

S12-B1-Q2

TYPE

Short answer

CORRECT ANSWER

4.2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q2

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 12 • BACKWARD 1 • Q2/5

How many centimeters are in 6 meters?

A. 60

B. 0.06

C. 6,000

D. 600

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q3

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 12 • BACKWARD 1 • Q3/5

357

QUESTION ID

S12-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 600

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q3

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 12 • BACKWARD 1 • Q3/5

True or false?

5 km = 5,000 m

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q4

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 12 • BACKWARD 1 • Q4/5

358

QUESTION ID

S12-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q4

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 12 • BACKWARD 1 • Q4/5

True or false?

250 cm = 25 m

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q5

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 12 • BACKWARD 1 • Q5/5

359

QUESTION ID

S12-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q5

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 12 • BACKWARD 1 • Q5/5

SET 12 • 8.EE.A.4

BACKWARD 2

Divide by powers of 10 • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2

Divide.

$3{,}840 \div 10^3$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 12 • BACKWARD 2 • Q1/5

361

QUESTION ID

S12-B2-Q1

TYPE

Short answer

CORRECT ANSWER

3.84

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 12 • BACKWARD 2 • Q1/5

Divide.

$75 \div 10^2$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 12 • BACKWARD 2 • Q2/5

362

QUESTION ID

S12-B2-Q2

TYPE

Short answer

CORRECT ANSWER

0.75

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 12 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Divide.

$6 \div 10^4$

A. 0.006

B. 0.00006

C. 60,000

D. 0.0006

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 12 • BACKWARD 2 • Q3/5

363

QUESTION ID

S12-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 0.0006

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 12 • BACKWARD 2 • Q3/5

True or false?

520 ÷ 10 = 52

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 12 • BACKWARD 2 • Q4/5

364

QUESTION ID

S12-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 12 • BACKWARD 2 • Q4/5

True or false?

$9 \div 10^3 = 0.09$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 12 • BACKWARD 2 • Q5/5

365

QUESTION ID

S12-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 12 • BACKWARD 2 • Q5/5

SET 12 • 8.EE.A.4

FORWARD 1

Convert units of very large and very small quantities • High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1

Convert 3.6×10^6 mm to kilometers.

Kilometers: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q1

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 12 • FORWARD 1 • Q1/5

367

QUESTION ID

S12-F1-Q1

TYPE

Short answer

CORRECT ANSWER

3.6 km

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q1

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 12 • FORWARD 1 • Q1/5

A server sends 4.5×10^9 bytes per minute.

Write this rate in gigabytes per second. (1 GB = 10^9 bytes)

Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q2

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 12 • FORWARD 1 • Q2/5

368

QUESTION ID

S12-F1-Q2

TYPE

Short answer

CORRECT ANSWER

0.075 GB per second

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S12-F1-Q2**

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 12 • FORWARD 1 • Q2/5

A virus is 1.2×10^{-7} m wide.

What is its width in nanometers? (1 nm = 10^{-9} m)

A. 1,200 nm

B. 120 nm

C. 12 nm

D. 0.12 nm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q3

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 12 • FORWARD 1 • Q3/5

369

QUESTION ID

S12-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 120 nm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q3

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 12 • FORWARD 1 • Q3/5

True or false?

7×10^5 g = 700 kg

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q4

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 12 • FORWARD 1 • Q4/5

370

QUESTION ID

S12-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q4

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 12 • FORWARD 1 • Q4/5

True or false?

2.5×10^{-3} km = 2.5 cm

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q5

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 12 • FORWARD 1 • Q5/5

371

QUESTION ID

S12-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 2.5×10^{-3} km = 2.5 m.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q5

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 12 • FORWARD 1 • Q5/5

SET 12 • 8.EE.A.4

FORWARD 2

Define the quantities and units a model needs • High School (Grade 10 in this bank) • Number & Quantity: Quantities • HSN-Q.A.2

You want to estimate how much water a city of 2×10^6 people uses each day.

Which quantities do you need, and with what units?

Quantities: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q1

High School (Grade 10 in this bank) • Number & Quantity: Quantities • HSN-Q.A.2 • Set 12 • FORWARD 2 • Q1/5

373

QUESTION ID

S12-F2-Q1

TYPE

Short answer

CORRECT ANSWER

The number of people and the water used per person per day (liters per person per day).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q1

High School (Grade 10 in this bank) • Number & Quantity: Quantities • HSN-Q.A.2 • Set 12 • FORWARD 2 • Q1/5

2×10^6 people each use about 3×10^2 liters of water per day.

About how many liters does the city use per day? Use scientific notation.

Liters: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q2

High School (Grade 10 in this bank) • Number & Quantity: Quantities • HSN-Q.A.2 • Set 12 • FORWARD 2 • Q2/5

374

QUESTION ID

S12-F2-Q2

TYPE

Short answer

CORRECT ANSWER

6×10^8 liters per day

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q2

High School (Grade 10 in this bank) • Number & Quantity: Quantities • HSN-Q.A.2 • Set 12 • FORWARD 2 • Q2/5

Which unit is the best choice for the mass of a large ship?

A. Metric tons

B. Grams

C. Ounces

D. Milligrams

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q3

High School (Grade 10 in this bank) • Number & Quantity: Quantities • HSN-Q.A.2 • Set 12 • FORWARD 2 • Q3/5

375

QUESTION ID

S12-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Metric tons

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q3

High School (Grade 10 in this bank) • Number & Quantity: Quantities • HSN-Q.A.2 • Set 12 • FORWARD 2 • Q3/5

True or false?

Choosing units that keep numbers between about 1 and 1,000 can make a model easier to read.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q4

High School (Grade 10 in this bank) • Number & Quantity: Quantities • HSN-Q.A.2 • Set 12 • FORWARD 2 • Q4/5

376

QUESTION ID

S12-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q4

High School (Grade 10 in this bank) • Number & Quantity: Quantities • HSN-Q.A.2 • Set 12 • FORWARD 2 • Q4/5

True or false?

Kilometers are the best unit for the width of a cell.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q5

High School (Grade 10 in this bank) • Number & Quantity: Quantities • HSN-Q.A.2 • Set 12 • FORWARD 2 • Q5/5

377

QUESTION ID

S12-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q5

High School (Grade 10 in this bank) • Number & Quantity: Quantities • HSN-Q.A.2 • Set 12 • FORWARD 2 • Q5/5

SET 13 • 8.EE.B.5 • EXPRESSIONS & EQUATIONS

Graph a proportional relationship and interpret the unit rate as the slope

MAIN	Grade 8 main question	**8.EE.B.5**
BACKWARD 1	Explain points on the graph of a proportional relationship	**7.RP.A.2.d**
BACKWARD 2	Find the constant of proportionality from a graph	**7.RP.A.2.b**
FORWARD 1	Calculate and interpret an average rate of change	**HSF-IF.B.6**
FORWARD 2	Compare properties of functions represented in different ways	**HSF-IF.C.9**

SET 13 • 8.EE.B.5

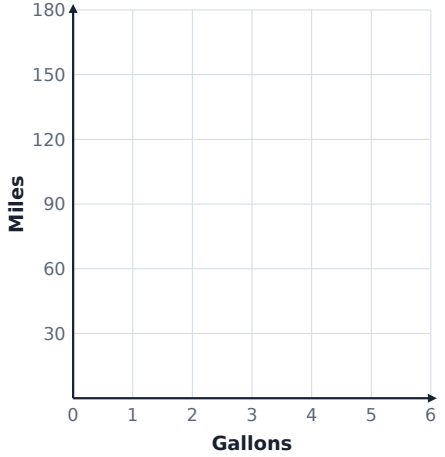
MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.B.5

A car uses gas at a constant rate: 3 gallons for every 90 miles.

Graph the relationship between gallons (x) and miles (y). What is the slope, and what does it mean?

Slope and meaning: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q1

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 13 • MAIN • Q1/5

380

QUESTION ID

S13-M-Q1

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: A straight line through (0, 0) and points such as (1, 30) and (3, 90)

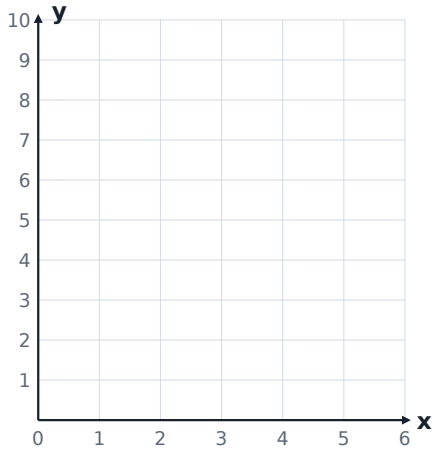
Written: 30; the car travels 30 miles per gallon.

GRADING NOTE Grade both: the line through the origin with slope 30, and the slope with its meaning.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q1

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 13 • MAIN • Q1/5

Graph y = 1.5x on the grid for x from 0 to 6.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q2

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 13 • MAIN • Q2/5

381

QUESTION ID

S13-M-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

A straight line through (0, 0), (2, 3), (4, 6), and (6, 9)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

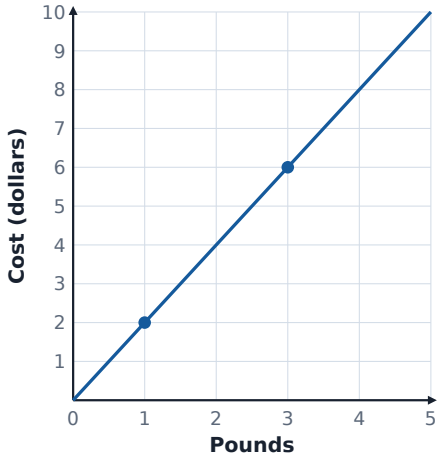
ID S13-M-Q2

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 13 • MAIN • Q2/5

The graph shows the cost of apples.

What is the slope, and what does it represent?

Slope and meaning: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q3

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 13 • MAIN • Q3/5

382

QUESTION ID

S13-M-Q3

TYPE

Short answer

CORRECT ANSWER

2; the apples cost $2 per pound.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q3

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 13 • MAIN • Q3/5

The graph of a proportional relationship passes through (4, 10).

What is the unit rate?

A. 14

B. 2.5

C. 0.4

D. 40

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q4

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 13 • MAIN • Q4/5

383

QUESTION ID

S13-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 2.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q4

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 13 • MAIN • Q4/5

True or false?

On graphs of distance (y) against time (x) that pass through the origin, a steeper line means a greater speed.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q5

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 13 • MAIN • Q5/5

384

QUESTION ID

S13-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q5

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 13 • MAIN • Q5/5

SET 13 • 8.EE.B.5

BACKWARD 1

Explain points on the graph of a proportional relationship • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d

A proportional graph of hours (x) and pay in dollars (y) passes through (1, 15).

What does this point mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 13 • BACKWARD 1 • Q1/5

386

QUESTION ID

S13-B1-Q1

TYPE

Short answer

CORRECT ANSWER

The pay is $15 for 1 hour (the unit rate).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 13 • BACKWARD 1 • Q1/5

A proportional graph shows cups of flour (x) and cookies (y).

What does the point (0, 0) mean?

A. Each cup makes 0 cookies.

B. 0 cups of flour make 0 cookies.

C. The recipe starts with some cookies.

D. There is no graph.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 13 • BACKWARD 1 • Q2/5

387

QUESTION ID

S13-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. 0 cups of flour make 0 cookies.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 13 • BACKWARD 1 • Q2/5

A proportional graph of minutes (x) and pages read (y) passes through (5, 20).

What does this point mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 13 • BACKWARD 1 • Q3/5

388

QUESTION ID

S13-B1-Q3

TYPE

Short answer

CORRECT ANSWER

20 pages are read in 5 minutes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 13 • BACKWARD 1 • Q3/5

True or false?

On the graph of a proportional relationship, the point (1, r) tells you the unit rate is r.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 13 • BACKWARD 1 • Q4/5

389

QUESTION ID

S13-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 13 • BACKWARD 1 • Q4/5

True or false?

On a graph of hours (x) and miles (y), the point (3, 12) means 3 miles in 12 hours.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 13 • BACKWARD 1 • Q5/5

390

QUESTION ID

S13-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It means 12 miles in 3 hours.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 13 • BACKWARD 1 • Q5/5

SET 13 • 8.EE.B.5

BACKWARD 2

Find the constant of proportionality from a graph • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b

A proportional graph passes through (0, 0) and (5, 20).

What is the constant of proportionality?

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 13 • BACKWARD 2 • Q1/5

392

QUESTION ID

S13-B2-Q1

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 13 • BACKWARD 2 • Q1/5

Which point shows k = 3 on the graph of a proportional relationship?

A. (3, 6)

B. (6, 2)

C. (3, 1)

D. (2, 6)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 13 • BACKWARD 2 • Q2/5

393

QUESTION ID

S13-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. (2, 6)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 13 • BACKWARD 2 • Q2/5

A proportional graph passes through (8, 2).

What is the constant of proportionality?

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 13 • BACKWARD 2 • Q3/5

394

QUESTION ID

S13-B2-Q3

TYPE

Short answer

CORRECT ANSWER

0.25

GRADING NOTE 1/4 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 13 • BACKWARD 2 • Q3/5

True or false?

A line through (0, 0) and (10, 25) has a constant of proportionality of 2.5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 13 • BACKWARD 2 • Q4/5

395

QUESTION ID

S13-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 13 • BACKWARD 2 • Q4/5

True or false?

A line through (0, 0) and (6, 3) has a constant of proportionality of 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 13 • BACKWARD 2 • Q5/5

396

QUESTION ID

S13-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $k = 3 \div 6 = 0.5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 13 • BACKWARD 2 • Q5/5

SET 13 • 8.EE.B.5

FORWARD 1

Calculate and interpret an average rate of change • High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6

$f(x) = 3x + 2$

What is the average rate of change of f from $x = 1$ to $x = 5$?

Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q1

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 13 • FORWARD 1 • Q1/5

398

QUESTION ID

S13-F1-Q1

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q1

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 13 • FORWARD 1 • Q1/5

The table shows a trip.

What is the average rate of change from 2 hours to 5 hours?

Hours	0	2	5
Miles	0	110	260

Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q2

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 13 • FORWARD 1 • Q2/5

399

QUESTION ID

S13-F1-Q2

TYPE

Short answer

CORRECT ANSWER

50 miles per hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q2

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 13 • FORWARD 1 • Q2/5

$h(t) = t^2$

What is the average rate of change of h from t = 1 to t = 4?

A. 5

B. 3

C. 16

D. 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q3

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 13 • FORWARD 1 • Q3/5

400

QUESTION ID

S13-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q3

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 13 • FORWARD 1 • Q3/5

True or false?

For a linear function, the average rate of change is the same over every interval.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q4

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 13 • FORWARD 1 • Q4/5

401

QUESTION ID

S13-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q4

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 13 • FORWARD 1 • Q4/5

True or false?

For $f(x) = x^2$, the average rate of change from $x = 0$ to $x = 2$ is 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q5

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 13 • FORWARD 1 • Q5/5

402

QUESTION ID

S13-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is $(4 - 0) \div 2 = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q5

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 13 • FORWARD 1 • Q5/5

SET 13 • 8.EE.B.5

FORWARD 2

Compare properties of functions represented in different ways • High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.9

f(x) = 2.5x. The graph of g is a line through (0, 0) and (4, 12).

Which function has the greater rate of change?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q1

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.9 • Set 13 • FORWARD 2 • Q1/5

404

QUESTION ID

S13-F2-Q1

TYPE

Short answer

CORRECT ANSWER

g (its rate is 3, and f's is 2.5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q1

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.9 • Set 13 • FORWARD 2 • Q1/5

f(x) = 40 + 5x. The table shows g.

Which function has the greater initial value, and which has the greater rate of change?

x	0	1	2
g(x)	30	37	44

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q2

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.9 • Set 13 • FORWARD 2 • Q2/5

405

QUESTION ID

S13-F2-Q2

TYPE

Short answer

CORRECT ANSWER

f has the greater initial value (40 > 30); g has the greater rate of change (7 > 5).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q2

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.9 • Set 13 • FORWARD 2 • Q2/5

f(x) = 2^x and g(x) = 10x.

For which x-value is f(x) > g(x)?

A. x = 3

B. x = 4

C. x = 6

D. x = 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q3

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.9 • Set 13 • FORWARD 2 • Q3/5

406

QUESTION ID

S13-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. x = 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q3

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.9 • Set 13 • FORWARD 2 • Q3/5

True or false?

A linear function with a slope of 4 eventually grows more slowly than f(x) = 2^x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q4

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.9 • Set 13 • FORWARD 2 • Q4/5

QUESTION ID

S13-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q4

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.9 • Set 13 • FORWARD 2 • Q4/5

True or false?

Two functions with the same y-intercept must be the same function.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q5

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.9 • Set 13 • FORWARD 2 • Q5/5

408

QUESTION ID

S13-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q5

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.9 • Set 13 • FORWARD 2 • Q5/5

SET 14 • 8.EE.B.5 • EXPRESSIONS & EQUATIONS

Compare two proportional relationships represented in different ways

MAIN	Grade 8 main question	**8.EE.B.5**
BACKWARD 1	Write an equation for a proportional relationship	**7.RP.A.2.c**
BACKWARD 2	Compute unit rates with fractions	**7.RP.A.1**
FORWARD 1	Create and compare linear equations in two variables	**HSA-CED.A.2**
FORWARD 2	Compare linear and exponential growth	**HSF-LE.A.3**

SET 14 • 8.EE.B.5

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.B.5

Runner A's distance is d = 6.2t (miles, hours). Runner B's graph is a line through (0, 0) and (2, 13).

Which runner is faster?

A. Runner A

B. Runner B

C. They run at the same speed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q1

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 14 • MAIN • Q1/5

411

QUESTION ID

S14-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. Runner B

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q1

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 14 • MAIN • Q1/5

Plan A costs c = 0.12m dollars for m minutes. The table shows Plan B.

Which plan costs less per minute, and by how much?

Minutes	100	250	400
Cost ($)	15	37.5	60

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q2

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 14 • MAIN • Q2/5

412

QUESTION ID

S14-M-Q2

TYPE

Short answer

CORRECT ANSWER

Plan A, by $0.03 per minute

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q2

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 14 • MAIN • Q2/5

Pump A fills 45 gallons in 3 minutes at a constant rate.
Pump B fills according to g = 14m.

Which pump is faster, and by how much?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q3

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 14 • MAIN • Q3/5

413

QUESTION ID

S14-M-Q3

TYPE

Short answer

CORRECT ANSWER

Pump A, by 1 gallon per minute

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q3

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 14 • MAIN • Q3/5

True or false?

The graph of y = 0.8x is less steep than the graph of a proportional relationship through (5, 5).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q4

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 14 • MAIN • Q4/5

414

QUESTION ID

S14-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q4

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 14 • MAIN • Q4/5

True or false?

A proportional relationship through (3, 7) has a greater unit rate than y = 2.5x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q5

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 14 • MAIN • Q5/5

415

QUESTION ID

S14-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 7 ÷ 3 ≈ 2.33, which is less than 2.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q5

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 14 • MAIN • Q5/5

SET 14 • 8.EE.B.5

BACKWARD 1

Write an equation for a proportional relationship • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c

Pencils cost $0.45 each.

Write an equation for the cost c of n pencils.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 14 • BACKWARD 1 • Q1/5

417

QUESTION ID

S14-B1-Q1

TYPE

Short answer

CORRECT ANSWER

c = 0.45n

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 14 • BACKWARD 1 • Q1/5

The table shows a proportional relationship.

Write an equation for y in terms of x.

x	2	4	6
y	7	14	21

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 14 • BACKWARD 1 • Q2/5

418

QUESTION ID

S14-B1-Q2

TYPE

Short answer

CORRECT ANSWER

y = 3.5x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 14 • BACKWARD 1 • Q2/5

A car travels 55 miles per hour.

Which equation gives the distance d after t hours?

A. $d = \frac{t}{55}$

B. $d = 55t$

C. $t = 55d$

D. $d = t + 55$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 14 • BACKWARD 1 • Q3/5

QUESTION ID

S14-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. d = 55t

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 14 • BACKWARD 1 • Q3/5

True or false?

A recipe uses 6 cups of water for every cup of rice. The equation $y = 6x$ gives the cups of water y for x cups of rice.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 14 • BACKWARD 1 • Q4/5

420

QUESTION ID

S14-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 14 • BACKWARD 1 • Q4/5

True or false?

A printer prints 30 pages in 2 minutes. The equation p = 30m gives the pages p printed in m minutes.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 14 • BACKWARD 1 • Q5/5

421

QUESTION ID

S14-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The equation is p = 15m.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 14 • BACKWARD 1 • Q5/5

SET 14 • 8.EE.B.5

BACKWARD 2

Compute unit rates with fractions • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1

Ty walks $\frac{3}{4}$ mile in $\frac{1}{4}$ hour.

How many miles per hour is this?

Miles per hour: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 14 • BACKWARD 2 • Q1/5

423

QUESTION ID

S14-B2-Q1

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 14 • BACKWARD 2 • Q1/5

$\frac{2}{5}$ pound of nuts costs $2.

What is the price per pound?

Price: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 14 • BACKWARD 2 • Q2/5

424

QUESTION ID

S14-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 14 • BACKWARD 2 • Q2/5

A hose fills $\frac{1}{2}$ gallon in $\frac{1}{6}$ minute.

How many gallons per minute is this?

A. $\frac{1}{12}$

B. 6

C. $\frac{1}{3}$

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 14 • BACKWARD 2 • Q3/5

425

QUESTION ID

S14-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 14 • BACKWARD 2 • Q3/5

True or false?

Walking $\frac{1}{3}$ mile in $\frac{1}{6}$ hour is a speed of 2 miles per hour.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 14 • BACKWARD 2 • Q4/5

426

QUESTION ID

S14-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 14 • BACKWARD 2 • Q4/5

True or false?

A drink uses $\frac{3}{4}$ cup of juice for every $\frac{1}{2}$ cup of water. That is $\frac{3}{8}$ cup of juice per cup of water.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 14 • BACKWARD 2 • Q5/5

427

QUESTION ID

S14-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is 3/2 cups of juice per cup of water.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 14 • BACKWARD 2 • Q5/5

SET 14 • 8.EE.B.5

FORWARD 1

Create and compare linear equations in two variables • High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2

Tank A has 20 gallons and gains 5 gallons per minute. Tank B starts empty and gains 7 gallons per minute.

Write an equation for each tank, with y gallons after x minutes.

Equations: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q1

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 14 • FORWARD 1 • Q1/5

429

QUESTION ID

S14-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Tank A: y = 5x + 20;

Tank B: y = 7x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q1

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 14 • FORWARD 1 • Q1/5

Tank A holds y = 5x + 20 gallons and Tank B holds y = 7x gallons after x minutes.

When do they hold the same amount?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q2

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 14 • FORWARD 1 • Q2/5

430

QUESTION ID

S14-F1-Q2

TYPE

Short answer

CORRECT ANSWER

After 10 minutes (70 gallons each)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q2

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 14 • FORWARD 1 • Q2/5

A car starts 30 miles from home and drives away at 50 miles per hour.

Which equation gives its distance d from home after t hours?

A. d = 50t − 30

B. d = 50t + 30

C. d = 30t + 50

D. d = 80t

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q3

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 14 • FORWARD 1 • Q3/5

431

QUESTION ID

S14-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. d = 50t + 30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q3

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 14 • FORWARD 1 • Q3/5

True or false?

The graphs of y = 5x + 20 and y = 7x intersect at (10, 70).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q4

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 14 • FORWARD 1 • Q4/5

QUESTION ID

S14-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q4

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 14 • FORWARD 1 • Q4/5

True or false?

$y = 3x + 4$ and $y = 3x - 2$ have different rates of change.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q5

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 14 • FORWARD 1 • Q5/5

433

QUESTION ID

S14-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Both have a rate of change of 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q5

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 14 • FORWARD 1 • Q5/5

SET 14 • 8.EE.B.5

FORWARD 2

Compare linear and exponential growth • High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3

f(x) = 100 + 50x and g(x) = 100(1.5)^x.

Which is greater at x = 5? Give both values.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q1

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3 • Set 14 • FORWARD 2 • Q1/5

435

QUESTION ID

S14-F2-Q1

TYPE

Short answer

CORRECT ANSWER

g: g(5) ≈ 759.4, while f(5) = 350

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q1

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3 • Set 14 • FORWARD 2 • Q1/5

Which function eventually has the greater values:
y = 1,000x or y = 2^x?

A. It depends only on the y-intercepts

B. y = 1,000x

C. y = 2^x

D. Neither; they are always equal

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q2

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3 • Set 14 • FORWARD 2 • Q2/5

436

QUESTION ID

S14-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. y = 2^x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q2

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3 • Set 14 • FORWARD 2 • Q2/5

Which function in the table is exponential? Explain.

x	0	1	2	3
f(x)	2	4	8	16
g(x)	2	4	6	8

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q3

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3 • Set 14 • FORWARD 2 • Q3/5

437

QUESTION ID

S14-F2-Q3

TYPE

Short answer

CORRECT ANSWER

f, because it doubles each time x increases by 1 (g increases by 2 each time).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q3

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3 • Set 14 • FORWARD 2 • Q3/5

True or false?

A quantity that increases exponentially eventually exceeds a quantity that increases linearly.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q4

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3 • Set 14 • FORWARD 2 • Q4/5

438

QUESTION ID

S14-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q4

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3 • Set 14 • FORWARD 2 • Q4/5

True or false?

At x = 2, 3^x is greater than 10x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q5

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3 • Set 14 • FORWARD 2 • Q5/5

439

QUESTION ID

S14-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $3^2 = 9$, which is less than 20.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q5

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3 • Set 14 • FORWARD 2 • Q5/5

SET 15 • 8.EE.B.6 • EXPRESSIONS & EQUATIONS

Use similar triangles to explain slope and derive y = mx and y = mx + b

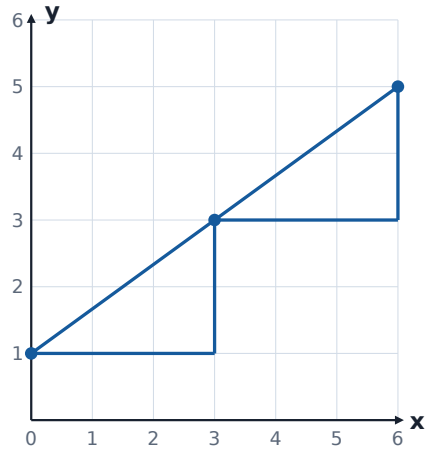
MAIN	Grade 8 main question	**8.EE.B.6**
BACKWARD 1	Use a scale factor to find a length in a scaled copy	**7.G.A.1**
BACKWARD 2	Write an equation from a table of values	**6.EE.C.9**
FORWARD 1	Use similarity transformations to decide whether triangles are sim…	**HSG-SRT.A.2**
FORWARD 2	Use slopes of parallel and perpendicular lines	**HSG-GPE.B.5**

SET 15 • 8.EE.B.6

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.B.6

Two slope triangles are drawn under the line. Explain why they give the same slope.



Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q1

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 15 • MAIN • Q1/5

442

QUESTION ID

S15-M-Q1

TYPE

Short answer

CORRECT ANSWER

The triangles are similar (they have the same angles), so their rise-to-run ratios are equal: 2/3 = 2/3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q1

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 15 • MAIN • Q1/5

A line passes through the origin and (4, 6). By similar triangles, any point (x, y) on the line has y ÷ x = 6 ÷ 4.

Write the equation of the line.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q2

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 15 • MAIN • Q2/5

443

QUESTION ID

S15-M-Q2

TYPE

Short answer

CORRECT ANSWER

y = 1.5x

GRADING NOTE y = (3/2)x is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q2

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 15 • MAIN • Q2/5

A line has a slope of 2 and crosses the y-axis at (0, 3).

Which equation describes it?

A. y = 2x

B. y = 2x + 3

C. y = 3x + 2

D. y = x + 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q3

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 15 • MAIN • Q3/5

QUESTION ID

S15-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. y = 2x + 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q3

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 15 • MAIN • Q3/5

A line crosses the y-axis at (0, −2) and passes through (5, 8).

Write its equation in the form y = mx + b.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q4

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 15 • MAIN • Q4/5

445

QUESTION ID

S15-M-Q4

TYPE

Short answer

CORRECT ANSWER

y = 2x − 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q4

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 15 • MAIN • Q4/5

True or false?

Any two slope triangles drawn on the same non-vertical line are similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q5

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 15 • MAIN • Q5/5

446

QUESTION ID

S15-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q5

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 15 • MAIN • Q5/5

SET 15 • 8.EE.B.6

BACKWARD 1

Use a scale factor to find a length in a scaled copy • Grade 7 • Geometry • 7.G.A.1

A right triangle has legs of 2 and 3. In a scaled copy, the leg of 2 becomes 8.

How long is the other leg in the copy?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 15 • BACKWARD 1 • Q1/5

448

QUESTION ID

S15-B1-Q1

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 15 • BACKWARD 1 • Q1/5

A right triangle has legs of 4 and 6. A scaled copy has a leg of 6 matching the leg of 4.

How long is the other leg in the copy?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q2

Grade 7 • Geometry • 7.G.A.1 • Set 15 • BACKWARD 1 • Q2/5

449

QUESTION ID

S15-B1-Q2

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q2

Grade 7 • Geometry • 7.G.A.1 • Set 15 • BACKWARD 1 • Q2/5

Which pair of right triangles are scaled copies of each other (matching the shorter legs)?

A. Legs 2 and 5; legs 4 and 7

B. Legs 2 and 5; legs 6 and 12

C. Legs 2 and 5; legs 3 and 6

D. Legs 2 and 5; legs 4 and 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 15 • BACKWARD 1 • Q3/5

450

QUESTION ID

S15-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. Legs 2 and 5; legs 4 and 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 15 • BACKWARD 1 • Q3/5

True or false?

A scaled copy with a scale factor of 3 has every length 3 times as long.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 15 • BACKWARD 1 • Q4/5

451

QUESTION ID

S15-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 15 • BACKWARD 1 • Q4/5

True or false?

Right triangles with legs 3 and 4 and legs 6 and 9 are scaled copies of each other.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 15 • BACKWARD 1 • Q5/5

452

QUESTION ID

S15-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 15 • BACKWARD 1 • Q5/5

SET 15 • 8.EE.B.6

BACKWARD 2

Write an equation from a table of values • Grade 6 • Expressions & Equations • 6.EE.C.9

Write an equation for y in terms of x.

x	1	2	3
y	6	7	8

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 15 • BACKWARD 2 • Q1/5

454

QUESTION ID

S15-B2-Q1

TYPE

Short answer

CORRECT ANSWER

y = x + 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 15 • BACKWARD 2 • Q1/5

Write an equation for y in terms of x.

x	1	2	3
y	3	6	9

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 15 • BACKWARD 2 • Q2/5

455

QUESTION ID

S15-B2-Q2

TYPE

Short answer

CORRECT ANSWER

y = 3x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 15 • BACKWARD 2 • Q2/5

Which equation matches the table?

x	0	1	2
y	1	3	5

A. y = 2x

B. y = x + 1

C. y = 3x

D. y = 2x + 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 15 • BACKWARD 2 • Q3/5

456

QUESTION ID

S15-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. y = 2x + 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 15 • BACKWARD 2 • Q3/5

True or false?

For y = 2x + 4, y = 10 when x = 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 15 • BACKWARD 2 • Q4/5

457

QUESTION ID

S15-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 15 • BACKWARD 2 • Q4/5

True or false?

For y = x + 7, y = 14 when x = 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 15 • BACKWARD 2 • Q5/5

458

QUESTION ID

S15-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 15 • BACKWARD 2 • Q5/5

SET 15 • 8.EE.B.6

FORWARD 1

Use similarity transformations to decide whether triangles are similar • High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

Triangle ABC has sides 3, 4, and 5. Triangle DEF has sides 6, 8, and 10.

Are the triangles similar? If so, what is the scale factor from ABC to DEF?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q1

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 15 • FORWARD 1 • Q1/5

460

QUESTION ID

S15-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Yes; the scale factor is 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q1

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 15 • FORWARD 1 • Q1/5

Explain how a dilation shows that two slope triangles on the same line are similar.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q2

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 15 • FORWARD 1 • Q2/5

461

QUESTION ID

S15-F1-Q2

TYPE

Short answer

CORRECT ANSWER

A dilation centered at a point on the line maps one triangle onto the other. Dilations keep angle measures, so the triangles are similar.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S15-F1-Q2**

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 15 • FORWARD 1 • Q2/5

Which information is enough to show that two triangles are similar?

A. One pair of equal sides

B. Two pairs of equal angles

C. Equal perimeters

D. One pair of equal angles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q3

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 15 • FORWARD 1 • Q3/5

462

QUESTION ID

S15-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. Two pairs of equal angles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q3

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 15 • FORWARD 1 • Q3/5

True or false?

If one triangle is a dilation of another, their corresponding sides are proportional.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q4

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 15 • FORWARD 1 • Q4/5

463

QUESTION ID

S15-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q4

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 15 • FORWARD 1 • Q4/5

True or false?

Any two right triangles are similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q5

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 15 • FORWARD 1 • Q5/5

464

QUESTION ID

S15-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q5

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 15 • FORWARD 1 • Q5/5

SET 15 • 8.EE.B.6

FORWARD 2

Use slopes of parallel and perpendicular lines • High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

Line p has a slope of $\frac{3}{4}$.

What is the slope of a line perpendicular to p?

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q1

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 15 • FORWARD 2 • Q1/5

466

QUESTION ID

S15-F2-Q1

TYPE

Short answer

CORRECT ANSWER

−4/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S15-F2-Q1**

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 15 • FORWARD 2 • Q1/5

Write an equation of the line through (0, 2) that is parallel to y = −3x + 7.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q2

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 15 • FORWARD 2 • Q2/5

467

QUESTION ID

S15-F2-Q2

TYPE

Short answer

CORRECT ANSWER

y = −3x + 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q2

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 15 • FORWARD 2 • Q2/5

Which line is perpendicular to $y = 2x - 5$?

A. $y = -2x - 5$

B. $y = 2x + 1$

C. $y = -\frac{1}{2}x + 1$

D. $y = \frac{1}{2}x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q3

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 15 • FORWARD 2 • Q3/5

468

QUESTION ID

S15-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $y = -(1/2)x + 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q3

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 15 • FORWARD 2 • Q3/5

True or false?

Parallel lines that are not vertical have equal slopes.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q4

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 15 • FORWARD 2 • Q4/5

469

QUESTION ID

S15-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q4

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 15 • FORWARD 2 • Q4/5

True or false?

Lines with slopes 3 and −3 are perpendicular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q5

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 15 • FORWARD 2 • Q5/5

470

QUESTION ID

S15-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Perpendicular slopes multiply to −1; 3 × (−3) = −9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S15-F2-Q5**

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 15 • FORWARD 2 • Q5/5

SET 16 • 8.EE.C.7.a • EXPRESSIONS & EQUATIONS

Decide whether a linear equation has one solution, no solution, or infinitely many

MAIN	Grade 8 main question	**8.EE.C.7.a**
BACKWARD 1	Expand linear expressions	**7.EE.A.1**
BACKWARD 2	Solve two-step equations	**7.EE.B.4.a**
FORWARD 1	Solve systems and recognize when they have no solution or infinit…	**HSA-REI.C.6**
FORWARD 2	Use the form $(x - p)^2 = q$ to find how many real solutions a quadra…	**HSA-REI.B.4.a**

SET 16 • 8.EE.C.7.a

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.C.7.a

How many solutions does this equation have?

$4(x + 3) = 4x + 12$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 16 • MAIN • Q1/5

473

QUESTION ID

S16-M-Q1

TYPE

Short answer

CORRECT ANSWER

Infinitely many solutions

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 16 • MAIN • Q1/5

How many solutions does this equation have?

$5x - 2 = 5x + 7$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 16 • MAIN • Q2/5

474

QUESTION ID

S16-M-Q2

TYPE

Short answer

CORRECT ANSWER

No solution

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 16 • MAIN • Q2/5

Which equation has exactly one solution?

A. 3(x + 1) = 3x + 3

B. 2x − 2x = 0

C. 3x + 1 = 2x + 9

D. 3x + 1 = 3x + 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 16 • MAIN • Q3/5

475

QUESTION ID

S16-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 3x + 1 = 2x + 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 16 • MAIN • Q3/5

What number in the blank makes the equation have no solution?

$6x + 5 = ___x + 1$

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 16 • MAIN • Q4/5

476

QUESTION ID

S16-M-Q4

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 16 • MAIN • Q4/5

True or false?

$2(x - 4) = 2x - 4$ has infinitely many solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 16 • MAIN • Q5/5

477

QUESTION ID

S16-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It simplifies to $-8 = -4$, so it has no solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 16 • MAIN • Q5/5

SET 16 • 8.EE.C.7.a

BACKWARD 1

Expand linear expressions • Grade 7 • Expressions & Equations • 7.EE.A.1

Expand.

4(x + 3)

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 16 • BACKWARD 1 • Q1/5

479

QUESTION ID

S16-B1-Q1

TYPE

Short answer

CORRECT ANSWER

4x + 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 16 • BACKWARD 1 • Q1/5

Expand.

$-2(3x - 5)$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 16 • BACKWARD 1 • Q2/5

480

QUESTION ID

S16-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$-6x + 10$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 16 • BACKWARD 1 • Q2/5

Which expression is equal to 5(2x − 1)?

A. 10x + 5

B. 10x − 1

C. 10x − 5

D. 7x − 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 16 • BACKWARD 1 • Q3/5

481

QUESTION ID

S16-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 10x − 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 16 • BACKWARD 1 • Q3/5

True or false?

$3(x + 2) = 3x + 6$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 16 • BACKWARD 1 • Q4/5

482

QUESTION ID

S16-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 16 • BACKWARD 1 • Q4/5

True or false?

$-(x - 4) = -x - 4$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 16 • BACKWARD 1 • Q5/5

483

QUESTION ID

S16-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 16 • BACKWARD 1 • Q5/5

SET 16 • 8.EE.C.7.a

BACKWARD 2

Solve two-step equations • Grade 7 • Expressions & Equations • 7.EE.B.4.a

Solve for x.

3x + 7 = 22

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 16 • BACKWARD 2 • Q1/5

485

QUESTION ID

S16-B2-Q1

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 16 • BACKWARD 2 • Q1/5

Solve for x.

$-2x + 4 = 10$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 16 • BACKWARD 2 • Q2/5

486

QUESTION ID

S16-B2-Q2

TYPE

Short answer

CORRECT ANSWER

−3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 16 • BACKWARD 2 • Q2/5

Solve for x.

$5x - 6 = 19$

A. $x = \frac{13}{5}$

B. $x = -5$

C. $x = 5$

D. $x = 25$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 16 • BACKWARD 2 • Q3/5

487

QUESTION ID

S16-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. x = 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 16 • BACKWARD 2 • Q3/5

True or false?

$x = 2$ is the solution of $4x + 1 = 9$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 16 • BACKWARD 2 • Q4/5

488

QUESTION ID

S16-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 16 • BACKWARD 2 • Q4/5

True or false?

$x = 3$ is the solution of $2x - 5 = 3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 16 • BACKWARD 2 • Q5/5

489

QUESTION ID

S16-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 16 • BACKWARD 2 • Q5/5

SET 16 • 8.EE.C.7.a

FORWARD 1

Solve systems and recognize when they have no solution or infinitely many • High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6

How many solutions does the system have?

$y = 2x + 3$

$y = 2x - 1$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 16 • FORWARD 1 • Q1/5

491

QUESTION ID

S16-F1-Q1

TYPE

Short answer

CORRECT ANSWER

No solution

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 16 • FORWARD 1 • Q1/5

Solve the system.

$x + y = 7$

$x - y = 1$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 16 • FORWARD 1 • Q2/5

492

QUESTION ID

S16-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$x = 4, y = 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 16 • FORWARD 1 • Q2/5

How many solutions does the system have?

$2x + 4y = 8$

$x + 2y = 4$

A. Two

B. Infinitely many

C. One

D. None

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 16 • FORWARD 1 • Q3/5

493

QUESTION ID

S16-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. Infinitely many

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 16 • FORWARD 1 • Q3/5

True or false?

A system of two different parallel lines has no solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 16 • FORWARD 1 • Q4/5

494

QUESTION ID

S16-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 16 • FORWARD 1 • Q4/5

True or false?

The system $y = x + 1$ and $y = -x + 1$ has no solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 16 • FORWARD 1 • Q5/5

495

QUESTION ID

S16-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The lines meet at (0, 1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 16 • FORWARD 1 • Q5/5

SET 16 • 8.EE.C.7.a

FORWARD 2

Use the form $(x - p)^2 = q$ to find how many real solutions a quadratic equation has • High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a

How many real solutions does $(x - 3)^2 = 0$ have? Give any solutions.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q1

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 16 • FORWARD 2 • Q1/5

497

QUESTION ID

S16-F2-Q1

TYPE

Short answer

CORRECT ANSWER

One: x = 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q1

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 16 • FORWARD 2 • Q1/5

How many real solutions does $(x + 1)^2 = -4$ have? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q2

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 16 • FORWARD 2 • Q2/5

498

QUESTION ID

S16-F2-Q2

TYPE

Short answer

CORRECT ANSWER

None. A square of a real number cannot be negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q2

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 16 • FORWARD 2 • Q2/5

Completing the square on $x^2 + 6x + 13 = 0$ gives $(x + 3)^2 = q$.

What is q, and how many real solutions are there?

A. q = 4; two real solutions

B. q = −4; one real solution

C. q = 13; two real solutions

D. q = −4; no real solutions

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q3

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 16 • FORWARD 2 • Q3/5

499

QUESTION ID

S16-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. q = −4; no real solutions

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q3

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 16 • FORWARD 2 • Q3/5

True or false?

$(x - 2)^2 = 9$ has two real solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q4

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 16 • FORWARD 2 • Q4/5

500

QUESTION ID

S16-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $x = 5$ and $x = -1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q4

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 16 • FORWARD 2 • Q4/5

True or false?

$x^2 - 6x + 9 = 0$ has two different real solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q5

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 16 • FORWARD 2 • Q5/5

501

QUESTION ID

S16-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is $(x - 3)^2 = 0$, so $x = 3$ is the only solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q5

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 16 • FORWARD 2 • Q5/5

SET 17 • 8.EE.C.7.b • EXPRESSIONS & EQUATIONS

Solve multistep linear equations by expanding and collecting like terms

MAIN	Grade 8 main question	**8.EE.C.7.b**
BACKWARD 1	Combine like terms	**7.EE.A.1**
BACKWARD 2	Solve two-step equations with rational numbers	**7.EE.B.4.a**
FORWARD 1	Solve linear equations with letter coefficients	**HSA-REI.B.3**
FORWARD 2	Justify each step of a solution and find an invalid step	**HSA-REI.A.1**

SET 17 • 8.EE.C.7.b

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.C.7.b

Solve for x.

$3(2x - 4) = 4x + 6$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • MAIN • Q1/5

504

QUESTION ID

S17-M-Q1

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • MAIN • Q1/5

Solve for x.

$5 - 2(x + 1) = 3x - 7$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • MAIN • Q2/5

505

QUESTION ID

S17-M-Q2

TYPE

Short answer

CORRECT ANSWER

2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • MAIN • Q2/5

Solve for x.

$4(x - 1) - (x + 2) = 2x + 3$

A. $x = 1$

B. $x = 5$

C. $x = 9$

D. $x = -9$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • MAIN • Q3/5

506

QUESTION ID

S17-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. x = 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • MAIN • Q3/5

Solve. Show each step.

$-3(x + 5) + 2x = 4(x - 10)$

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • MAIN • Q4/5

507

QUESTION ID

S17-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

5

GRADING NOTE REQUIRED WORK: Any valid sequence of steps. Example: $-3x - 15 + 2x = 4x - 40$; $-x - 15 = 4x - 40$; $25 = 5x$; $x = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • MAIN • Q4/5

True or false?

$x = -1$ is the solution of $2(3x + 1) = 4x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • MAIN • Q5/5

508

QUESTION ID

S17-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • MAIN • Q5/5

SET 17 • 8.EE.C.7.b

BACKWARD 1

Combine like terms • Grade 7 • Expressions & Equations • 7.EE.A.1

Simplify.

$6x - 4 + 2x + 9$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 17 • BACKWARD 1 • Q1/5

510

QUESTION ID

S17-B1-Q1

TYPE

Short answer

CORRECT ANSWER

8x + 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 17 • BACKWARD 1 • Q1/5

Simplify.

$-3a + 7 - a$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 17 • BACKWARD 1 • Q2/5

511

QUESTION ID

S17-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$-4a + 7$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 17 • BACKWARD 1 • Q2/5

Simplify.

$2(x + 3) + 5x$

A. $10x + 6$

B. $2x + 11$

C. $7x + 3$

D. $7x + 6$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 17 • BACKWARD 1 • Q3/5

512

QUESTION ID

S17-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 7x + 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 17 • BACKWARD 1 • Q3/5

True or false?

$4y - y = 3y$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 17 • BACKWARD 1 • Q4/5

513

QUESTION ID

S17-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 17 • BACKWARD 1 • Q4/5

True or false?

$5n + 2 - 2n = 3n$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 17 • BACKWARD 1 • Q5/5

514

QUESTION ID

S17-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $5n + 2 - 2n = 3n + 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 17 • BACKWARD 1 • Q5/5

SET 17 • 8.EE.C.7.b

BACKWARD 2

Solve two-step equations with rational numbers • Grade 7 • Expressions & Equations • 7.EE.B.4.a

Solve for x.

$6x - 5 = 31$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 17 • BACKWARD 2 • Q1/5

516

QUESTION ID

S17-B2-Q1

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 17 • BACKWARD 2 • Q1/5

Solve for x.

$\frac{x}{4} + 3 = 7$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 17 • BACKWARD 2 • Q2/5

517

QUESTION ID

S17-B2-Q2

TYPE

Short answer

CORRECT ANSWER

16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 17 • BACKWARD 2 • Q2/5

Solve for x.

$-3x + 8 = -10$

A. $x = -6$

B. $x = -\frac{2}{3}$

C. $x = \frac{2}{3}$

D. $x = 6$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 17 • BACKWARD 2 • Q3/5

518

QUESTION ID

S17-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. x = 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 17 • BACKWARD 2 • Q3/5

True or false?

$x = -2$ is the solution of $5x + 12 = 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 17 • BACKWARD 2 • Q4/5

519

QUESTION ID

S17-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 17 • BACKWARD 2 • Q4/5

True or false?

$x = 4$ is the solution of $3x - 2 = 14$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 17 • BACKWARD 2 • Q5/5

520

QUESTION ID

S17-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 17 • BACKWARD 2 • Q5/5

SET 17 • 8.EE.C.7.b

FORWARD 1

Solve linear equations with letter coefficients • High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

Solve for x. Assume a ≠ 3.

ax + 2 = 3x + 8

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 17 • FORWARD 1 • Q1/5

522

QUESTION ID

S17-F1-Q1

TYPE

Short answer

CORRECT ANSWER

6/(a − 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S17-F1-Q1**

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 17 • FORWARD 1 • Q1/5

Solve for x. Assume k ≠ 0.

$k(x - 4) = 2k$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 17 • FORWARD 1 • Q2/5

523

QUESTION ID

S17-F1-Q2

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S17-F1-Q2**

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 17 • FORWARD 1 • Q2/5

Solve for x. Assume b ≠ 0.

$bx - 5b = 2b$

A. $x = 7b$

B. $x = \frac{7}{b}$

C. $x = 7$

D. $x = 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 17 • FORWARD 1 • Q3/5

524

QUESTION ID

S17-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. x = 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 17 • FORWARD 1 • Q3/5

True or false?

If $c \neq 1$, the solution of $cx + 1 = x + 1$ is $x = 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 17 • FORWARD 1 • Q4/5

525

QUESTION ID

S17-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 17 • FORWARD 1 • Q4/5

True or false?

The solution of $mx = m + 4$ is $x = 5$ for every nonzero value of m.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 17 • FORWARD 1 • Q5/5

526

QUESTION ID

S17-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $x = (m + 4)/m$, which is 5 only when $m = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 17 • FORWARD 1 • Q5/5

SET 17 • 8.EE.C.7.b

FORWARD 2

Justify each step of a solution and find an invalid step • High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Solve 2(x − 3) + 4 = 5x − 11. Give a reason for each step.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q1

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • FORWARD 2 • Q1/5

528

QUESTION ID

S17-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

3

GRADING NOTE REQUIRED WORK: Every step with a valid reason. Other valid orders of steps are accepted. A correct answer without reasons is not full credit. Example: distributive property (2x − 6 + 4 = 5x − 11); combine like terms (2x − 2 = 5x − 11); subtract 2x and add 11 to both sides (9 = 3x); divide both sides by 3 (x = 3).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q1

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • FORWARD 2 • Q1/5

Dee solved $4 - 2(x - 1) = 8$ like this:

Step 1: $4 - 2x - 2 = 8$

Step 2: $2 - 2x = 8$

Step 3: $x = -3$

Which step is wrong? Correct it and give the solution.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q2

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • FORWARD 2 • Q2/5

529

QUESTION ID

S17-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Step 1: $-2(x - 1) = -2x + 2$, so $6 - 2x = 8$ and $x = -1$.

GRADING NOTE Grade both: Step 1 with the correction, and $x = -1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q2

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • FORWARD 2 • Q2/5

Which reason justifies rewriting $3x - 5 = 7$ as $3x = 12$?

A. Addition property of equality

B. Multiplication property of equality

C. Commutative property of addition

D. Distributive property

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q3

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • FORWARD 2 • Q3/5

530

QUESTION ID

S17-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Addition property of equality

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q3

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • FORWARD 2 • Q3/5

True or false?

Each step of a valid solution gives an equation with the same solutions as the original equation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q4

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • FORWARD 2 • Q4/5

531

QUESTION ID

S17-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q4

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • FORWARD 2 • Q4/5

True or false?

Multiplying both sides of an equation by 0 keeps the same solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q5

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • FORWARD 2 • Q5/5

532

QUESTION ID

S17-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q5

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • FORWARD 2 • Q5/5

SET 18 • 8.EE.C.7.b • EXPRESSIONS & EQUATIONS

Write and solve linear equations with rational coefficients for real-world problems

MAIN	Grade 8 main question	**8.EE.C.7.b**
BACKWARD 1	Solve word problems that lead to px + q = r	**7.EE.B.4.a**
BACKWARD 2	Multiply rational numbers in context	**7.NS.A.3**
FORWARD 1	Solve systems of linear equations from real-world problems	**HSA-REI.C.6**
FORWARD 2	Create equations and inequalities in one variable to solve proble…	**HSA-CED.A.1**

SET 18 • 8.EE.C.7.b

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.C.7.b

Gym A charges $30 to join plus $12.50 per month. Gym B charges $55 to join plus $10 per month.

Write and solve an equation to find when the total costs are equal.

Equation and answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • MAIN • Q1/5

535

QUESTION ID

S18-M-Q1

TYPE

Short answer

CORRECT ANSWER

The equation 30 + 12.5m = 55 + 10m and the answer m = 10 months

GRADING NOTE Both parts are required: a valid equation (or system) and the correct solution. Any valid model is accepted, but the solution must also be correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • MAIN • Q1/5

Plant A is $6\frac{1}{2}$ inches tall and grows $\frac{3}{4}$ inch per week.

Plant B is 10 inches tall and grows $\frac{1}{4}$ inch per week.

Write and solve an equation to find when they are the same height.

Equation and answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • MAIN • Q2/5

536

QUESTION ID

S18-M-Q2

TYPE

Short answer

CORRECT ANSWER

The equation 6.5 + 0.75w = 10 + 0.25w and the answer w = 7 weeks

GRADING NOTE Both parts are required: a valid equation (or system) and the correct solution. Any valid model is accepted, but the solution must also be correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • MAIN • Q2/5

Solve for x.

$\frac{2}{3}x - 4 = \frac{1}{2}x + 1$

A. x = 18

B. $x = \frac{30}{7}$

C. x = 30

D. x = 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • MAIN • Q3/5

537

QUESTION ID

S18-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. x = 30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • MAIN • Q3/5

Solve for x.

$0.4(x + 15) = 0.25x + 9$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • MAIN • Q4/5

538

QUESTION ID

S18-M-Q4

TYPE

Short answer

CORRECT ANSWER

20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • MAIN • Q4/5

True or false?

$x = 12$ is the solution of $\frac{3}{4}x + 2 = \frac{1}{2}x + 5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • MAIN • Q5/5

539

QUESTION ID

S18-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • MAIN • Q5/5

SET 18 • 8.EE.C.7.b

BACKWARD 1

Solve word problems that lead to px + q = r • Grade 7 • Expressions & Equations • 7.EE.B.4.a

Ana has $40. She buys some $6.50 tickets and has $1 left.

How many tickets did she buy?

Tickets: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 18 • BACKWARD 1 • Q1/5

541

QUESTION ID

S18-B1-Q1

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 18 • BACKWARD 1 • Q1/5

A rental costs $20 plus $8 per hour. The total is $68.

For how many hours was it rented?

Hours: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 18 • BACKWARD 1 • Q2/5

542

QUESTION ID

S18-B1-Q2

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 18 • BACKWARD 1 • Q2/5

Solve for x.

0.5x + 3 = 11

A. x = 7

B. x = 16

C. x = 28

D. x = 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 18 • BACKWARD 1 • Q3/5

543

QUESTION ID

S18-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. x = 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 18 • BACKWARD 1 • Q3/5

True or false?

$x = 9$ is the solution of $\frac{1}{3}x + 2 = 5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 18 • BACKWARD 1 • Q4/5

544

QUESTION ID

S18-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 18 • BACKWARD 1 • Q4/5

True or false?

$x = 4$ is the solution of $2.5x - 1 = 11$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 18 • BACKWARD 1 • Q5/5

545

QUESTION ID

S18-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The solution is $x = 4.8$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 18 • BACKWARD 1 • Q5/5

SET 18 • 8.EE.C.7.b

BACKWARD 2

Multiply rational numbers in context • Grade 7 • The Number System • 7.NS.A.3

A plant grows $\frac{3}{4}$ inch per week.

How much does it grow in 8 weeks?

Growth: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q1

Grade 7 • The Number System • 7.NS.A.3 • Set 18 • BACKWARD 2 • Q1/5

547

QUESTION ID

S18-B2-Q1

TYPE

Short answer

CORRECT ANSWER

6 inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q1

Grade 7 • The Number System • 7.NS.A.3 • Set 18 • BACKWARD 2 • Q1/5

A gym charges $12.50 per month.

How much do 10 months cost?

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q2

Grade 7 • The Number System • 7.NS.A.3 • Set 18 • BACKWARD 2 • Q2/5

548

QUESTION ID

S18-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$125

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q2

Grade 7 • The Number System • 7.NS.A.3 • Set 18 • BACKWARD 2 • Q2/5

A cyclist rides 0.4 mile per minute for 15 minutes.

How far does she ride?

A. 6 miles

B. 60 miles

C. 4 miles

D. 15.4 miles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q3

Grade 7 • The Number System • 7.NS.A.3 • Set 18 • BACKWARD 2 • Q3/5

549

QUESTION ID

S18-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 6 miles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q3

Grade 7 • The Number System • 7.NS.A.3 • Set 18 • BACKWARD 2 • Q3/5

True or false?

$\frac{2}{3}$ of 9 feet is 6 feet.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q4

Grade 7 • The Number System • 7.NS.A.3 • Set 18 • BACKWARD 2 • Q4/5

550

QUESTION ID

S18-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q4

Grade 7 • The Number System • 7.NS.A.3 • Set 18 • BACKWARD 2 • Q4/5

True or false?

0.25 of a 12-hour day is 4 hours.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q5

Grade 7 • The Number System • 7.NS.A.3 • Set 18 • BACKWARD 2 • Q5/5

551

QUESTION ID

S18-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 0.25 × 12 = 3 hours.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q5

Grade 7 • The Number System • 7.NS.A.3 • Set 18 • BACKWARD 2 • Q5/5

SET 18 • 8.EE.C.7.b

FORWARD 1

Solve systems of linear equations from real-world problems • High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6

Solve the system.

$y = 12.5x + 30$

$y = 10x + 55$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 18 • FORWARD 1 • Q1/5

553

QUESTION ID

S18-F1-Q1

TYPE

Short answer

CORRECT ANSWER

x = 10, y = 155

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S18-F1-Q1**

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 18 • FORWARD 1 • Q1/5

Adult tickets cost $8 and student tickets cost $5. 120 tickets were sold for $750.

How many adult tickets were sold?

Adult tickets: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 18 • FORWARD 1 • Q2/5

554

QUESTION ID

S18-F1-Q2

TYPE

Short answer

CORRECT ANSWER

50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 18 • FORWARD 1 • Q2/5

Solve the system.

$2x + 3y = 12$

$x - y = 1$

A. $x = 1, y = 0$

B. $x = 3, y = 2$

C. $x = 4, y = 3$

D. $x = 2, y = 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 18 • FORWARD 1 • Q3/5

555

QUESTION ID

S18-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. x = 3, y = 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 18 • FORWARD 1 • Q3/5

True or false?

$x = 2, y = 5$ is a solution of both $y = 3x - 1$ and $y = x + 3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 18 • FORWARD 1 • Q4/5

556

QUESTION ID

S18-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 18 • FORWARD 1 • Q4/5

True or false?

The system x + y = 10 and 2x + 2y = 15 has exactly one solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 18 • FORWARD 1 • Q5/5

557

QUESTION ID

S18-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The lines are parallel, so there is no solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 18 • FORWARD 1 • Q5/5

SET 18 • 8.EE.C.7.b

FORWARD 2

Create equations and inequalities in one variable to solve problems • High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.1

Plan A costs $25 plus $0.15 per text. Plan B costs $40 with unlimited texts.

Write and solve an inequality to find when Plan A costs less.

Inequality and answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q1

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.1 • Set 18 • FORWARD 2 • Q1/5

559

QUESTION ID

S18-F2-Q1

TYPE

Short answer

CORRECT ANSWER

The inequality 25 + 0.15t < 40 and the answer t < 100 texts (0 to 99 texts)

GRADING NOTE Both parts are required: a valid inequality and the correct solution. Any valid model is accepted, but the solution must also be correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q1

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.1 • Set 18 • FORWARD 2 • Q1/5

Two trains leave a station at the same time in opposite directions, at 55 and 65 miles per hour.

Write and solve an equation to find when they are 300 miles apart.

Equation and answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q2

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.1 • Set 18 • FORWARD 2 • Q2/5

560

QUESTION ID

S18-F2-Q2

TYPE

Short answer

CORRECT ANSWER

The equation 55t + 65t = 300 and the answer t = 2.5 hours

GRADING NOTE Both parts are required: a valid equation (or system) and the correct solution. Any valid model is accepted, but the solution must also be correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q2

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.1 • Set 18 • FORWARD 2 • Q2/5

A rectangle's length is 3 more than twice its width w. Its perimeter is 54.

Which equation can find w?

A. 2(2w) + 3 = 54

B. w + 2w + 3 = 54

C. 2w + 2(2w + 3) = 54

D. w(2w + 3) = 54

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q3

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.1 • Set 18 • FORWARD 2 • Q3/5

561

QUESTION ID

S18-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C.

2w + 2(2w + 3) = 54

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q3

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.1 • Set 18 • FORWARD 2 • Q3/5

True or false?

A rectangle has a length of 2w + 3 and a perimeter of 54. Its width w is 8.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q4

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.1 • Set 18 • FORWARD 2 • Q4/5

562

QUESTION ID

S18-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q4

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.1 • Set 18 • FORWARD 2 • Q4/5

True or false?

The inequality $25 + 0.15t < 40$ is true when $t = 120$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q5

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.1 • Set 18 • FORWARD 2 • Q5/5

563

QUESTION ID

S18-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $25 + 18 = 43$, which is not less than 40.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q5

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.1 • Set 18 • FORWARD 2 • Q5/5

SET 19 • 8.EE.C.8.a • EXPRESSIONS & EQUATIONS

Solve a system of two linear equations by graphing

MAIN	Grade 8 main question	**8.EE.C.8.a**
BACKWARD 1	Make a table and graph an equation	**6.EE.C.9**
BACKWARD 2	Graph points in all four quadrants	**6.NS.C.8**
FORWARD 1	Solve f(x) = g(x) from the intersection of two graphs	**HSA-REI.D.11**
FORWARD 2	Graph the solution set of a system of linear inequalities	**HSA-REI.D.12**

SET 19 • 8.EE.C.8.a

MAIN

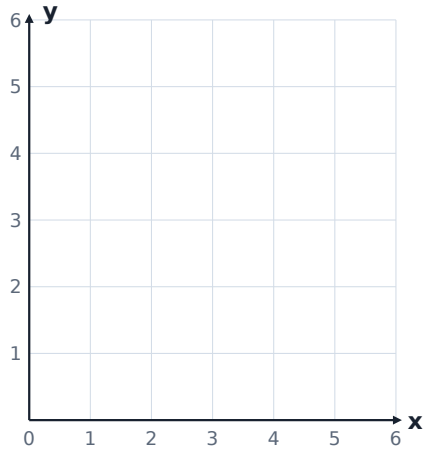
Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.C.8.a

ANSWER KEY Reduced copy of the question page, with its answer

Graph y = x + 1 and
y = −x + 5 on the grid.

What is the solution of the system?

Solution: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 19 • MAIN • Q1/5

566

QUESTION ID
S19-M-Q1

TYPE
Drawing + written answer

CORRECT ANSWER

Drawing: The line y = x + 1 through (0, 1) and (4, 5), and the line y = −x + 5 through (0, 5) and (5, 0), crossing at (2, 3)

Written: (2, 3)

GRADING NOTE Grade both: the two lines drawn correctly, and the solution (2, 3).

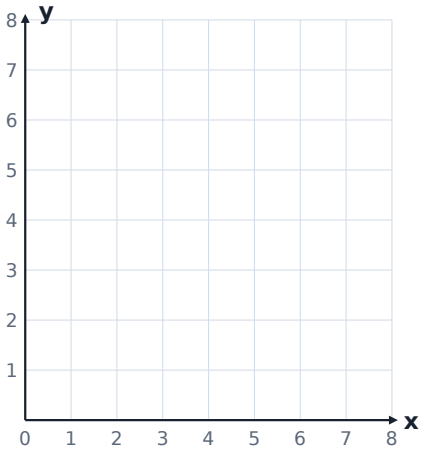
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 19 • MAIN • Q1/5

Graph y = 0.5x + 1 and y = −x + 7 on the grid.

What is the solution of the system?

Solution: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 19 • MAIN • Q2/5

567

QUESTION ID

S19-M-Q2

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: The line y = 0.5x + 1 through (0, 1) and (6, 4), and the line y = −x + 7 through (0, 7) and (7, 0), crossing at (4, 3)

Written: (4, 3)

GRADING NOTE Grade both: the two lines drawn correctly, and the solution (4, 3).

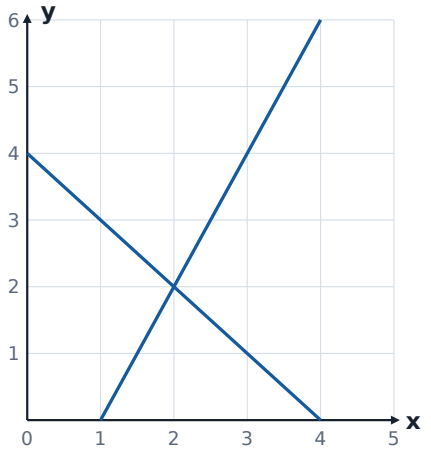
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 19 • MAIN • Q2/5

The graph shows the lines $y = -x + 4$ and $y = 2x - 2$.

What is the solution of the system? Check it in both equations.

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 19 • MAIN • Q3/5

568

QUESTION ID

S19-M-Q3

TYPE

Short answer

CORRECT ANSWER

(2, 2): 2 = −2 + 4 and 2 = 2(2) − 2.

GRADING NOTE Grade both: (2, 2) and the check in both equations.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 19 • MAIN • Q3/5

The graphs of the two equations in a system are parallel lines.

What does this tell you about the system?

A. It has no solution.

B. It has infinitely many solutions.

C. It has one solution.

D. Its solution is the origin.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 19 • MAIN • Q4/5

569

QUESTION ID

S19-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. It has no solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 19 • MAIN • Q4/5

True or false?

When the graphs of two linear equations are two different lines that cross, the crossing point is the only solution of the system.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 19 • MAIN • Q5/5

570

QUESTION ID

S19-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q5

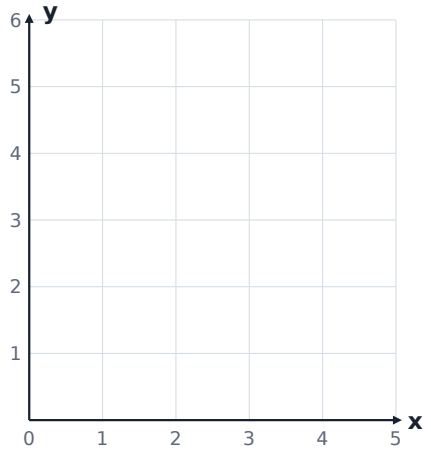
Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 19 • MAIN • Q5/5

SET 19 • 8.EE.C.8.a

BACKWARD 1

Make a table and graph an equation • Grade 6 • Expressions & Equations • 6.EE.C.9

Make a table for y = x + 2 with x = 0, 1, 2, and 3. Graph the points on the grid.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 19 • BACKWARD 1 • Q1/5

572

QUESTION ID

S19-B1-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Points at (0, 2), (1, 3), (2, 4), and (3, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 19 • BACKWARD 1 • Q1/5

What is the value of y when x = 4 in y = 3x?

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 19 • BACKWARD 1 • Q2/5

573

QUESTION ID

S19-B1-Q2

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 19 • BACKWARD 1 • Q2/5

Which point is on the graph of y = x + 3?

A. (1, 3)

B. (3, 0)

C. (5, 2)

D. (2, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 19 • BACKWARD 1 • Q3/5

574

QUESTION ID

S19-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (2, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 19 • BACKWARD 1 • Q3/5

True or false?

(4, 8) is on the graph of y = 2x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 19 • BACKWARD 1 • Q4/5

575

QUESTION ID

S19-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 19 • BACKWARD 1 • Q4/5

True or false?

(0, 1) is on the graph of y = x + 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 19 • BACKWARD 1 • Q5/5

576

QUESTION ID

S19-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q5

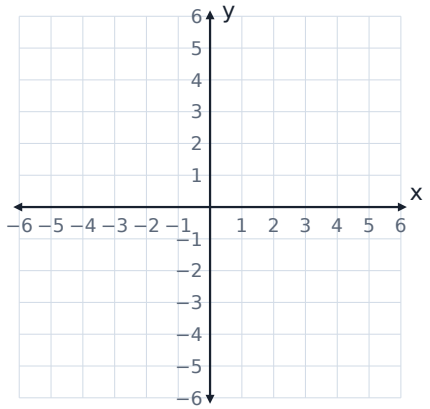
Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 19 • BACKWARD 1 • Q5/5

SET 19 • 8.EE.C.8.a

BACKWARD 2

Graph points in all four quadrants • Grade 6 • The Number System • 6.NS.C.8

Plot A(2, −3) and B(−4, 1) on the coordinate plane.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 19 • BACKWARD 2 • Q1/5

578

QUESTION ID

S19-B2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

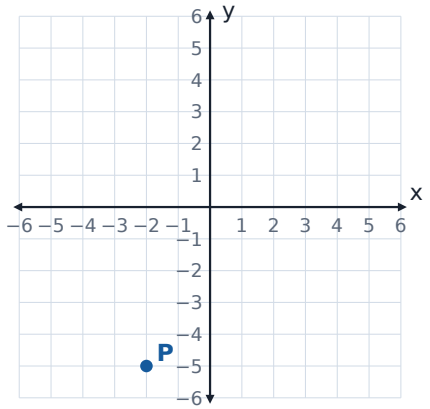
Point A at (2, −3) and point B at (−4, 1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 19 • BACKWARD 2 • Q1/5

What are the coordinates of point P?



Coordinates: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 19 • BACKWARD 2 • Q2/5

579

QUESTION ID

S19-B2-Q2

TYPE

Short answer

CORRECT ANSWER

(−2, −5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 19 • BACKWARD 2 • Q2/5

Which point is in Quadrant II?

A. (3, −4)

B. (−3, −4)

C. (3, 4)

D. (−3, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 19 • BACKWARD 2 • Q3/5

580

QUESTION ID

S19-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (−3, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 19 • BACKWARD 2 • Q3/5

True or false?

(0, −2) is on the y-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 19 • BACKWARD 2 • Q4/5

581

QUESTION ID

S19-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 19 • BACKWARD 2 • Q4/5

True or false?

(5, −1) is in Quadrant III.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 19 • BACKWARD 2 • Q5/5

582

QUESTION ID

S19-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE (5, −1) is in Quadrant IV.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 19 • BACKWARD 2 • Q5/5

SET 19 • 8.EE.C.8.a

FORWARD 1

Solve f(x) = g(x) from the intersection of two graphs • High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11

The graphs of f(x) = x + 1 and g(x) = 7 − x intersect at (3, 4).

What is the solution of x + 1 = 7 − x?

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 19 • FORWARD 1 • Q1/5

584

QUESTION ID

S19-F1-Q1

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S19-F1-Q1**

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 19 • FORWARD 1 • Q1/5

Use a table of values for x = 0, 1, 2, and 3 to solve 2x + 1 = 7 − x.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 19 • FORWARD 1 • Q2/5

585

QUESTION ID

S19-F1-Q2

TYPE

Short answer

CORRECT ANSWER

x = 2 (both sides equal 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S19-F1-Q2**

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 19 • FORWARD 1 • Q2/5

The graphs of $f(x) = x^2$ and $g(x) = x + 2$ intersect at (2, 4) and (−1, 1).

What are the solutions of $x^2 = x + 2$?

A. $x = 2$ only

B. $x = -1$ only

C. $x = 4$ and $x = 1$

D. $x = 2$ and $x = -1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 19 • FORWARD 1 • Q3/5

586

QUESTION ID

S19-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. x = 2 and x = −1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 19 • FORWARD 1 • Q3/5

True or false?

If the graphs of y = f(x) and y = g(x) intersect at (5, 9), then f(5) = g(5) = 9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 19 • FORWARD 1 • Q4/5

587

QUESTION ID

S19-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 19 • FORWARD 1 • Q4/5

True or false?

If the graphs of f and g never meet, then f(x) = g(x) has exactly one solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 19 • FORWARD 1 • Q5/5

588

QUESTION ID

S19-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It has no solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 19 • FORWARD 1 • Q5/5

SET 19 • 8.EE.C.8.a

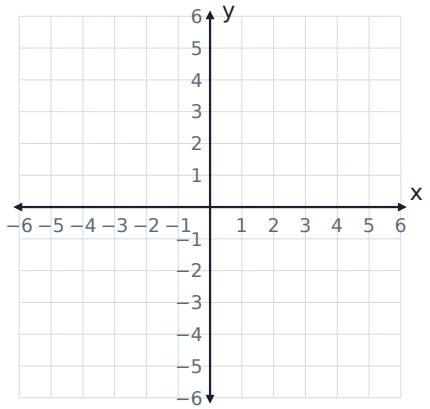
FORWARD 2

Graph the solution set of a system of linear inequalities • High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12

Graph the system on the coordinate plane.

$y \geq x$

$y < 4$



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q1

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 19 • FORWARD 2 • Q1/5

590

QUESTION ID

S19-F2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

A solid line y = x and a dashed line y = 4, with the region above or on y = x and below y = 4 shaded

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q1

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 19 • FORWARD 2 • Q1/5

Is (1, 3) a solution of the system $y > 2x$ and $y < x + 3$? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q2

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 19 • FORWARD 2 • Q2/5

591

QUESTION ID

S19-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Yes. 3 > 2(1) and 3 < 1 + 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q2

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 19 • FORWARD 2 • Q2/5

Which point is a solution of the system $y \leq -x + 4$ and $y > 1$?

A. (1, 2)

B. (4, 3)

C. (3, 2)

D. (0, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q3

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 19 • FORWARD 2 • Q3/5

592

QUESTION ID

S19-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. (1, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q3

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 19 • FORWARD 2 • Q3/5

True or false?

The boundary line of $y < 2x + 1$ is drawn dashed.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q4

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 19 • FORWARD 2 • Q4/5

593

QUESTION ID

S19-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q4

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 19 • FORWARD 2 • Q4/5

True or false?

The solution set of a system of two inequalities is every point that satisfies at least one of them.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q5

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 19 • FORWARD 2 • Q5/5

QUESTION ID

S19-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A solution must satisfy both inequalities (the overlap of the two regions).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q5

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 19 • FORWARD 2 • Q5/5

SET 20 • 8.EE.C.8.b • EXPRESSIONS & EQUATIONS

Solve a system of two linear equations algebraically

MAIN	Grade 8 main question	**8.EE.C.8.b**
BACKWARD 1	Evaluate expressions for given values	**6.EE.A.2.c**
BACKWARD 2	Solve equations with the distributive property	**7.EE.B.4.a**
FORWARD 1	Explain why adding equations (elimination) keeps the same solut…	**HSA-REI.C.5**
FORWARD 2	Solve a system of a linear equation and a quadratic equation	**HSA-REI.C.7**

SET 20 • 8.EE.C.8.b

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.C.8.b

Solve the system. Show your work.

$y = 3x - 2$

$2x + y = 13$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 20 • MAIN • Q1/5

597

QUESTION ID

S20-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

x = 3, y = 7

GRADING NOTE REQUIRED WORK: Any valid method (substitution, elimination, or another), with the steps shown. Example: substitute $y = 3x - 2$ into $2x + y = 13$ to get $5x - 2 = 13$, so $x = 3$ and $y = 7$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 20 • MAIN • Q1/5

Solve the system. Show your work.

$x + 2y = 10$

$3x - 2y = 6$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 20 • MAIN • Q2/5

598

QUESTION ID

S20-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

x = 4, y = 3

GRADING NOTE REQUIRED WORK: Any valid method, with the steps shown. Example: add the equations to get $4x = 16$, so $x = 4$; then $4 + 2y = 10$, so $y = 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 20 • MAIN • Q2/5

Solve the system.

$y = x + 4$

$y = -2x + 1$

A. $x = 1, y = 5$

B. $x = 3, y = 7$

C. $x = -1, y = 5$

D. $x = -1, y = 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 20 • MAIN • Q3/5

599

QUESTION ID

S20-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. x = −1, y = 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 20 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve the system.

$4x + 3y = 18$

$2x - y = 4$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 20 • MAIN • Q4/5

600

QUESTION ID

S20-M-Q4

TYPE

Short answer

CORRECT ANSWER

x = 3, y = 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 20 • MAIN • Q4/5

True or false?

$x = 2, y = -1$ is a solution of both $3x + y = 5$ and $x - y = 3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 20 • MAIN • Q5/5

601

QUESTION ID

S20-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 20 • MAIN • Q5/5

SET 20 • 8.EE.C.8.b

BACKWARD 1

Evaluate expressions for given values • Grade 6 • Expressions & Equations • 6.EE.A.2.c

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate 2x + y when x = 3 and y = 7.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 20 • BACKWARD 1 • Q1/5

603

QUESTION ID

S20-B1-Q1

TYPE

Short answer

CORRECT ANSWER

13

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 20 • BACKWARD 1 • Q1/5

Evaluate 3x − 2 when x = 4.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 20 • BACKWARD 1 • Q2/5

604

QUESTION ID

S20-B1-Q2

TYPE

Short answer

CORRECT ANSWER

10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 20 • BACKWARD 1 • Q2/5

Evaluate x + 2y when x = 4 and y = 3.

A. 9

B. 10

C. 11

D. 14

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 20 • BACKWARD 1 • Q3/5

605

QUESTION ID

S20-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 20 • BACKWARD 1 • Q3/5

True or false?

When x = 2, the value of 5x − 1 is 9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 20 • BACKWARD 1 • Q4/5

606

QUESTION ID

S20-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 20 • BACKWARD 1 • Q4/5

True or false?

When y = 3, the value of 2y + 4 is 9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 20 • BACKWARD 1 • Q5/5

607

QUESTION ID

S20-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 20 • BACKWARD 1 • Q5/5

SET 20 • 8.EE.C.8.b

BACKWARD 2

Solve equations with the distributive property • Grade 7 • Expressions & Equations • 7.EE.B.4.a

Solve for x.

$5x - 2 = 13$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 20 • BACKWARD 2 • Q1/5

609

QUESTION ID

S20-B2-Q1

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 20 • BACKWARD 2 • Q1/5

Solve for x.

$2x + 3(x - 2) = 9$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 20 • BACKWARD 2 • Q2/5

610

QUESTION ID

S20-B2-Q2

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 20 • BACKWARD 2 • Q2/5

Solve for x.

$4x + 2 = -10$

A. $x = 3$

B. $x = -2$

C. $x = -3$

D. $x = 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 20 • BACKWARD 2 • Q3/5

611

QUESTION ID

S20-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. x = −3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 20 • BACKWARD 2 • Q3/5

True or false?

$x = 4$ is the solution of $3x - 5 = 7$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 20 • BACKWARD 2 • Q4/5

612

QUESTION ID

S20-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 20 • BACKWARD 2 • Q4/5

True or false?

$x = -1$ is the solution of $2x + 9 = 5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 20 • BACKWARD 2 • Q5/5

613

QUESTION ID

S20-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The solution is $x = -2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 20 • BACKWARD 2 • Q5/5

SET 20 • 8.EE.C.8.b

FORWARD 1

Explain why adding equations (elimination) keeps the same solutions • High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5

Add the equations x + y = 9 and x − y = 3.

What new equation do you get, and what is x?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 20 • FORWARD 1 • Q1/5

615

QUESTION ID

S20-F1-Q1

TYPE

Short answer

CORRECT ANSWER

2x = 12; x = 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 20 • FORWARD 1 • Q1/5

Multiply $x + 2y = 7$ by 3, then add the result to $-3x + y = 0$.

What equation results?

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 20 • FORWARD 1 • Q2/5

616

QUESTION ID

S20-F1-Q2

TYPE

Short answer

CORRECT ANSWER

7y = 21 (so y = 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 20 • FORWARD 1 • Q2/5

In a system, one equation is replaced by its sum with a multiple of the other equation.

What happens to the solutions?

A. The solutions always change.

B. The number of solutions doubles.

C. All solutions are removed.

D. The solutions stay the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 20 • FORWARD 1 • Q3/5

617

QUESTION ID

S20-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. The solutions stay the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 20 • FORWARD 1 • Q3/5

True or false?

The system $x + y = 5$ and $x - y = 1$ has the same solution as the system $x + y = 5$ and $2x = 6$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 20 • FORWARD 1 • Q4/5

618

QUESTION ID

S20-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 20 • FORWARD 1 • Q4/5

True or false?

Multiplying one equation of a system by 0 keeps the same solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 20 • FORWARD 1 • Q5/5

619

QUESTION ID

S20-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 20 • FORWARD 1 • Q5/5

SET 20 • 8.EE.C.8.b

FORWARD 2

Solve a system of a linear equation and a quadratic equation • High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7

ANSWER KEY Reduced copy of the question page, with its answer

Solve the system. Give all solutions.

$y = x^2$

$y = 4$

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q1

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 20 • FORWARD 2 • Q1/5

621

QUESTION ID

S20-F2-Q1

TYPE

Short answer

CORRECT ANSWER

(2, 4) and (−2, 4)

GRADING NOTE Both solutions are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q1

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 20 • FORWARD 2 • Q1/5

Solve the system. Give all solutions.

$y = x^2 - 1$

$y = x + 1$

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q2

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 20 • FORWARD 2 • Q2/5

622

QUESTION ID

S20-F2-Q2

TYPE

Short answer

CORRECT ANSWER

(2, 3) and (−1, 0)

GRADING NOTE Both solutions are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S20-F2-Q2**

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 20 • FORWARD 2 • Q2/5

How many solutions does the system have?

$y = x^2 + 3$

$y = 1$

A. Infinitely many

B. Two

C. One

D. None

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q3

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 20 • FORWARD 2 • Q3/5

623

QUESTION ID

S20-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. None

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q3

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 20 • FORWARD 2 • Q3/5

True or false?

The line $y = 2x$ and the parabola $y = x^2$ meet at (0, 0) and (2, 4).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q4

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 20 • FORWARD 2 • Q4/5

624

QUESTION ID

S20-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q4

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 20 • FORWARD 2 • Q4/5

True or false?

A line and a parabola always meet at exactly two points.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q5

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 20 • FORWARD 2 • Q5/5

625

QUESTION ID

S20-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q5

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 20 • FORWARD 2 • Q5/5

SET 21 • 8.EE.C.8.b • EXPRESSIONS & EQUATIONS

Solve simple systems by inspection

MAIN	Grade 8 main question	**8.EE.C.8.b**
BACKWARD 1	Recognize equivalent expressions	**7.EE.A.1**
BACKWARD 2	Decide whether a value makes two equations true	**6.EE.B.5**
FORWARD 1	Solve or classify systems of linear equations	**HSA-REI.C.6**
FORWARD 2	Classify lines as parallel, perpendicular, or neither from their equ..	**HSG-GPE.B.5**

SET 21 • 8.EE.C.8.b

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.C.8.b

Without solving, explain why this system has no solution.

$3x + 2y = 5$

$3x + 2y = 6$

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 21 • MAIN • Q1/5

628

QUESTION ID

S21-M-Q1

TYPE

Short answer

CORRECT ANSWER

The same expression, 3x + 2y, cannot equal both 5 and 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 21 • MAIN • Q1/5

Without solving, how many solutions does this system have? Explain.

$y = 4x - 1$

$y = 4x + 3$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 21 • MAIN • Q2/5

629

QUESTION ID

S21-M-Q2

TYPE

Short answer

CORRECT ANSWER

None. The lines have the same slope and different y-intercepts, so they are parallel.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 21 • MAIN • Q2/5

Which system has infinitely many solutions?

A. $y = x$ and $y = -x$

B. $x + y = 4$ and $x + y = 8$

C. $x + y = 4$ and $x - y = 4$

D. $x + y = 4$ and $2x + 2y = 8$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 21 • MAIN • Q3/5

630

QUESTION ID

S21-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $x + y = 4$ and $2x + 2y = 8$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 21 • MAIN • Q3/5

Without solving, how many solutions does this system have? Explain.

$y = -2x + 5$

$y = 3x + 5$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 21 • MAIN • Q4/5

631

QUESTION ID

S21-M-Q4

TYPE

Short answer

CORRECT ANSWER

One. The slopes are different, so the lines cross once (at (0, 5)).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 21 • MAIN • Q4/5

True or false?

The system $2x - y = 7$ and $6x - 3y = 21$ has no solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 21 • MAIN • Q5/5

632

QUESTION ID

S21-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The second equation is 3 times the first, so there are infinitely many solutions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 21 • MAIN • Q5/5

SET 21 • 8.EE.C.8.b

BACKWARD 1

Recognize equivalent expressions • Grade 7 • Expressions & Equations • 7.EE.A.1

Is 2x + 2y equivalent to 2(x + y)? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 21 • BACKWARD 1 • Q1/5

634

QUESTION ID

S21-B1-Q1

TYPE

Short answer

CORRECT ANSWER

Yes. By the distributive property, 2(x + y) = 2x + 2y.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 21 • BACKWARD 1 • Q1/5

Write 3(2x + y) without parentheses.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 21 • BACKWARD 1 • Q2/5

635

QUESTION ID

S21-B1-Q2

TYPE

Short answer

CORRECT ANSWER

6x + 3y

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 21 • BACKWARD 1 • Q2/5

Which expression is equivalent to 4x + 6y?

A. 6(4x + y)

B. 4(x + 6y)

C. 2(4x + 6y)

D. 2(2x + 3y)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 21 • BACKWARD 1 • Q3/5

636

QUESTION ID

S21-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 2(2x + 3y)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 21 • BACKWARD 1 • Q3/5

True or false?

$5x - 5y = 5(x - y)$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 21 • BACKWARD 1 • Q4/5

637

QUESTION ID

S21-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 21 • BACKWARD 1 • Q4/5

True or false?

$2(x + 3) = 2x + 3$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 21 • BACKWARD 1 • Q5/5

638

QUESTION ID

S21-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 21 • BACKWARD 1 • Q5/5

SET 21 • 8.EE.C.8.b

BACKWARD 2

Decide whether a value makes two equations true • Grade 6 • Expressions & Equations • 6.EE.B.5

Is x = 2 a solution of both x + 3 = 5 and 2x = 4?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 21 • BACKWARD 2 • Q1/5

640

QUESTION ID

S21-B2-Q1

TYPE

Short answer

CORRECT ANSWER

Yes

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 21 • BACKWARD 2 • Q1/5

Which of the values 1, 2, and 3 makes both 2x = 6 and x + 1 = 4 true?

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 21 • BACKWARD 2 • Q2/5

641

QUESTION ID

S21-B2-Q2

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 21 • BACKWARD 2 • Q2/5

Which value makes both x + 2 = 7 and 3x = 15 true?

A. 2

B. 7

C. 3

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 21 • BACKWARD 2 • Q3/5

642

QUESTION ID

S21-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 21 • BACKWARD 2 • Q3/5

True or false?

No value of x makes both x + 1 = 3 and x + 1 = 4 true.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 21 • BACKWARD 2 • Q4/5

643

QUESTION ID

S21-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 21 • BACKWARD 2 • Q4/5

True or false?

$x = 4$ makes both $x - 1 = 3$ and $2x = 6$ true.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 21 • BACKWARD 2 • Q5/5

644

QUESTION ID

S21-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 21 • BACKWARD 2 • Q5/5

SET 21 • 8.EE.C.8.b

FORWARD 1

Solve or classify systems of linear equations • High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6

Solve the system, or say that it has no solution or infinitely many.

$2x + y = 3$

$4x + 2y = 6$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 21 • FORWARD 1 • Q1/5

646

QUESTION ID

S21-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Infinitely many solutions (the second equation is 2 times the first)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 21 • FORWARD 1 • Q1/5

Solve the system.

$3x - y = 5$

$x + y = 7$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 21 • FORWARD 1 • Q2/5

647

QUESTION ID

S21-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$x = 3, y = 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 21 • FORWARD 1 • Q2/5

Which value of k makes the system have infinitely many solutions?

$x + ky = 4$

$2x + 6y = 8$

A. 4

B. 2

C. 6

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 21 • FORWARD 1 • Q3/5

QUESTION ID

S21-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 21 • FORWARD 1 • Q3/5

True or false?

If the two equations of a system describe the same line, every point on the line is a solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 21 • FORWARD 1 • Q4/5

649

QUESTION ID

S21-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 21 • FORWARD 1 • Q4/5

True or false?

The system y = 3x + 1 and y = 3x + 1 has no solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 21 • FORWARD 1 • Q5/5

650

QUESTION ID

S21-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It has infinitely many solutions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 21 • FORWARD 1 • Q5/5

SET 21 • 8.EE.C.8.b

FORWARD 2

Classify lines as parallel, perpendicular, or neither from their equations • High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

Are the lines y = 2x + 1 and 4x − 2y = 6 parallel, perpendicular, or neither? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q1

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 21 • FORWARD 2 • Q1/5

652

QUESTION ID

S21-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Parallel; both have slope 2 (and different y-intercepts).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q1

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 21 • FORWARD 2 • Q1/5

Are the lines $3x + y = 2$ and $x - 3y = 9$ parallel, perpendicular, or neither? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q2

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 21 • FORWARD 2 • Q2/5

653

QUESTION ID

S21-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Perpendicular; the slopes are −3 and 1/3, and their product is −1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q2

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 21 • FORWARD 2 • Q2/5

One line passes through (0, 0) and (2, 4). Another passes through (1, 1) and (3, 5).

How are the lines related?

A. Perpendicular

B. The same line

C. Parallel

D. Neither

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q3

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 21 • FORWARD 2 • Q3/5

654

QUESTION ID

S21-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Parallel

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q3

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 21 • FORWARD 2 • Q3/5

True or false?

Lines with slopes $\frac{2}{3}$ and $-\frac{3}{2}$ are perpendicular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q4

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 21 • FORWARD 2 • Q4/5

655

QUESTION ID

S21-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q4

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 21 • FORWARD 2 • Q4/5

True or false?

The lines $y = x$ and $y = -x$ are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q5

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 21 • FORWARD 2 • Q5/5

656

QUESTION ID

S21-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They are perpendicular.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q5

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 21 • FORWARD 2 • Q5/5

SET 22 • 8.EE.C.8.c • EXPRESSIONS & EQUATIONS

Write and solve systems of linear equations for real-world problems

MAIN	Grade 8 main question	**8.EE.C.8.c**
BACKWARD 1	Solve word problems with one unknown	**7.EE.B.4.a**
BACKWARD 2	Use variables to represent quantities in a situation	**6.EE.B.6**
FORWARD 1	Create equations in two variables to model and solve problems	**HSA-CED.A.2**
FORWARD 2	Represent constraints with a system of inequalities	**HSA-CED.A.3**

SET 22 • 8.EE.C.8.c

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.C.8.c

A school sells 200 tickets. Adult tickets cost $6 and student tickets cost $4. The sales total $940.

Write a system and solve it to find how many of each ticket were sold.

System and answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 22 • MAIN • Q1/5

659

QUESTION ID

S22-M-Q1

TYPE

Short answer

CORRECT ANSWER

The system a + s = 200 and 6a + 4s = 940, and the answer 70 adult and 130 student tickets

GRADING NOTE Both parts are required: a valid equation (or system) and the correct solution. Any valid model is accepted, but the solution must also be correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 22 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Plan A costs $15 plus $0.05 per minute. Plan B costs $0.10 per minute.

Write a system and solve it to find when the plans cost the same.

System and answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 22 • MAIN • Q2/5

660

QUESTION ID

S22-M-Q2

TYPE

Short answer

CORRECT ANSWER

The system c = 15 + 0.05m and c = 0.10m, and the answer 300 minutes (both cost $30)

GRADING NOTE Both parts are required: a valid equation (or system) and the correct solution. Any valid model is accepted, but the solution must also be correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 22 • MAIN • Q2/5

Jo has 25 coins, all dimes and quarters, worth $4.00 in all.

Which system models this?

A. $d - q = 25$ and $0.10d + 0.25q = 4$

B. $d + q = 25$ and $0.10d + 0.25q = 4$

C. $d + q = 4$ and $0.10d + 0.25q = 25$

D. $d + q = 25$ and $0.25d + 0.10q = 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 22 • MAIN • Q3/5

661

QUESTION ID

S22-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $d + q = 25$ and $0.10d + 0.25q = 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 22 • MAIN • Q3/5

Jo has 25 coins, all dimes and quarters, worth $4.00 in all: d + q = 25 and 0.10d + 0.25q = 4.

How many of each coin does Jo have?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 22 • MAIN • Q4/5

662

QUESTION ID

S22-M-Q4

TYPE

Short answer

CORRECT ANSWER

15 dimes and 10 quarters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 22 • MAIN • Q4/5

True or false?

Two numbers have a sum of 40 and a difference of 12. The larger number is 26.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 22 • MAIN • Q5/5

663

QUESTION ID

S22-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 22 • MAIN • Q5/5

SET 22 • 8.EE.C.8.c

BACKWARD 1

Solve word problems with one unknown • Grade 7 • Expressions & Equations • 7.EE.B.4.a

A taxi charges $3 plus $2 per mile. A ride costs $19.

How many miles long was the ride?

Miles: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 22 • BACKWARD 1 • Q1/5

665

QUESTION ID

S22-B1-Q1

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 22 • BACKWARD 1 • Q1/5

Concert tickets cost $9 each, plus a $5 fee for the whole order. An order costs $50.

How many tickets were bought?

Tickets: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 22 • BACKWARD 1 • Q2/5

666

QUESTION ID

S22-B1-Q2

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 22 • BACKWARD 1 • Q2/5

The sum of a number and 3 times the number is 48.

What is the number?

A. 16

B. 45

C. 12

D. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 22 • BACKWARD 1 • Q3/5

667

QUESTION ID

S22-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 22 • BACKWARD 1 • Q3/5

True or false?

If 2x + 7 = 25, then x = 9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 22 • BACKWARD 1 • Q4/5

668

QUESTION ID

S22-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 22 • BACKWARD 1 • Q4/5

True or false?

Items cost $4 each, and shipping is $6. A $30 order has 7 items.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 22 • BACKWARD 1 • Q5/5

669

QUESTION ID

S22-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 4n + 6 = 30 gives n = 6 items.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 22 • BACKWARD 1 • Q5/5

SET 22 • 8.EE.C.8.c

BACKWARD 2

Use variables to represent quantities in a situation • Grade 6 • Expressions & Equations • 6.EE.B.6

ANSWER KEY Reduced copy of the question page, with its answer

Write an expression for the value, in dollars, of d dimes.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 22 • BACKWARD 2 • Q1/5

671

QUESTION ID

S22-B2-Q1

TYPE

Short answer

CORRECT ANSWER

0.10d

GRADING NOTE 0.1d is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 22 • BACKWARD 2 • Q1/5

Adult tickets cost $6.

Write an expression for the cost of a adult tickets.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 22 • BACKWARD 2 • Q2/5

672

QUESTION ID

S22-B2-Q2

TYPE

Short answer

CORRECT ANSWER

6a

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 22 • BACKWARD 2 • Q2/5

There are 25 coins: d dimes and the rest quarters.

Which expression gives the number of quarters?

A. d − 25

B. 25d

C. 25 − d

D. d + 25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 22 • BACKWARD 2 • Q3/5

673

QUESTION ID

S22-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 25 − d

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 22 • BACKWARD 2 • Q3/5

True or false?

q quarters are worth 0.25q dollars.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 22 • BACKWARD 2 • Q4/5

674

QUESTION ID

S22-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 22 • BACKWARD 2 • Q4/5

True or false?

There are s students and 3 more teachers than students. There are 3s teachers.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 22 • BACKWARD 2 • Q5/5

675

QUESTION ID

S22-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE There are s + 3 teachers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 22 • BACKWARD 2 • Q5/5

SET 22 • 8.EE.C.8.c

FORWARD 1

Create equations in two variables to model and solve problems • High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2

A company's cost is C = 2,000 + 15x and its revenue is R = 40x for x items.

Write and solve an equation to find the break-even number of items.

Equation and answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q1

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 22 • FORWARD 1 • Q1/5

677

QUESTION ID

S22-F1-Q1

TYPE

Short answer

CORRECT ANSWER

The equation 2,000 + 15x = 40x and the answer x = 80 items

GRADING NOTE Both parts are required: a valid equation (or system) and the correct solution. Any valid model is accepted, but the solution must also be correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q1

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 22 • FORWARD 1 • Q1/5

A boat travels 24 miles downstream in 2 hours and 24 miles back upstream in 3 hours. Let b be the boat's speed in still water and c the speed of the current.

Write a system of two equations.

System: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q2

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 22 • FORWARD 1 • Q2/5

678

QUESTION ID

S22-F1-Q2

TYPE

Short answer

CORRECT ANSWER

2(b + c) = 24 and
3(b − c) = 24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q2

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 22 • FORWARD 1 • Q2/5

A boat's trips give the system $2(b + c) = 24$ and $3(b - c) = 24$.

What is the boat's speed b in still water?

A. 10 miles per hour

B. 2 miles per hour

C. 12 miles per hour

D. 8 miles per hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q3

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 22 • FORWARD 1 • Q3/5

679

QUESTION ID

S22-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 10 miles per hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q3

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 22 • FORWARD 1 • Q3/5

True or false?

For the system 2(b + c) = 24 and 3(b − c) = 24, the current's speed c is 2 miles per hour.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q4

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 22 • FORWARD 1 • Q4/5

680

QUESTION ID

S22-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q4

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 22 • FORWARD 1 • Q4/5

True or false?

The break-even point is where revenue is twice the cost.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q5

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 22 • FORWARD 1 • Q5/5

681

QUESTION ID
S22-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE At break-even, revenue equals cost.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q5

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 22 • FORWARD 1 • Q5/5

SET 22 • 8.EE.C.8.c

FORWARD 2

Represent constraints with a system of inequalities • High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.3

A bakery has 30 pounds of flour. Each batch of bread uses 2 pounds and each batch of muffins uses 0.5 pound.

Write an inequality for b batches of bread and m batches of muffins.

Inequality: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q1

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.3 • Set 22 • FORWARD 2 • Q1/5

683

QUESTION ID

S22-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$2b + 0.5m \leq 30$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q1

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.3 • Set 22 • FORWARD 2 • Q1/5

A bakery has at most 20 hours of oven time. Each batch of bread takes 1 hour and each batch of muffins takes 2 hours.

Write an inequality for b batches of bread and m batches of muffins.

Inequality: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q2

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.3 • Set 22 • FORWARD 2 • Q2/5

684

QUESTION ID

S22-F2-Q2

TYPE

Short answer

CORRECT ANSWER

b + 2m ≤ 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q2

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.3 • Set 22 • FORWARD 2 • Q2/5

A bakery's limits are $2b + 0.5m \leq 30$ (flour) and $b + 2m \leq 20$ (oven hours).

Can it make 10 batches of bread and 5 batches of muffins?

A. No; both limits are broken.

B. No; the flour runs out.

C. Yes; both limits are met.

D. No; the oven time runs out.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q3

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.3 • Set 22 • FORWARD 2 • Q3/5

685

QUESTION ID

S22-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Yes; both limits are met.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q3

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.3 • Set 22 • FORWARD 2 • Q3/5

True or false?

In a model of batches of bread b and muffins m, the constraints $b \geq 0$ and $m \geq 0$ make sense because batches cannot be negative.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q4

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.3 • Set 22 • FORWARD 2 • Q4/5

686

QUESTION ID

S22-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q4

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.3 • Set 22 • FORWARD 2 • Q4/5

True or false?

The point (12, 6) meets both limits $2b + 0.5m \leq 30$ and $b + 2m \leq 20$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q5

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.3 • Set 22 • FORWARD 2 • Q5/5

687

QUESTION ID
S22-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $12 + 2(6) = 24$, which is more than 20.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q5

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.3 • Set 22 • FORWARD 2 • Q5/5

SET 23 • 8.F.A.1 • FUNCTIONS

Decide whether a relation is a function

MAIN	Grade 8 main question	**8.F.A.1**
BACKWARD 1	Find the output of a rule	**6.EE.C.9**
BACKWARD 2	Read points in the first quadrant	**5.G.A.2**
FORWARD 1	Use function notation and find the domain and range	**HSF-IF.A.1**
FORWARD 2	Find the inverse of a linear function	**HSF-BF.B.4.a**

SET 23 • 8.F.A.1

MAIN

Grade 8 main question • Grade 8 • Functions • 8.F.A.1

True or false?

The table shows y as a function of x.

x	1	2	3	4
y	5	5	7	9

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q1

Grade 8 • Functions • 8.F.A.1 • Set 23 • MAIN • Q1/5

690

QUESTION ID

S23-M-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q1

Grade 8 • Functions • 8.F.A.1 • Set 23 • MAIN • Q1/5

Is the relation in the table a function? Explain.

x	2	3	3	5
y	4	6	8	10

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q2

Grade 8 • Functions • 8.F.A.1 • Set 23 • MAIN • Q2/5

691

QUESTION ID

S23-M-Q2

TYPE

Short answer

CORRECT ANSWER

No. The input 3 has two different outputs (6 and 8).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q2

Grade 8 • Functions • 8.F.A.1 • Set 23 • MAIN • Q2/5

Which set of ordered pairs is a function?

A. (5, 1), (6, 2), (5, 3)

B. (0, 0), (0, 1), (1, 1)

C. (1, 2), (1, 3), (2, 4)

D. (1, 4), (2, 4), (3, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q3

Grade 8 • Functions • 8.F.A.1 • Set 23 • MAIN • Q3/5

692

QUESTION ID

S23-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (1, 4), (2, 4), (3, 4)

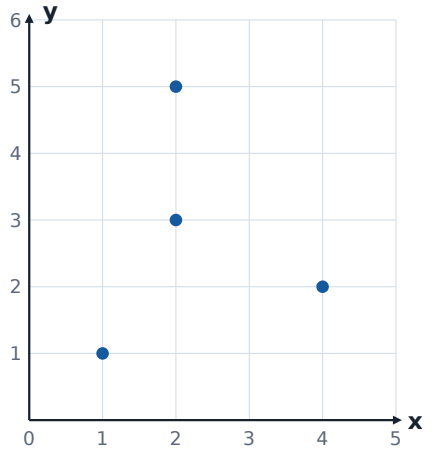
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q3

Grade 8 • Functions • 8.F.A.1 • Set 23 • MAIN • Q3/5

Do the points on the graph show y as a function of x? Explain.

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q4

Grade 8 • Functions • 8.F.A.1 • Set 23 • MAIN • Q4/5

693

QUESTION ID

S23-M-Q4

TYPE

Short answer

CORRECT ANSWER

No. The input x = 2 has two outputs (3 and 5).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q4

Grade 8 • Functions • 8.F.A.1 • Set 23 • MAIN • Q4/5

True or false?

A rule assigns each month of the year to the students in a class born in that month. This rule is a function.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q5

Grade 8 • Functions • 8.F.A.1 • Set 23 • MAIN • Q5/5

694

QUESTION ID

S23-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE One month can be paired with several students, so an input has more than one output.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q5

Grade 8 • Functions • 8.F.A.1 • Set 23 • MAIN • Q5/5

SET 23 • 8.F.A.1

BACKWARD 1

Find the output of a rule • Grade 6 • Expressions & Equations • 6.EE.C.9

The rule is y = 2x + 1.

What is y when x = 4?

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 23 • BACKWARD 1 • Q1/5

696

QUESTION ID

S23-B1-Q1

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 23 • BACKWARD 1 • Q1/5

The table follows a pattern.

What is y when x = 4?

x	1	2	3
y	4	7	10

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 23 • BACKWARD 1 • Q2/5

697

QUESTION ID

S23-B1-Q2

TYPE

Short answer

CORRECT ANSWER

13

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 23 • BACKWARD 1 • Q2/5

The rule is: output = input × 3.

What is the output for an input of 7?

A. 4

B. 73

C. 21

D. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 23 • BACKWARD 1 • Q3/5

698

QUESTION ID

S23-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 21

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 23 • BACKWARD 1 • Q3/5

True or false?

For $y = x - 5$, an input of $x = 9$ gives $y = 4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 23 • BACKWARD 1 • Q4/5

699

QUESTION ID

S23-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 23 • BACKWARD 1 • Q4/5

True or false?

For y = 4x, an input of x = 3 gives y = 7.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 23 • BACKWARD 1 • Q5/5

700

QUESTION ID

S23-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q5

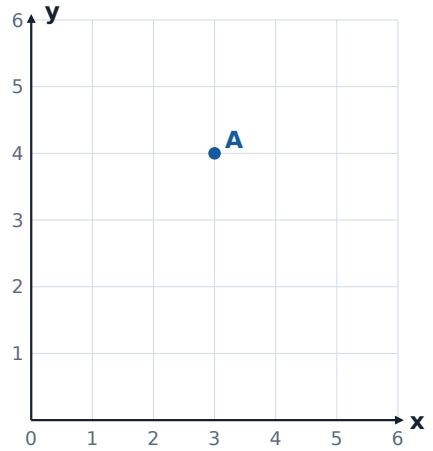
Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 23 • BACKWARD 1 • Q5/5

SET 23 • 8.F.A.1

BACKWARD 2

Read points in the first quadrant • Grade 5 • Geometry • 5.G.A.2

What are the coordinates of point A?



Coordinates: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q1

Grade 5 • Geometry • 5.G.A.2 • Set 23 • BACKWARD 2 • Q1/5

QUESTION ID

S23-B2-Q1

TYPE

Short answer

CORRECT ANSWER

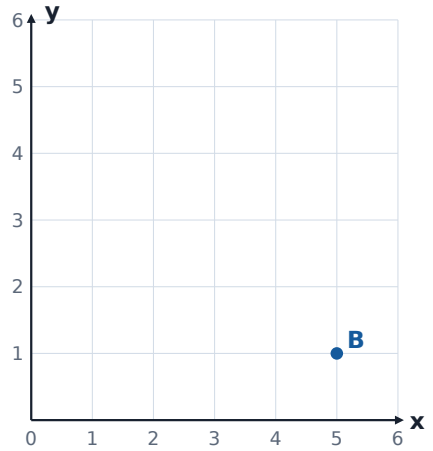
(3, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q1

Grade 5 • Geometry • 5.G.A.2 • Set 23 • BACKWARD 2 • Q1/5

What are the coordinates of point B?



Coordinates: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q2

Grade 5 • Geometry • 5.G.A.2 • Set 23 • BACKWARD 2 • Q2/5

703

QUESTION ID

S23-B2-Q2

TYPE

Short answer

CORRECT ANSWER

(5, 1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q2

Grade 5 • Geometry • 5.G.A.2 • Set 23 • BACKWARD 2 • Q2/5

Which point is 2 units to the right of the origin and 5 units up?

A. (5, 5)

B. (5, 2)

C. (2, 2)

D. (2, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q3

Grade 5 • Geometry • 5.G.A.2 • Set 23 • BACKWARD 2 • Q3/5

704

QUESTION ID

S23-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (2, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q3

Grade 5 • Geometry • 5.G.A.2 • Set 23 • BACKWARD 2 • Q3/5

True or false?

The point (0, 3) is on the y-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q4

Grade 5 • Geometry • 5.G.A.2 • Set 23 • BACKWARD 2 • Q4/5

705

QUESTION ID

S23-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q4

Grade 5 • Geometry • 5.G.A.2 • Set 23 • BACKWARD 2 • Q4/5

True or false?

The point (4, 0) is 4 units up from the origin.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q5

Grade 5 • Geometry • 5.G.A.2 • Set 23 • BACKWARD 2 • Q5/5

706

QUESTION ID

S23-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q5

Grade 5 • Geometry • 5.G.A.2 • Set 23 • BACKWARD 2 • Q5/5

SET 23 • 8.F.A.1

FORWARD 1

Use function notation and find the domain and range • High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.A.1

A function f is made of the pairs (1, 3), (2, 5), and (4, 9).

What is its domain?

Domain: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q1

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.A.1 • Set 23 • FORWARD 1 • Q1/5

708

QUESTION ID

S23-F1-Q1

TYPE

Short answer

CORRECT ANSWER

1, 2, and 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q1

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.A.1 • Set 23 • FORWARD 1 • Q1/5

A function f is made of the pairs (1, 3), (2, 5), and (4, 9).

What is its range?

Range: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q2

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.A.1 • Set 23 • FORWARD 1 • Q2/5

709

QUESTION ID

S23-F1-Q2

TYPE

Short answer

CORRECT ANSWER

3, 5, and 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q2

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.A.1 • Set 23 • FORWARD 1 • Q2/5

The output of f is 7 when the input is 2.

Which statement says this in function notation?

A. $f = 2 \times 7$

B. $f(7) = 2$

C. $2f = 7$

D. $f(2) = 7$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q3

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.A.1 • Set 23 • FORWARD 1 • Q3/5

710

QUESTION ID

S23-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. f(2) = 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q3

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.A.1 • Set 23 • FORWARD 1 • Q3/5

True or false?

If $f(x) = x^2$, then $f(-3) = 9$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q4

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.A.1 • Set 23 • FORWARD 1 • Q4/5

711

QUESTION ID

S23-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q4

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.A.1 • Set 23 • FORWARD 1 • Q4/5

True or false?

A function can assign two different outputs to the same input.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q5

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.A.1 • Set 23 • FORWARD 1 • Q5/5

712

QUESTION ID

S23-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q5

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.A.1 • Set 23 • FORWARD 1 • Q5/5

SET 23 • 8.F.A.1

FORWARD 2

Find the inverse of a linear function • High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.4.a

$f(x) = 2x + 3$

Write the inverse function.

$f^{-1}(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q1

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.4.a • Set 23 • FORWARD 2 • Q1/5

714

QUESTION ID

S23-F2-Q1

TYPE

Short answer

CORRECT ANSWER

(x − 3)/2

GRADING NOTE $\frac{x}{2} - \frac{3}{2}$ is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S23-F2-Q1**

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.4.a • Set 23 • FORWARD 2 • Q1/5

$f(x) = 2x + 3$

Solve $f(x) = 11$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q2

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.4.a • Set 23 • FORWARD 2 • Q2/5

715

QUESTION ID

S23-F2-Q2

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q2

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.4.a • Set 23 • FORWARD 2 • Q2/5

$g(x) = \frac{x}{5}$

What is the inverse function?

A. $g^{-1}(x) = x + 5$

B. $g^{-1}(x) = x - 5$

C. $g^{-1}(x) = 5x$

D. $g^{-1}(x) = \frac{5}{x}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q3

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.4.a • Set 23 • FORWARD 2 • Q3/5

716

QUESTION ID

S23-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $g^{-1}(x) = 5x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q3

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.4.a • Set 23 • FORWARD 2 • Q3/5

True or false?

If $f(2) = 7$ and f has an inverse, then $f^{-1}(7) = 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q4

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.4.a • Set 23 • FORWARD 2 • Q4/5

717

QUESTION ID

S23-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q4

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.4.a • Set 23 • FORWARD 2 • Q4/5

True or false?

Every function has an inverse function.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q5

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.4.a • Set 23 • FORWARD 2 • Q5/5

718

QUESTION ID

S23-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $f(x) = x^2$ gives the same output for 2 and −2, so it has no inverse on all real numbers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q5

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.4.a • Set 23 • FORWARD 2 • Q5/5

SET 24 • 8.F.A.1 • FUNCTIONS

Graph a function as the set of ordered pairs of inputs and outputs

MAIN	Grade 8 main question	**8.F.A.1**
BACKWARD 1	Make ordered pairs from an equation	**6.EE.C.9**
BACKWARD 2	Graph points from a pattern	**5.G.A.2**
FORWARD 1	Choose a domain that fits a function's situation	**HSF-IF.B.5**
FORWARD 2	Graph quadratic functions and show their key features	**HSF-IF.C.7.a**

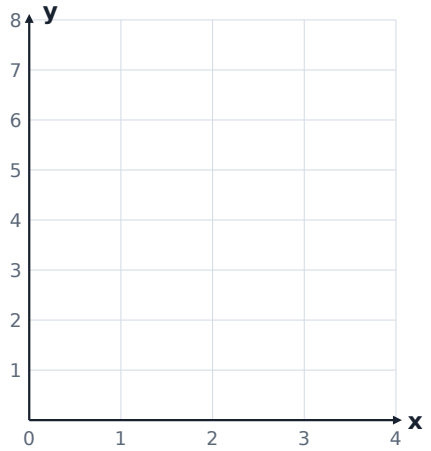
SET 24 • 8.F.A.1

MAIN

Grade 8 main question • Grade 8 • Functions • 8.F.A.1

The function y = 2x + 1 has the inputs 0, 1, 2, and 3.

Plot the ordered pairs (input, output) on the grid.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q1

Grade 8 • Functions • 8.F.A.1 • Set 24 • MAIN • Q1/5

721

QUESTION ID

S24-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Points at (0, 1), (1, 3), (2, 5), and (3, 7)

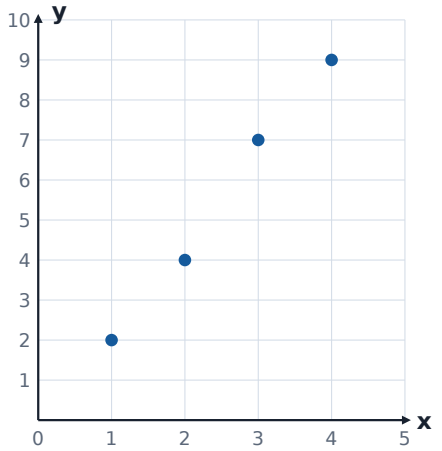
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q1

Grade 8 • Functions • 8.F.A.1 • Set 24 • MAIN • Q1/5

The points show a function.

What is the output when the input is 3?



Output: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q2

Grade 8 • Functions • 8.F.A.1 • Set 24 • MAIN • Q2/5

722

QUESTION ID

S24-M-Q2

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q2

Grade 8 • Functions • 8.F.A.1 • Set 24 • MAIN • Q2/5

The graph of a function contains the point (4, 10).

Which statement is true?

A. When the input is 4, the output is 10.

B. When the input is 10, the output is 4.

C. The graph has only one point.

D. The function adds 4 to every input.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q3

Grade 8 • Functions • 8.F.A.1 • Set 24 • MAIN • Q3/5

723

QUESTION ID

S24-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. When the input is 4, the output is 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q3

Grade 8 • Functions • 8.F.A.1 • Set 24 • MAIN • Q3/5

The rule is "output = input squared."

Write the ordered pairs (input, output) for the inputs −2, −1, 0, 1, and 2.

Ordered pairs: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q4

Grade 8 • Functions • 8.F.A.1 • Set 24 • MAIN • Q4/5

724

QUESTION ID

S24-M-Q4

TYPE

Short answer

CORRECT ANSWER

(−2, 4), (−1, 1), (0, 0), (1, 1), (2, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q4

Grade 8 • Functions • 8.F.A.1 • Set 24 • MAIN • Q4/5

True or false?

The graph of a function can contain both (2, 5) and (2, 8).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q5

Grade 8 • Functions • 8.F.A.1 • Set 24 • MAIN • Q5/5

725

QUESTION ID

S24-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The input 2 cannot have two outputs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q5

Grade 8 • Functions • 8.F.A.1 • Set 24 • MAIN • Q5/5

SET 24 • 8.F.A.1

BACKWARD 1

Make ordered pairs from an equation • Grade 6 • Expressions & Equations • 6.EE.C.9

For y = 3x, what are the values of y when x = 0, 1, and 2?

Values: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 24 • BACKWARD 1 • Q1/5

727

QUESTION ID

S24-B1-Q1

TYPE

Short answer

CORRECT ANSWER

0, 3, and 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 24 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

For y = x + 4, what is y when x = 5?

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 24 • BACKWARD 1 • Q2/5

728

QUESTION ID

S24-B1-Q2

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 24 • BACKWARD 1 • Q2/5

Which pair (x, y) fits y = 5x?

A. (2, 7)

B. (10, 2)

C. (2, 10)

D. (5, 1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 24 • BACKWARD 1 • Q3/5

729

QUESTION ID

S24-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. (2, 10)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 24 • BACKWARD 1 • Q3/5

True or false?

The pair (3, 9) fits y = 3x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 24 • BACKWARD 1 • Q4/5

730

QUESTION ID

S24-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 24 • BACKWARD 1 • Q4/5

True or false?

The pair (4, 6) fits y = x + 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 24 • BACKWARD 1 • Q5/5

731

QUESTION ID

S24-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q5

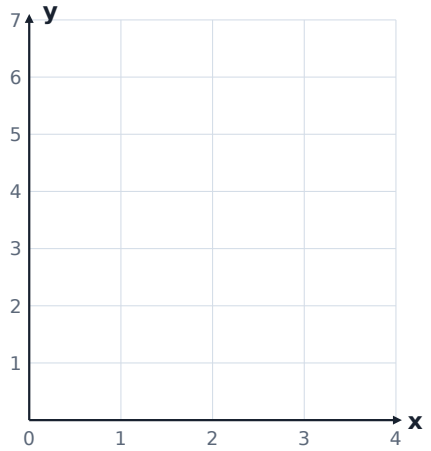
Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 24 • BACKWARD 1 • Q5/5

SET 24 • 8.F.A.1

BACKWARD 2

Graph points from a pattern • Grade 5 • Geometry • 5.G.A.2

Plot the points (1, 2), (2, 4), and (3, 6) on the grid.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q1

Grade 5 • Geometry • 5.G.A.2 • Set 24 • BACKWARD 2 • Q1/5

733

QUESTION ID

S24-B2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Points at (1, 2), (2, 4), and (3, 6)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q1

Grade 5 • Geometry • 5.G.A.2 • Set 24 • BACKWARD 2 • Q1/5

The y-values start at 1 and increase by 2: (1, 1), (2, 3), (3, 5), ...

What is the point for x = 4?

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q2

Grade 5 • Geometry • 5.G.A.2 • Set 24 • BACKWARD 2 • Q2/5

734

QUESTION ID

S24-B2-Q2

TYPE

Short answer

CORRECT ANSWER

(4, 7)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q2

Grade 5 • Geometry • 5.G.A.2 • Set 24 • BACKWARD 2 • Q2/5

Which point is 3 units to the right of the origin and 1 unit up?

A. (3, 1)

B. (3, 3)

C. (1, 3)

D. (1, 1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q3

Grade 5 • Geometry • 5.G.A.2 • Set 24 • BACKWARD 2 • Q3/5

735

QUESTION ID

S24-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. (3, 1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q3

Grade 5 • Geometry • 5.G.A.2 • Set 24 • BACKWARD 2 • Q3/5

True or false?

The point (2, 6) is 2 units to the right of the origin and 6 units up.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q4

Grade 5 • Geometry • 5.G.A.2 • Set 24 • BACKWARD 2 • Q4/5

736

QUESTION ID

S24-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q4

Grade 5 • Geometry • 5.G.A.2 • Set 24 • BACKWARD 2 • Q4/5

True or false?

To plot (5, 2), you move 5 units up and 2 units to the right.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q5

Grade 5 • Geometry • 5.G.A.2 • Set 24 • BACKWARD 2 • Q5/5

737

QUESTION ID

S24-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q5

Grade 5 • Geometry • 5.G.A.2 • Set 24 • BACKWARD 2 • Q5/5

SET 24 • 8.F.A.1

FORWARD 1

Choose a domain that fits a function's situation • High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.5

A function gives the cost of n concert tickets at $12 each. A buyer may buy at most 8 tickets.

What is a sensible domain?

Domain: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q1

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.5 • Set 24 • FORWARD 1 • Q1/5

739

QUESTION ID

S24-F1-Q1

TYPE

Short answer

CORRECT ANSWER

The whole numbers 0 through 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q1

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.5 • Set 24 • FORWARD 1 • Q1/5

h(t) gives the height of a ball t seconds after it is thrown, until it lands at t = 3.

What is a sensible domain?

Domain: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q2

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.5 • Set 24 • FORWARD 1 • Q2/5

740

QUESTION ID

S24-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$0 \leq t \leq 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q2

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.5 • Set 24 • FORWARD 1 • Q2/5

p is the number of people in a car that seats 5.

Which domain makes sense for p?

A. $-5 \leq p \leq 5$

B. 0, 1, 2, 3, 4, 5

C. All real numbers

D. $p \geq 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q3

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.5 • Set 24 • FORWARD 1 • Q3/5

741

QUESTION ID

S24-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 0, 1, 2, 3, 4, 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q3

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.5 • Set 24 • FORWARD 1 • Q3/5

True or false?

For the area of a square as a function of its side length, the domain contains only positive numbers.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q4

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.5 • Set 24 • FORWARD 1 • Q4/5

742

QUESTION ID

S24-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q4

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.5 • Set 24 • FORWARD 1 • Q4/5

True or false?

A function that gives the number of buses needed for n students should have all real numbers as its domain.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q5

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.5 • Set 24 • FORWARD 1 • Q5/5

743

QUESTION ID

S24-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q5

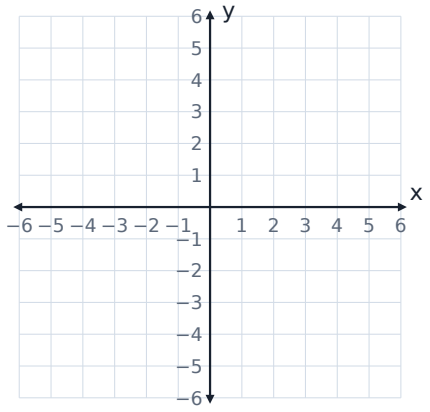
High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.5 • Set 24 • FORWARD 1 • Q5/5

SET 24 • 8.F.A.1

FORWARD 2

Graph quadratic functions and show their key features • High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.a

Graph $y = x^2 - 4$ on the coordinate plane. Mark the vertex and the x-intercepts.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q1

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 24 • FORWARD 2 • Q1/5

745

QUESTION ID

S24-F2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

A U-shaped parabola with its vertex at (0, −4), crossing the x-axis at (−2, 0) and (2, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q1

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 24 • FORWARD 2 • Q1/5

What is the vertex of the graph of $y = (x - 1)^2 + 2$?

Vertex: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q2

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 24 • FORWARD 2 • Q2/5

746

QUESTION ID

S24-F2-Q2

TYPE

Short answer

CORRECT ANSWER

(1, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q2

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 24 • FORWARD 2 • Q2/5

Where does the graph of $y = x^2 - 9$ cross the x-axis?

A. At (3, 0) and (−3, 0)

B. At (9, 0) and (−9, 0)

C. At (0, −9)

D. At (3, 0) only

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q3

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 24 • FORWARD 2 • Q3/5

747

QUESTION ID

S24-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. At (3, 0) and (−3, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q3

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 24 • FORWARD 2 • Q3/5

True or false?

The graph of $y = -x^2$ opens downward.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q4

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 24 • FORWARD 2 • Q4/5

748

QUESTION ID

S24-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q4

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 24 • FORWARD 2 • Q4/5

True or false?

The graph of $y = 2x + 3$ has a vertex.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q5

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 24 • FORWARD 2 • Q5/5

749

QUESTION ID

S24-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q5

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 24 • FORWARD 2 • Q5/5

SET 25 • 8.F.A.2 • FUNCTIONS

Compare two functions represented in different ways

MAIN	Grade 8 main question	**8.F.A.2**
BACKWARD 1	Find a unit rate from a table, graph, or equation	**7.RP.A.2.b**
BACKWARD 2	Find the starting value of a relationship	**6.EE.C.9**
FORWARD 1	Compare average rates of change of linear and nonlinear function…	**HSF-IF.B.6**
FORWARD 2	Describe the effect of transformations on the graph of a function	**HSF-BF.B.3**

SET 25 • 8.F.A.2

MAIN

Grade 8 main question • Grade 8 • Functions • 8.F.A.2

f(x) = 3x + 2. The table shows g.

Which function has the greater rate of change?

x	0	2	4
g(x)	1	9	17

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q1

Grade 8 • Functions • 8.F.A.2 • Set 25 • MAIN • Q1/5

752

QUESTION ID

S25-M-Q1

TYPE

Short answer

CORRECT ANSWER

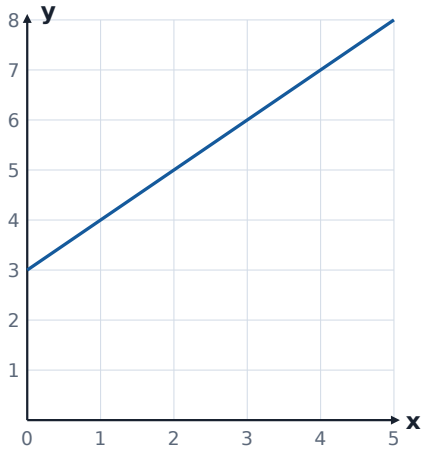
g (its rate is 4; f's is 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q1

Grade 8 • Functions • 8.F.A.2 • Set 25 • MAIN • Q1/5

The graph shows f.
$g(x) = -2x + 5$.

Which function has the greater initial value?



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q2

Grade 8 • Functions • 8.F.A.2 • Set 25 • MAIN • Q2/5

753

QUESTION ID

S25-M-Q2

TYPE

Short answer

CORRECT ANSWER

g (5 is greater than 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q2

Grade 8 • Functions • 8.F.A.2 • Set 25 • MAIN • Q2/5

Function A is y = 0.5x + 10. Function B is a line through (0, 4) and (2, 8).

Which statement is true?

A. B has the greater rate of change; A has the greater initial value.

B. B has the greater rate of change and the greater initial value.

C. A has the greater rate of change and the greater initial value.

D. They have the same rate of change.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q3

Grade 8 • Functions • 8.F.A.2 • Set 25 • MAIN • Q3/5

754

QUESTION ID

S25-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. B has the greater rate of change; A has the greater initial value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q3

Grade 8 • Functions • 8.F.A.2 • Set 25 • MAIN • Q3/5

Car A travels d = 45t miles in t hours. The table shows Car B.

Which car is faster, and by how much?

Hours	1	2	3
Miles	50	100	150

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q4

Grade 8 • Functions • 8.F.A.2 • Set 25 • MAIN • Q4/5

755

QUESTION ID

S25-M-Q4

TYPE

Short answer

CORRECT ANSWER

Car B, by 5 miles per hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q4

Grade 8 • Functions • 8.F.A.2 • Set 25 • MAIN • Q4/5

True or false?

A function with x-values 0, 1, 2 and y-values 6, 4, 2 decreases faster than $y = -x + 6$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q5

Grade 8 • Functions • 8.F.A.2 • Set 25 • MAIN • Q5/5

QUESTION ID

S25-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q5

Grade 8 • Functions • 8.F.A.2 • Set 25 • MAIN • Q5/5

SET 25 • 8.F.A.2

BACKWARD 1

Find a unit rate from a table, graph, or equation • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b

The table shows a proportional relationship.

What is the unit rate?

x	2	4	6
y	5	10	15

Unit rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 25 • BACKWARD 1 • Q1/5

758

QUESTION ID

S25-B1-Q1

TYPE

Short answer

CORRECT ANSWER

2.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 25 • BACKWARD 1 • Q1/5

What is the unit rate in y = 7x?

Unit rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 25 • BACKWARD 1 • Q2/5

759

QUESTION ID

S25-B1-Q2

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 25 • BACKWARD 1 • Q2/5

A proportional graph passes through (0, 0) and (3, 12).

What is the unit rate?

A. 36

B. 12

C. 3

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 25 • BACKWARD 1 • Q3/5

760

QUESTION ID

S25-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 25 • BACKWARD 1 • Q3/5

True or false?

A proportional table with the pairs (1, 9) and (2, 18) has a unit rate of 9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 25 • BACKWARD 1 • Q4/5

761

QUESTION ID

S25-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 25 • BACKWARD 1 • Q4/5

True or false?

The unit rate in y = 0.5x is 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 25 • BACKWARD 1 • Q5/5

762

QUESTION ID

S25-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 25 • BACKWARD 1 • Q5/5

SET 25 • 8.F.A.2

BACKWARD 2

Find the starting value of a relationship • Grade 6 • Expressions & Equations • 6.EE.C.9

ANSWER KEY Reduced copy of the question page, with its answer

For y = 4x + 7, what is y when x = 0?

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 25 • BACKWARD 2 • Q1/5

764

QUESTION ID

S25-B2-Q1

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 25 • BACKWARD 2 • Q1/5

For the table, what is y when x = 0?

x	0	1	2
y	9	11	13

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 25 • BACKWARD 2 • Q2/5

765

QUESTION ID

S25-B2-Q2

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 25 • BACKWARD 2 • Q2/5

Which equation gives y = 5 when x = 0?

A. y = 5x + 2

B. y = 2x + 5

C. y = 5x

D. y = x − 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 25 • BACKWARD 2 • Q3/5

766

QUESTION ID

S25-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. y = 2x + 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 25 • BACKWARD 2 • Q3/5

True or false?

For y = 3x + 1, y = 1 when x = 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 25 • BACKWARD 2 • Q4/5

767

QUESTION ID

S25-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 25 • BACKWARD 2 • Q4/5

True or false?

For y = 6x, y = 6 when x = 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 25 • BACKWARD 2 • Q5/5

768

QUESTION ID

S25-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 25 • BACKWARD 2 • Q5/5

SET 25 • 8.F.A.2

FORWARD 1

Compare average rates of change of linear and nonlinear functions over intervals • High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6

$f(x) = x^2$. The table shows g.

Find the average rate of change of each function from $x = 0$ to $x = 3$.

x	0	1	2	3
g(x)	0	3	6	9

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q1

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 25 • FORWARD 1 • Q1/5

770

QUESTION ID

S25-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Both are 3: f goes from 0 to 9 and g goes from 0 to 9 over 3 units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q1

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 25 • FORWARD 1 • Q1/5

f(x) = 2^x and g(x) = 3x + 5.

Find the average rate of change of each function from x = 0 to x = 4. Which is greater?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q2

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 25 • FORWARD 1 • Q2/5

771

QUESTION ID

S25-F1-Q2

TYPE

Short answer

CORRECT ANSWER

f:
(16 − 1) ÷ 4 = 3.75;
g: 3. f has the greater average rate of change.

GRADING NOTE Both rates and the comparison are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q2

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 25 • FORWARD 1 • Q2/5

$f(x) = x^2$ and $g(x) = 4x$.

Which statement about the average rates of change from $x = 1$ to $x = 3$ is true?

A. f's rate is greater.

B. f's rate is 9 and g's rate is 12.

C. g's rate is greater.

D. They are equal (both are 4).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q3

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 25 • FORWARD 1 • Q3/5

772

QUESTION ID

S25-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. They are equal (both are 4).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q3

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 25 • FORWARD 1 • Q3/5

True or false?

The average rate of change of $f(x) = -x^2 + 4$ from $x = 0$ to $x = 2$ is -2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q4

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 25 • FORWARD 1 • Q4/5

773

QUESTION ID

S25-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q4

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 25 • FORWARD 1 • Q4/5

True or false?

The average rate of change of a linear function depends on which interval you use.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q5

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 25 • FORWARD 1 • Q5/5

774

QUESTION ID

S25-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A linear function has the same rate over every interval.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q5

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 25 • FORWARD 1 • Q5/5

SET 25 • 8.F.A.2

FORWARD 2

Describe the effect of transformations on the graph of a function • High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.3

How does the graph of g(x) = f(x) + 3 compare with the graph of f?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q1

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 25 • FORWARD 2 • Q1/5

776

QUESTION ID

S25-F2-Q1

TYPE

Short answer

CORRECT ANSWER

It is the graph of f shifted up 3 units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q1

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 25 • FORWARD 2 • Q1/5

How does the graph of $g(x) = (x - 2)^2$ compare with the graph of $y = x^2$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q2

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 25 • FORWARD 2 • Q2/5

777

QUESTION ID

S25-F2-Q2

TYPE

Short answer

CORRECT ANSWER

It is shifted 2 units to the right.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q2

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 25 • FORWARD 2 • Q2/5

Which function shifts the graph of y = |x| down 4 units?

A. y = |x| − 4

B. y = |x + 4|

C. y = 4|x|

D. y = |x − 4|

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q3

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 25 • FORWARD 2 • Q3/5

778

QUESTION ID

S25-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. y = |x| − 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q3

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 25 • FORWARD 2 • Q3/5

True or false?

The graph of $y = -f(x)$ is the reflection of the graph of $y = f(x)$ across the x-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q4

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 25 • FORWARD 2 • Q4/5

779

QUESTION ID

S25-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q4

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 25 • FORWARD 2 • Q4/5

True or false?

The graph of y = f(x + 5) is the graph of y = f(x) shifted 5 units to the right.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q5

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 25 • FORWARD 2 • Q5/5

780

QUESTION ID

S25-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is shifted 5 units to the left.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q5

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 25 • FORWARD 2 • Q5/5

SET 26 • 8.F.A.3 • FUNCTIONS

Decide whether a function is linear or nonlinear

MAIN	Grade 8 main question	**8.F.A.3**
BACKWARD 1	Decide whether a table shows a proportional relationship	**7.RP.A.2.a**
BACKWARD 2	Find how much y changes for each change in x	**6.EE.C.9**
FORWARD 1	Show that linear functions grow by equal differences and expone..	**HSF-LE.A.1.a**
FORWARD 2	Use factoring to find the zeros and maximum of a quadratic functi..	**HSF-IF.C.8.a**

SET 26 • 8.F.A.3

MAIN

Grade 8 main question • Grade 8 • Functions • 8.F.A.3

Which equation is a linear function?

A. $y = 4x - 7$

B. $y = x^2 + 1$

C. $y = 2^x$

D. $y = \frac{3}{x}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q1

Grade 8 • Functions • 8.F.A.3 • Set 26 • MAIN • Q1/5

783

QUESTION ID

S26-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. y = 4x − 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q1

Grade 8 • Functions • 8.F.A.3 • Set 26 • MAIN • Q1/5

Is the function in the table linear? Explain.

x	0	1	2	3
y	1	2	5	10

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q2

Grade 8 • Functions • 8.F.A.3 • Set 26 • MAIN • Q2/5

784

QUESTION ID

S26-M-Q2

TYPE

Short answer

CORRECT ANSWER

No. y increases by 1, then 3, then 5, so the rate of change is not constant.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q2

Grade 8 • Functions • 8.F.A.3 • Set 26 • MAIN • Q2/5

$A = s^2$ gives the area of a square with side length s.

Is A a linear function of s? Use three points to support your answer.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q3

Grade 8 • Functions • 8.F.A.3 • Set 26 • MAIN • Q3/5

785

QUESTION ID

S26-M-Q3

TYPE

Short answer

CORRECT ANSWER

No. (1, 1), (2, 4), and (3, 9) are not on one line: the area increases by 3, then by 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q3

Grade 8 • Functions • 8.F.A.3 • Set 26 • MAIN • Q3/5

Is the function in the table linear? Explain.

x	0	2	4	6
y	7	4	1	−2

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q4

Grade 8 • Functions • 8.F.A.3 • Set 26 • MAIN • Q4/5

786

QUESTION ID

S26-M-Q4

TYPE

Short answer

CORRECT ANSWER

Yes. y decreases by 3 each time x increases by 2, a constant rate of change of −1.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q4

Grade 8 • Functions • 8.F.A.3 • Set 26 • MAIN • Q4/5

True or false?

y = 3(x + 2) is a linear function.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q5

Grade 8 • Functions • 8.F.A.3 • Set 26 • MAIN • Q5/5

787

QUESTION ID

S26-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q5

Grade 8 • Functions • 8.F.A.3 • Set 26 • MAIN • Q5/5

SET 26 • 8.F.A.3

BACKWARD 1

Decide whether a table shows a proportional relationship • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a

True or false?

The table shows a proportional relationship.

x	1	2	3
y	4	8	12

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 26 • BACKWARD 1 • Q1/5

789

QUESTION ID

S26-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 26 • BACKWARD 1 • Q1/5

Is the relationship in the table proportional? Explain.

x	1	2	3
y	3	5	7

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 26 • BACKWARD 1 • Q2/5

790

QUESTION ID

S26-B1-Q2

TYPE

Short answer

CORRECT ANSWER

No. The ratios 3, 2.5, and about 2.33 are not equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 26 • BACKWARD 1 • Q2/5

Which table shows a proportional relationship?

A. x: 0, 1, 2 and y: 1, 2, 3

B. x: 1, 2, 3 and y: 1, 4, 9

C. x: 1, 2, 3 and y: 2, 3, 4

D. x: 2, 4, 6 and y: 3, 6, 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 26 • BACKWARD 1 • Q3/5

791

QUESTION ID

S26-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. x: 2, 4, 6 and y: 3, 6, 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 26 • BACKWARD 1 • Q3/5

The pairs (5, 2) and (10, 4) are from a proportional relationship.

What is the constant of proportionality (y per x)?

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 26 • BACKWARD 1 • Q4/5

792

QUESTION ID

S26-B1-Q4

TYPE

Short answer

CORRECT ANSWER

0.4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 26 • BACKWARD 1 • Q4/5

True or false?

y = x + 1 is a proportional relationship.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 26 • BACKWARD 1 • Q5/5

793

QUESTION ID

S26-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 26 • BACKWARD 1 • Q5/5

SET 26 • 8.F.A.3

BACKWARD 2

Find how much y changes for each change in x • Grade 6 • Expressions & Equations • 6.EE.C.9

How much does y increase each time x increases by 1?

x	0	1	2	3
y	5	8	11	14

Increase: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 26 • BACKWARD 2 • Q1/5

795

QUESTION ID

S26-B2-Q1

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 26 • BACKWARD 2 • Q1/5

How much does y increase each time x increases by 2?

x	0	2	4
y	1	5	9

Increase: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 26 • BACKWARD 2 • Q2/5

796

QUESTION ID

S26-B2-Q2

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 26 • BACKWARD 2 • Q2/5

In y = 6x, how much does y increase when x increases by 1?

A. 1

B. 7

C. 6

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 26 • BACKWARD 2 • Q3/5

797

QUESTION ID

S26-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 26 • BACKWARD 2 • Q3/5

True or false?

In the table with x-values 1, 2, 3 and y-values 10, 20, 30, y increases by 10 each time x increases by 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 26 • BACKWARD 2 • Q4/5

798

QUESTION ID

S26-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 26 • BACKWARD 2 • Q4/5

True or false?

In y = x + 9, y increases by 9 each time x increases by 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 26 • BACKWARD 2 • Q5/5

799

QUESTION ID

S26-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 26 • BACKWARD 2 • Q5/5

SET 26 • 8.F.A.3

FORWARD 1

Show that linear functions grow by equal differences and exponential functions by equal factors • High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a